

Ray Tracing

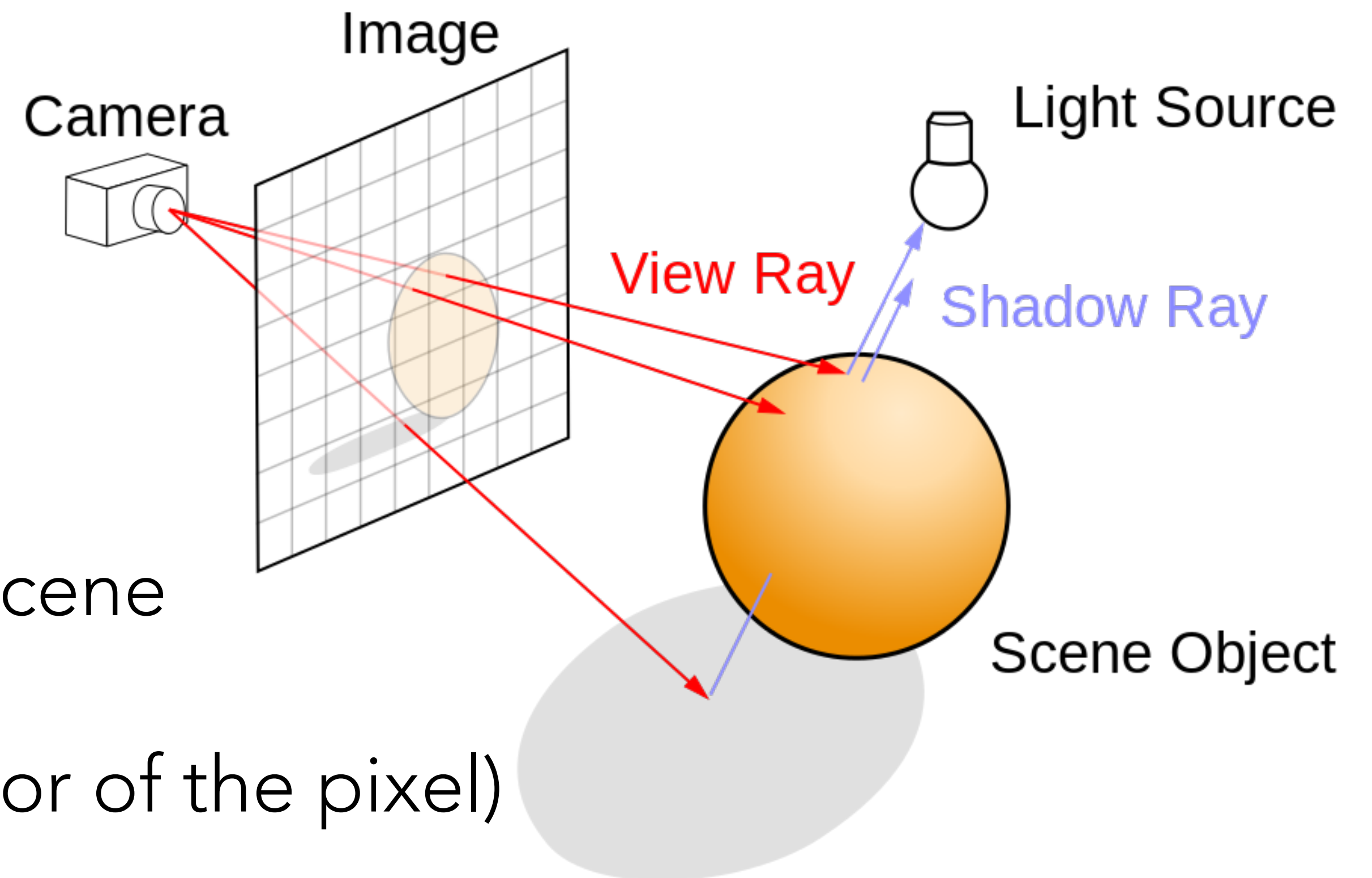


Assignment 2

- **Due: October 8 before class (10 am Pacific)**
- Should be able to start with Ex.1 after this class

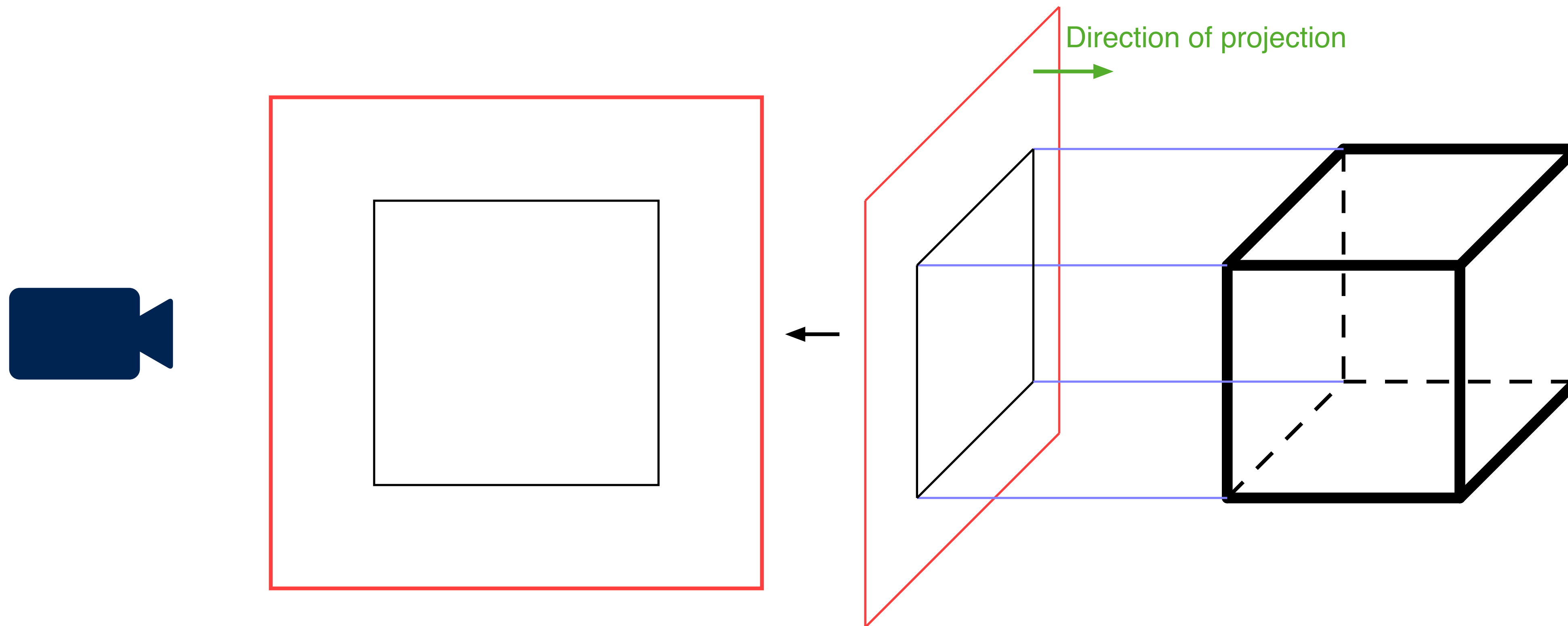
Basic Raytracing

1. Generation of rays
2. Intersection with objects in the scene
3. Shading (computation of the color of the pixel)

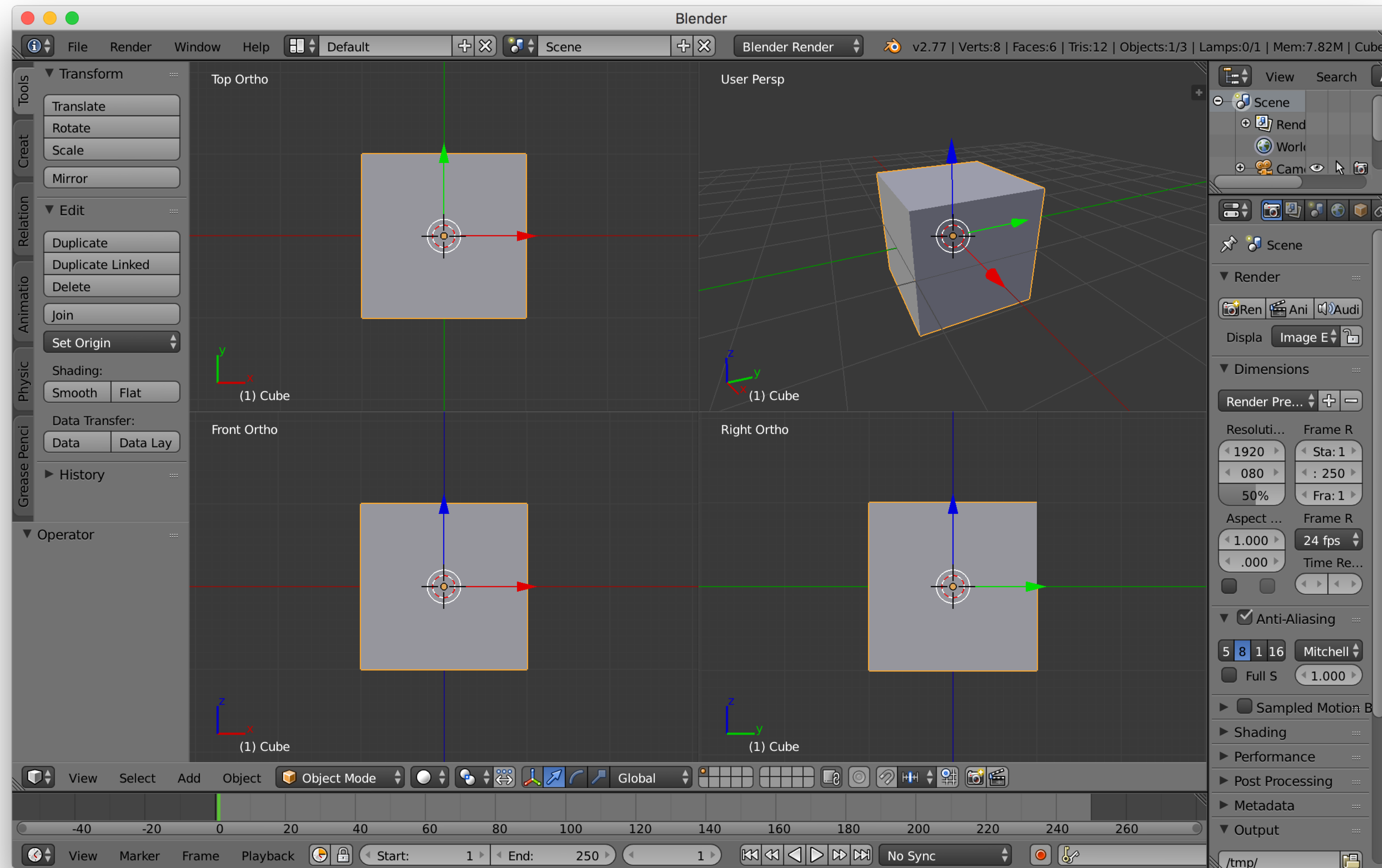


By Henrik - Own work, GFDL, <https://commons.wikimedia.org/w/index.php?curid=3869326>

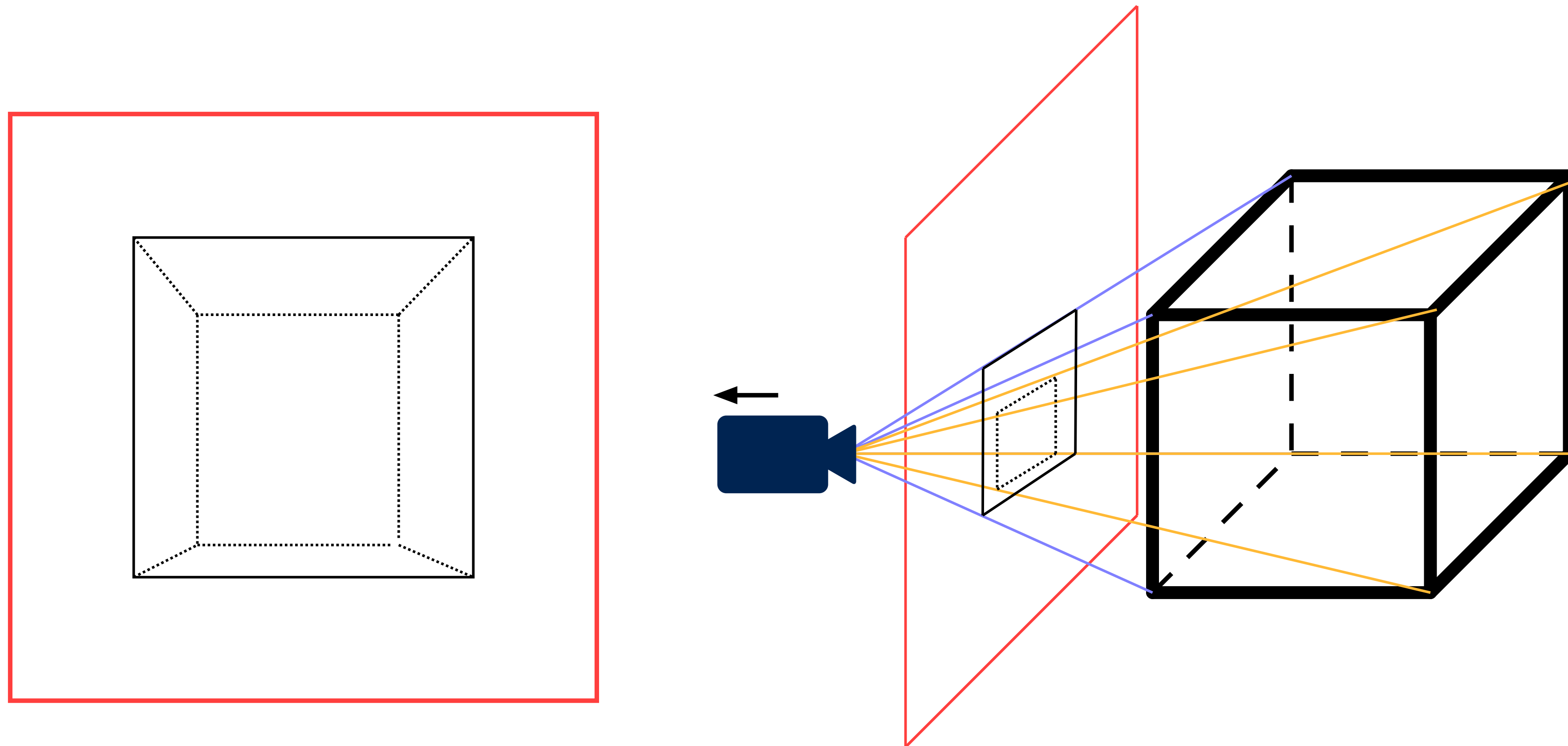
Projection - Parallel; Orthographic



Commonly used in modeling tools

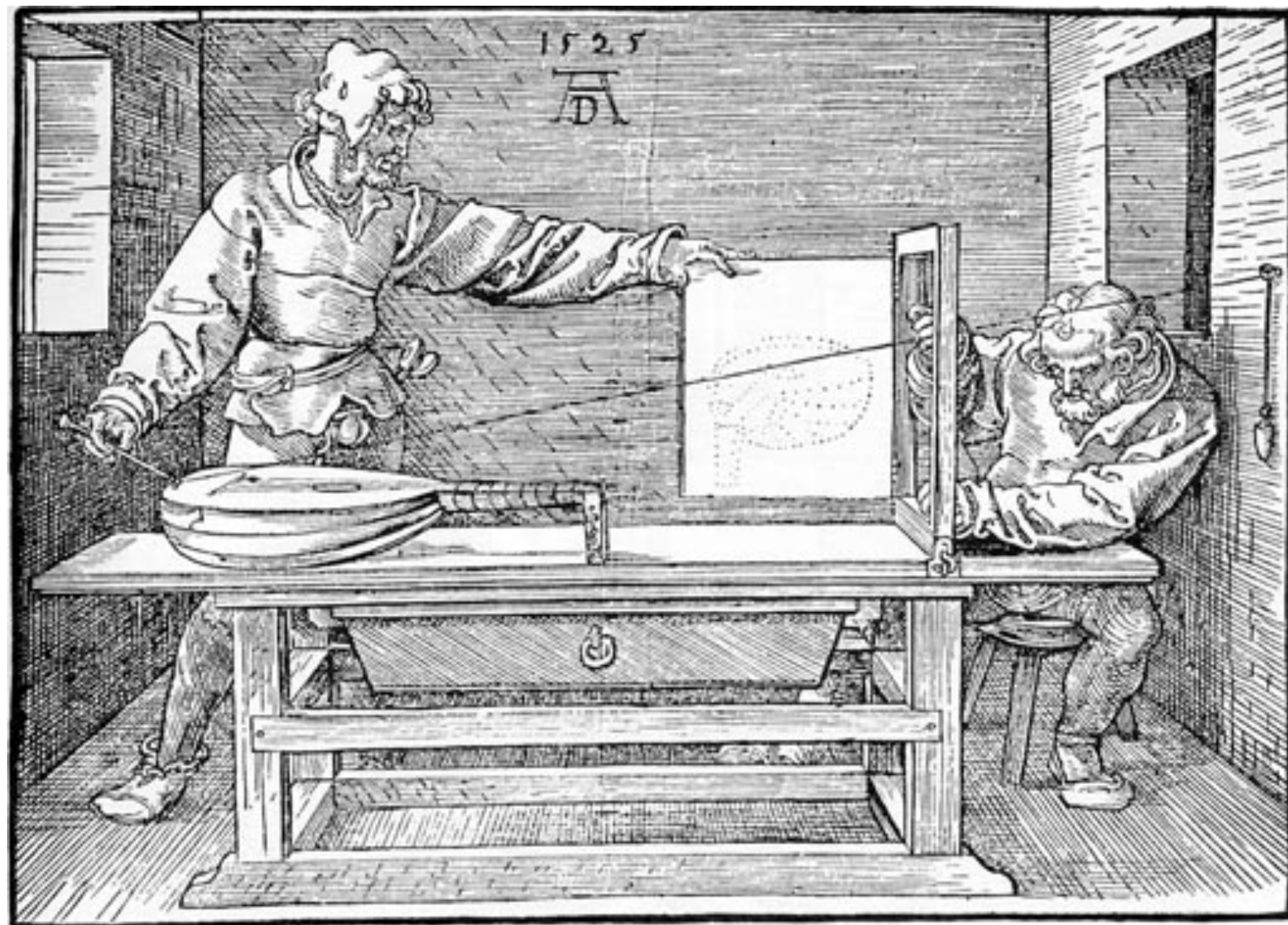


Perspective Projection

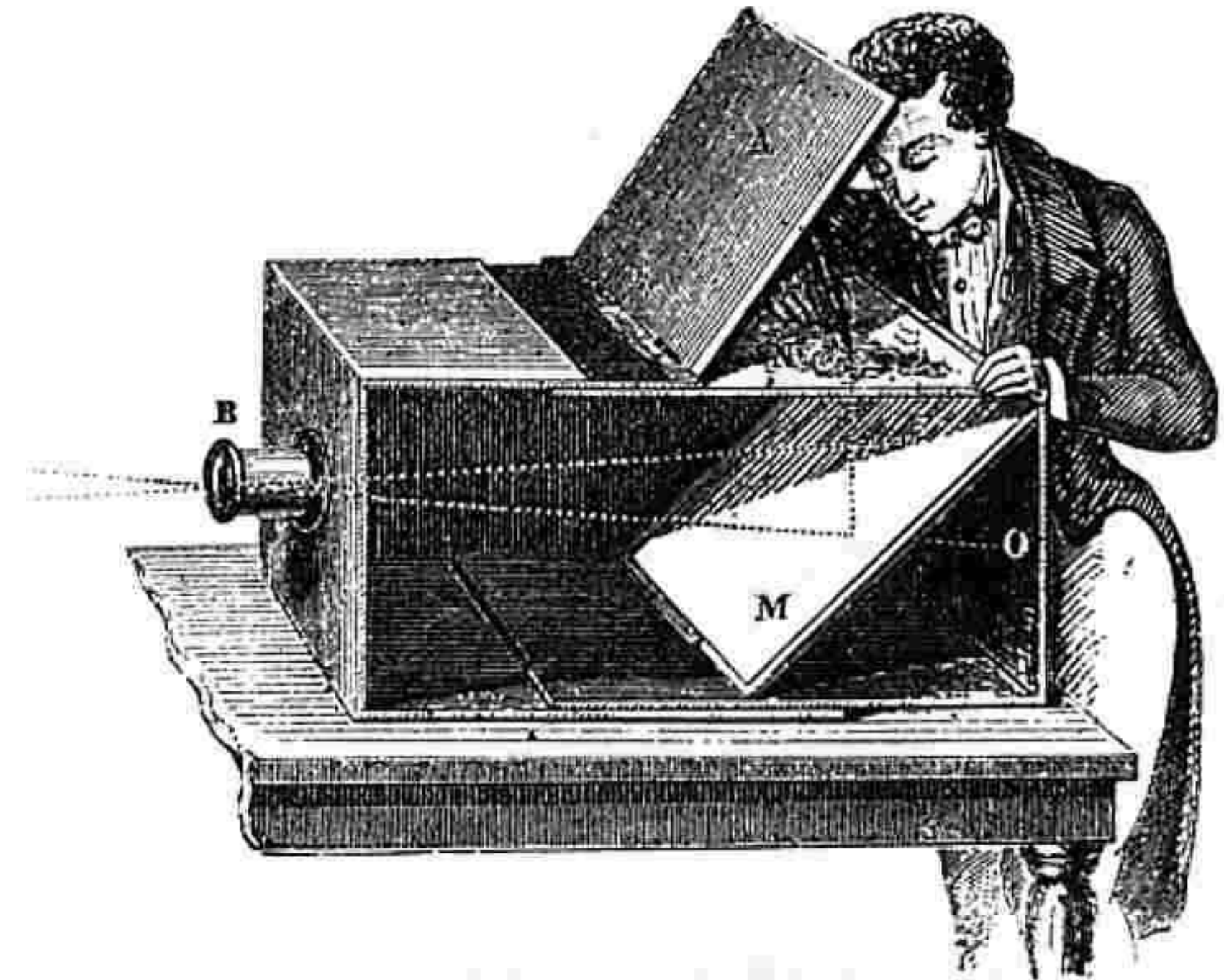


Each ray has a different direction!

Perspective Projection

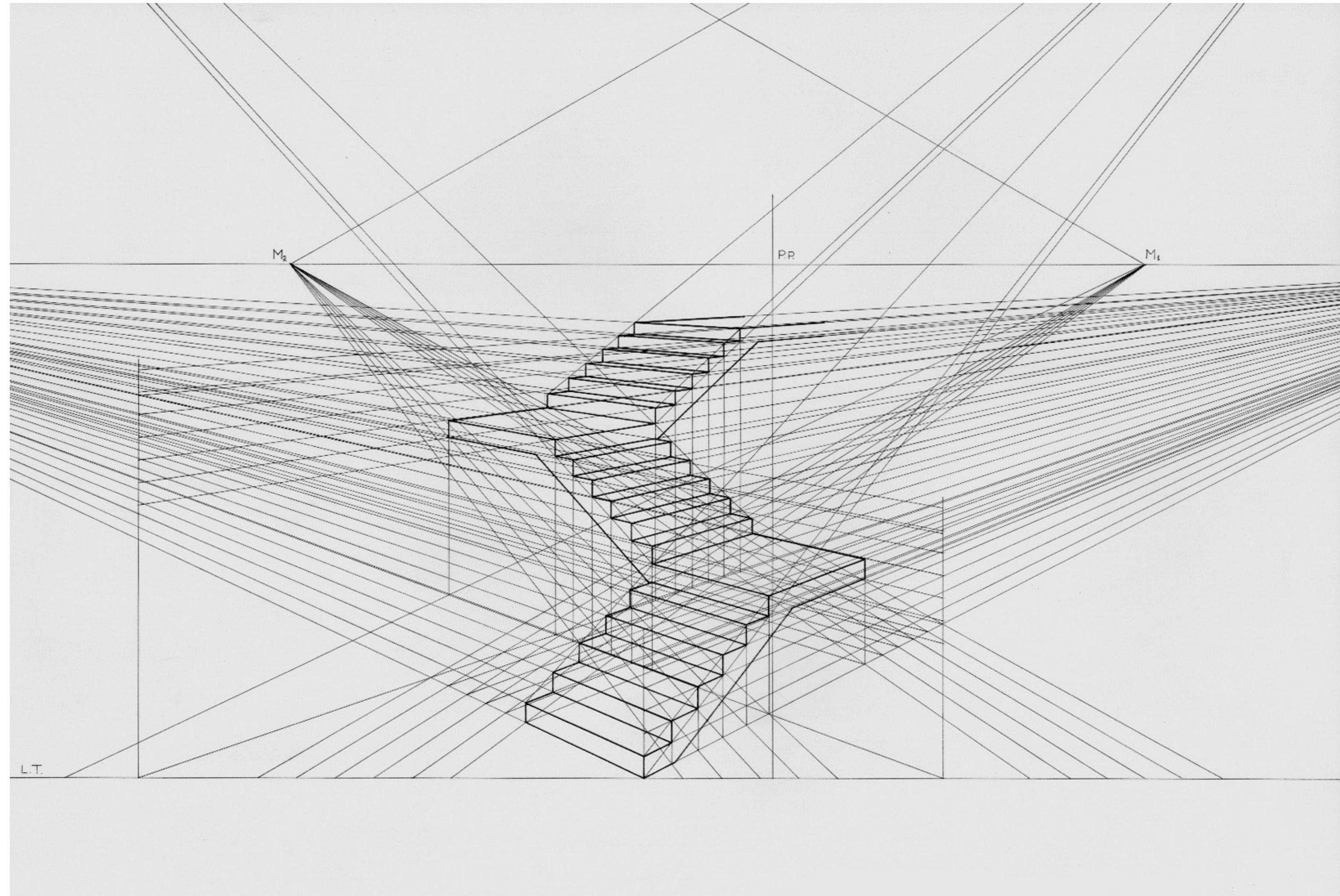


Albrecht Dürer, The Draughtsman of the Lute (1525)



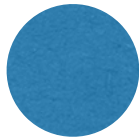
Camera Obscura box in 18th century,
https://commons.wikimedia.org/wiki/File:Camera_Obscura_box18thCentury.jpg

Two and Three Point Perspectives



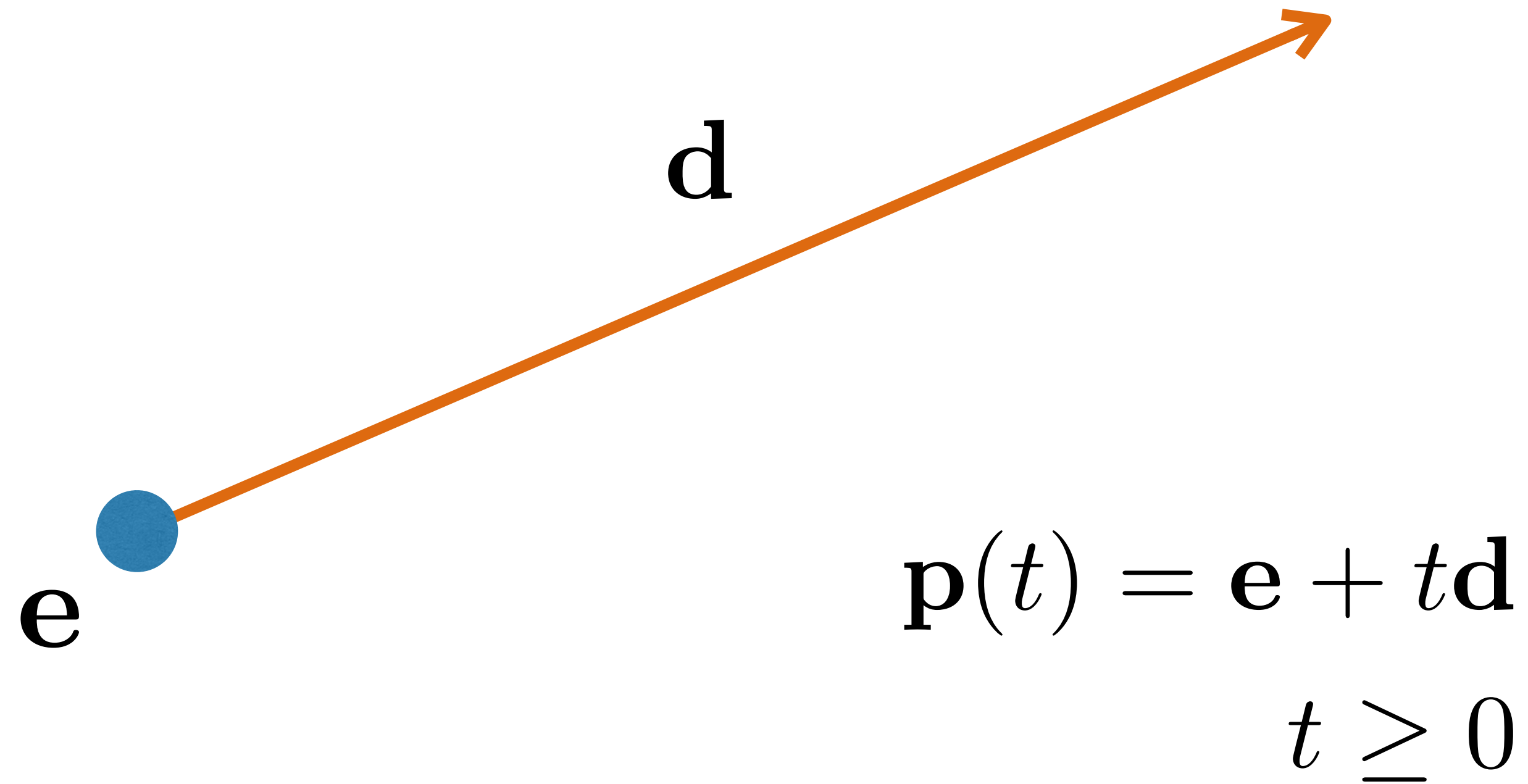
1. Compute Rays

$\mathbf{d}$

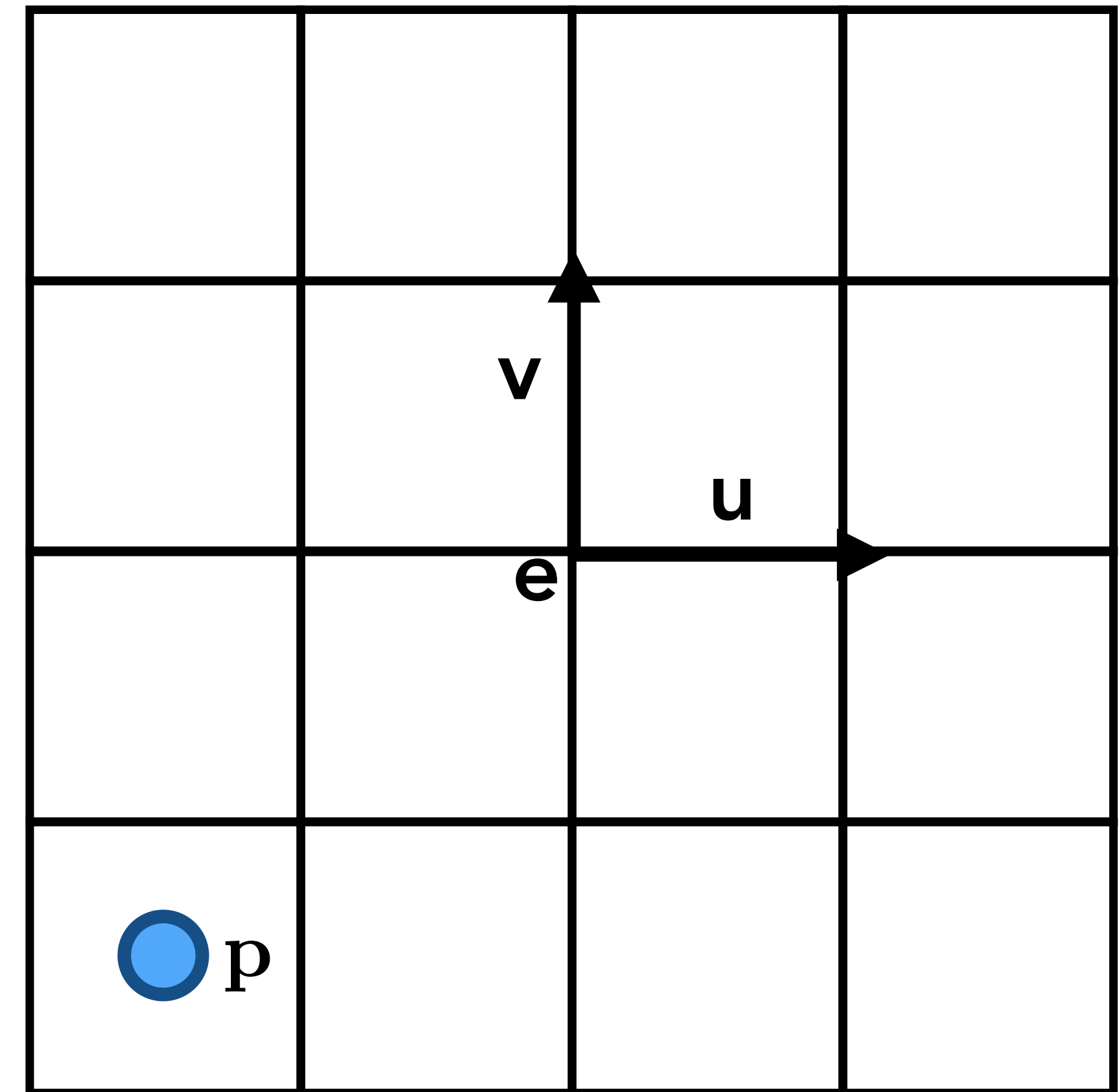
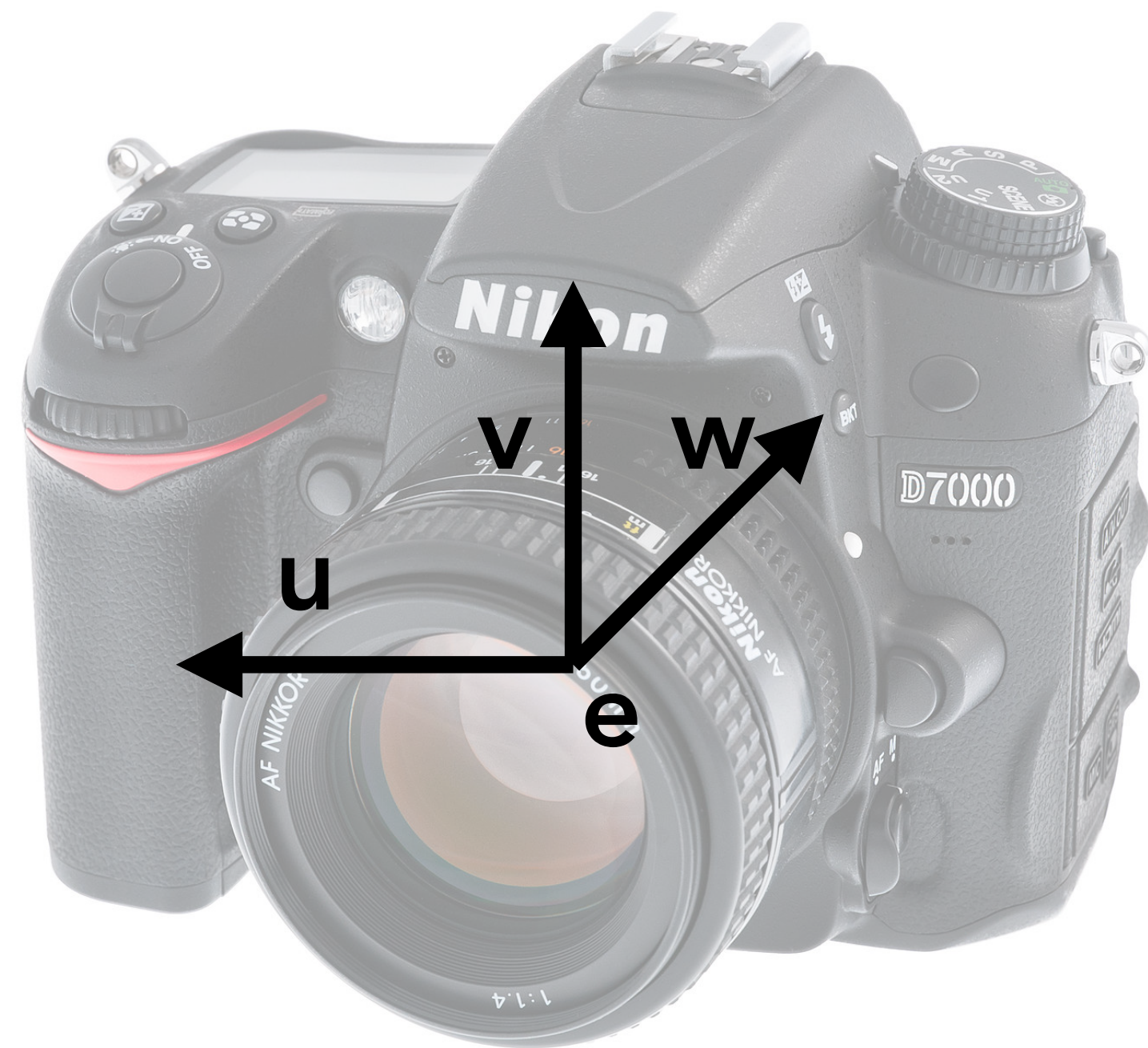
$\mathbf{e}$ 

$$\mathbf{p}(t) = \mathbf{e} + t\mathbf{d}$$
$$t \geq 0$$

1. Compute Rays



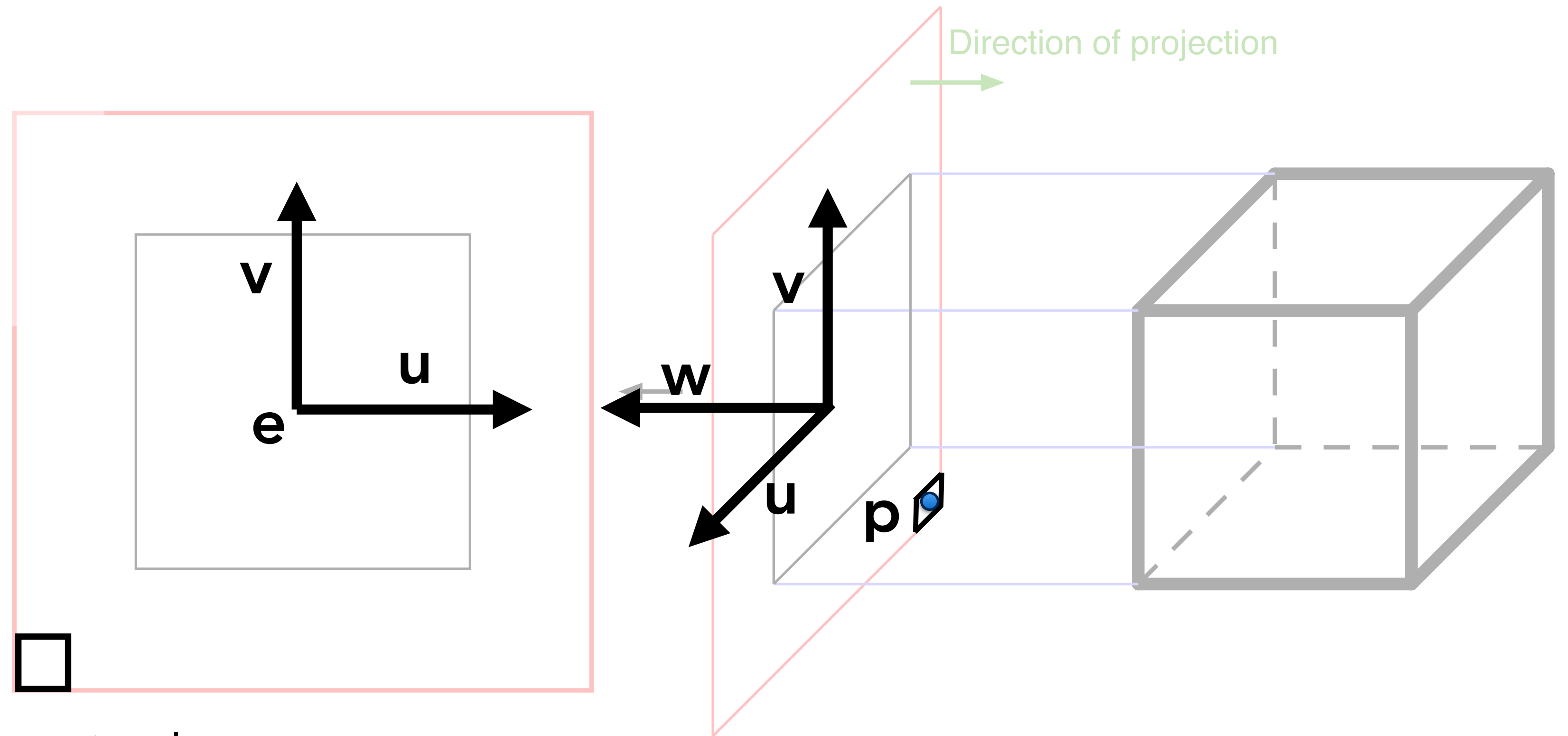
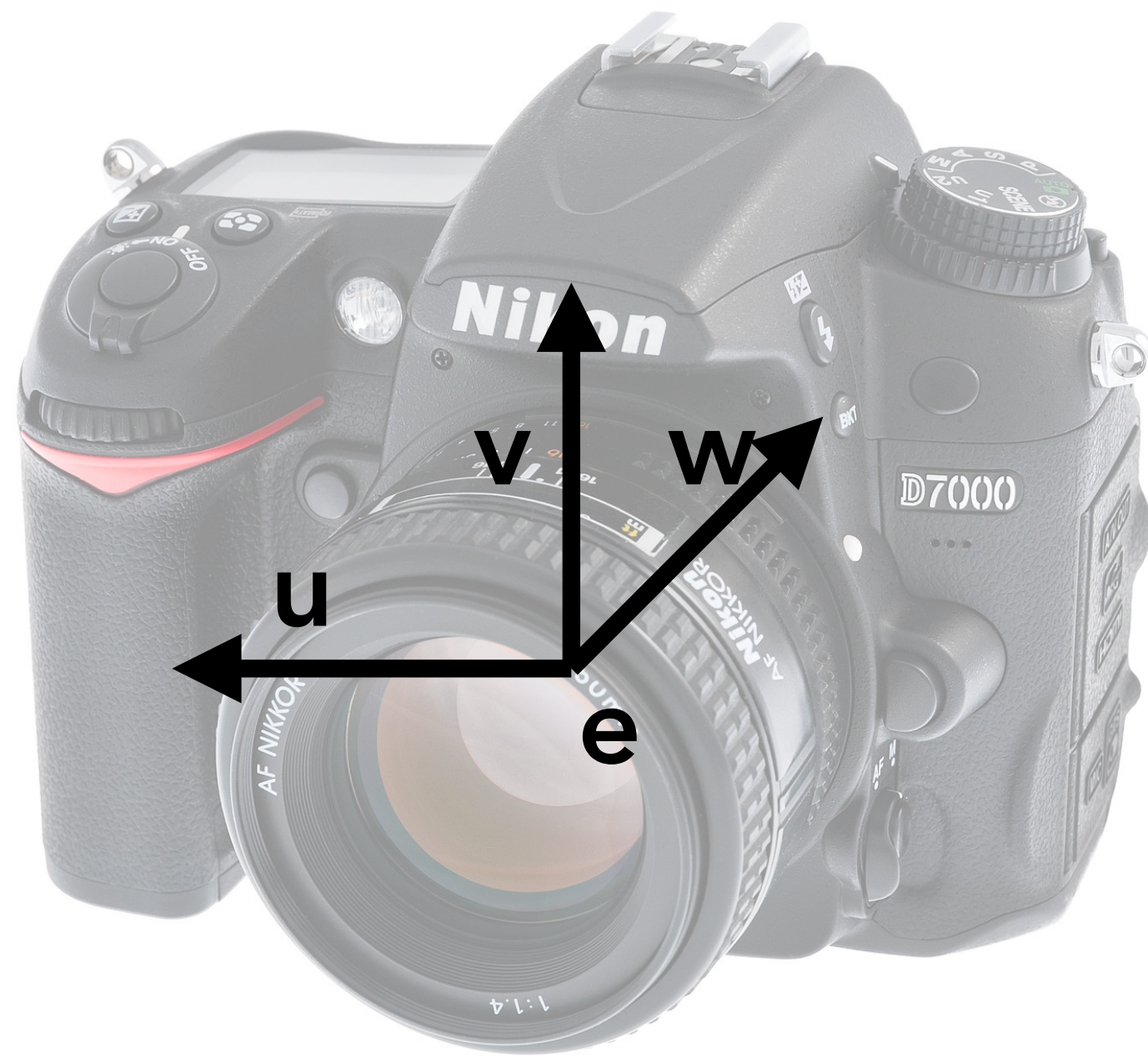
1. Compute Rays



e is the origin of the reference system
p is a sample position

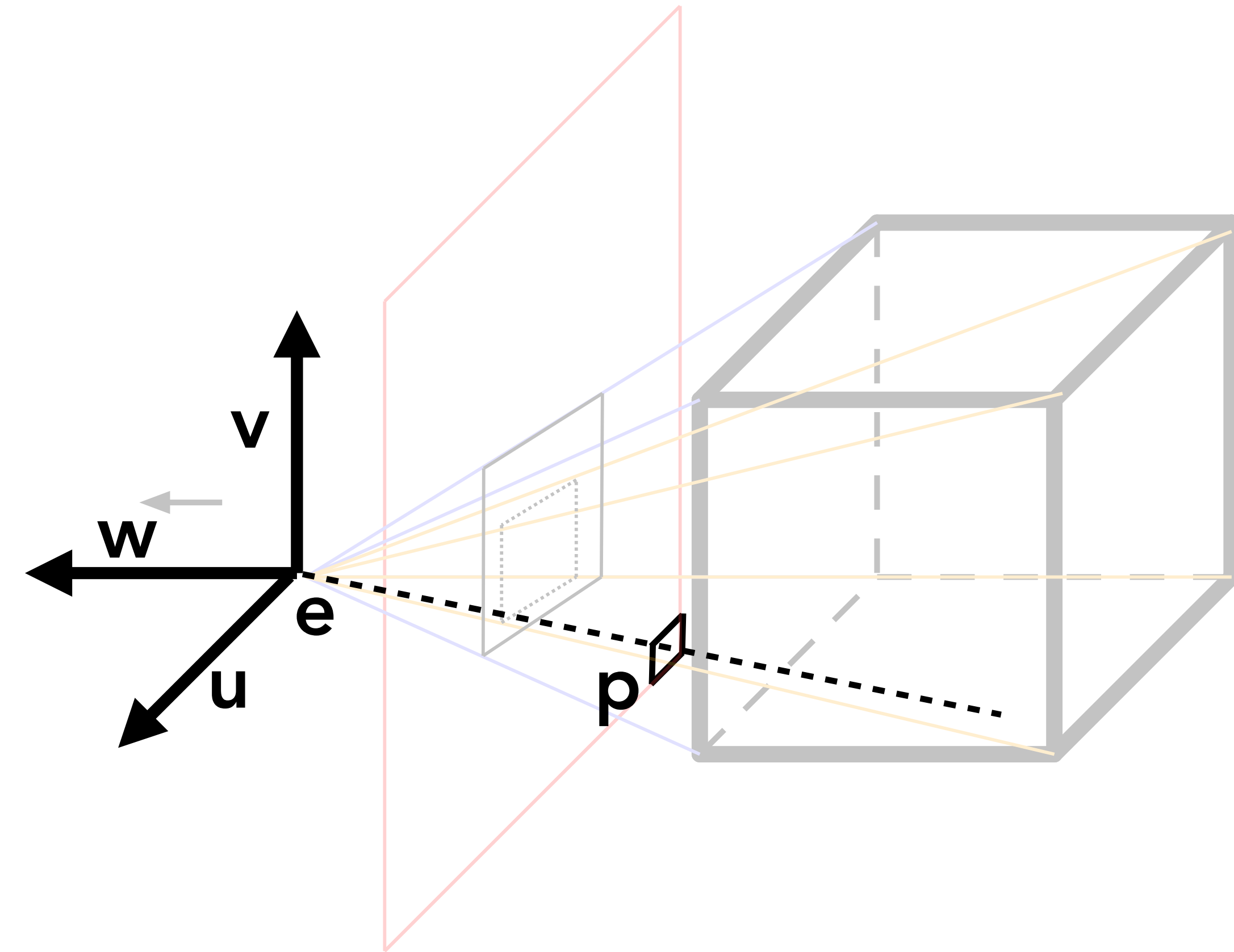
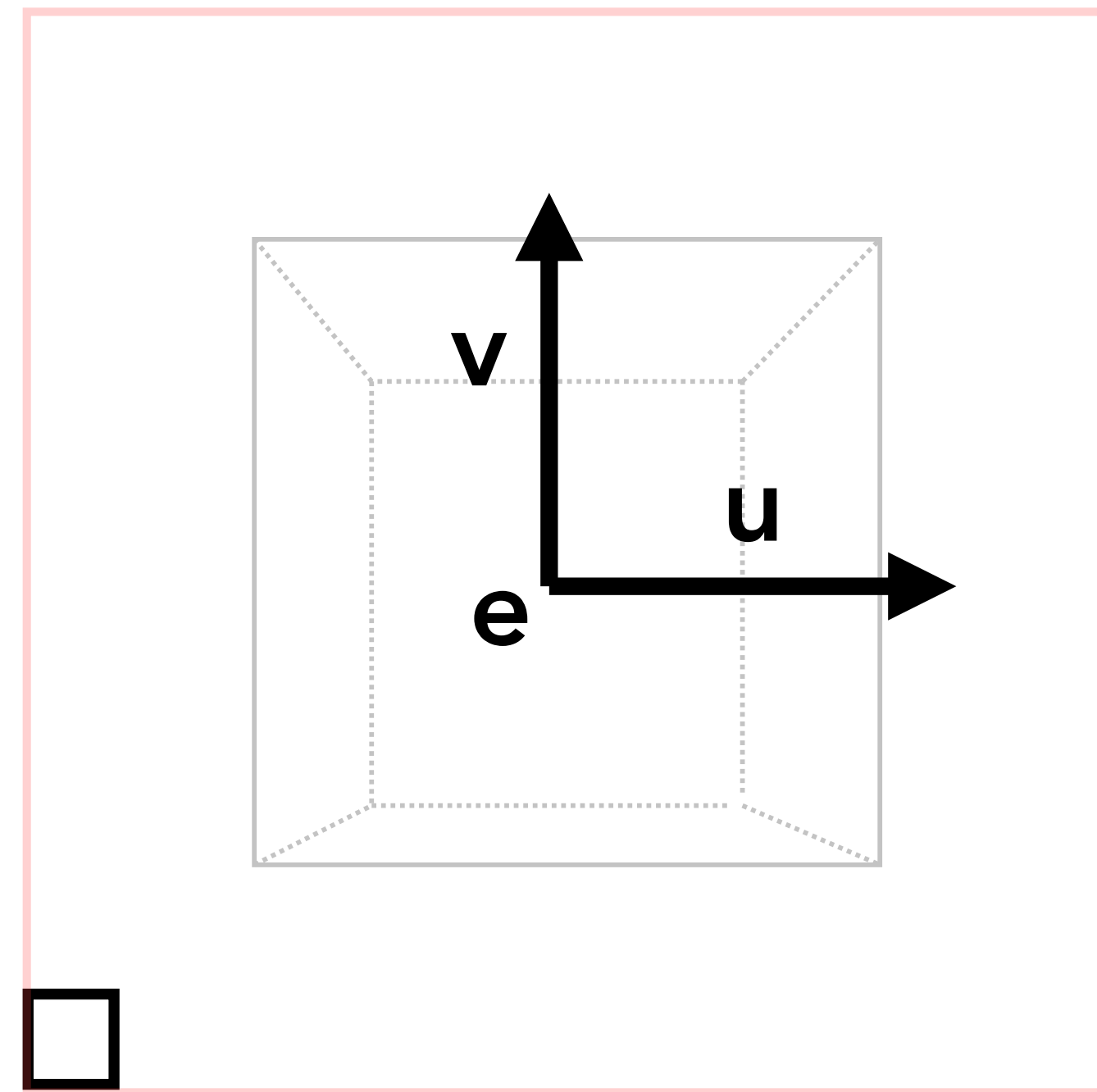
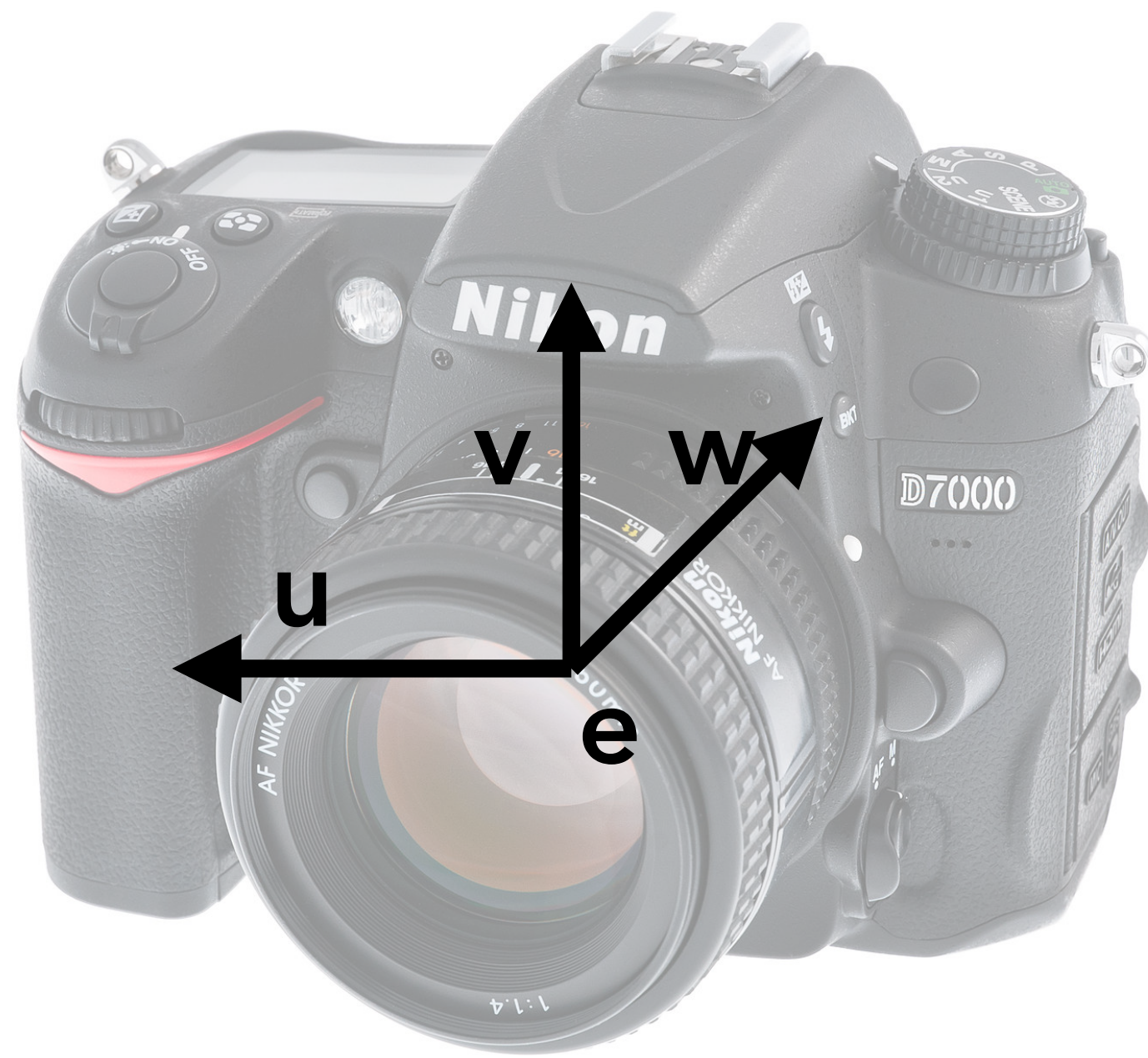
$$\mathbf{p} = \mathbf{e} + u\mathbf{u} + v\mathbf{v} + w\mathbf{w}$$

Orthographic



- For the ray assigned to pixel $\mathbf{p}$:
 - Origin: $\mathbf{p}$
 - Direction: $-\mathbf{w}$

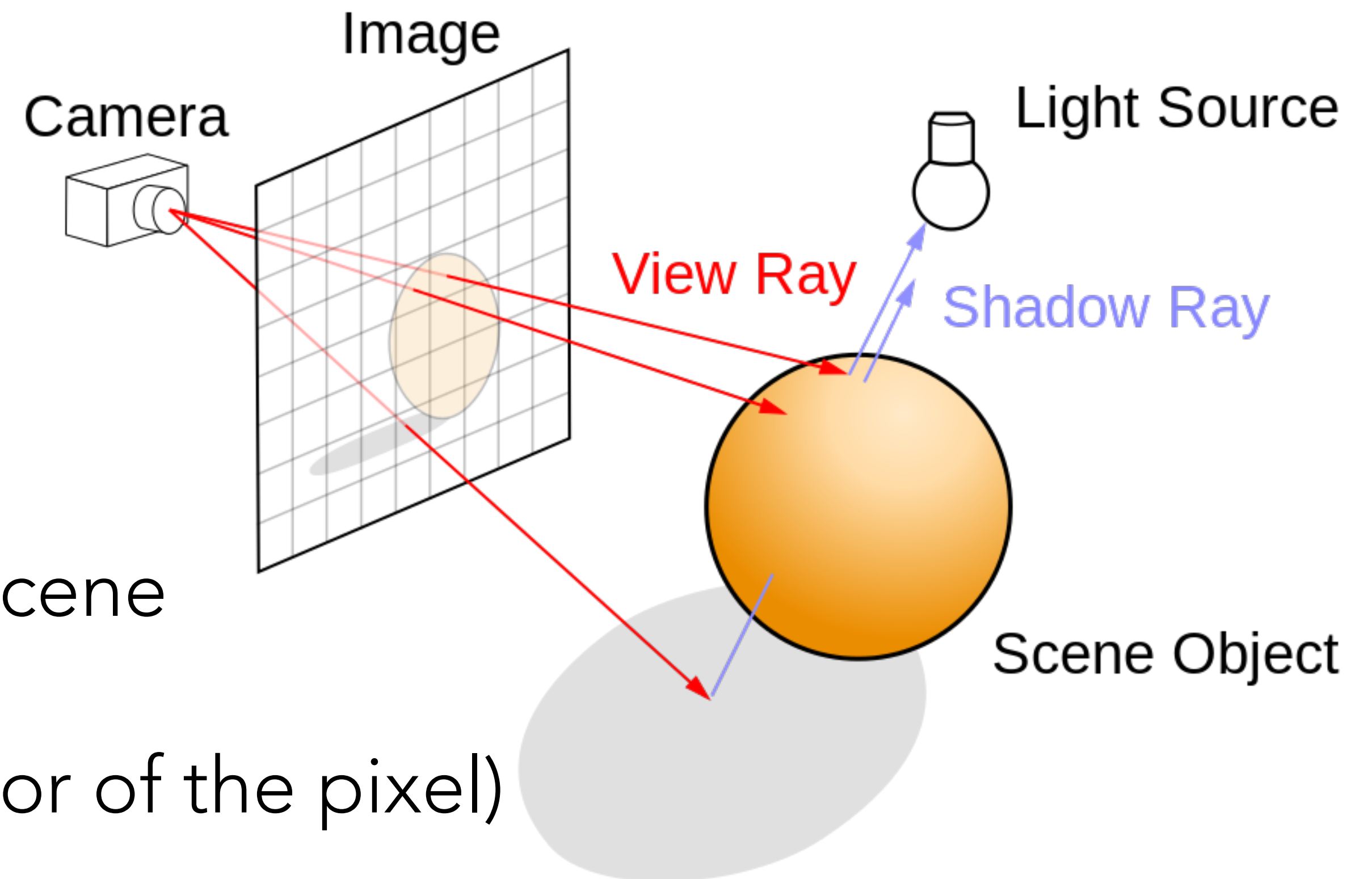
Perspective



- For the ray assigned to pixel **p**:
 - Origin: **e**
 - Direction: **p - e**

Basic Raytracing

1. Generation of rays
2. Intersection with objects in the scene
3. Shading (computation of the color of the pixel)



By Henrik - Own work, GFDL, <https://commons.wikimedia.org/w/index.php?curid=3869326>

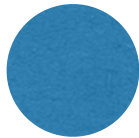
Intersections

- This is an expensive operation
- There is a large literature, a good overview is here: <http://www.realtimerendering.com/intersections.html>
- We will study two very useful cases:
 - Spheres
 - Triangles (by combining many triangles you can approximate complex surfaces)

Ray-Sphere Intersection

- We have a ray in parametric form:

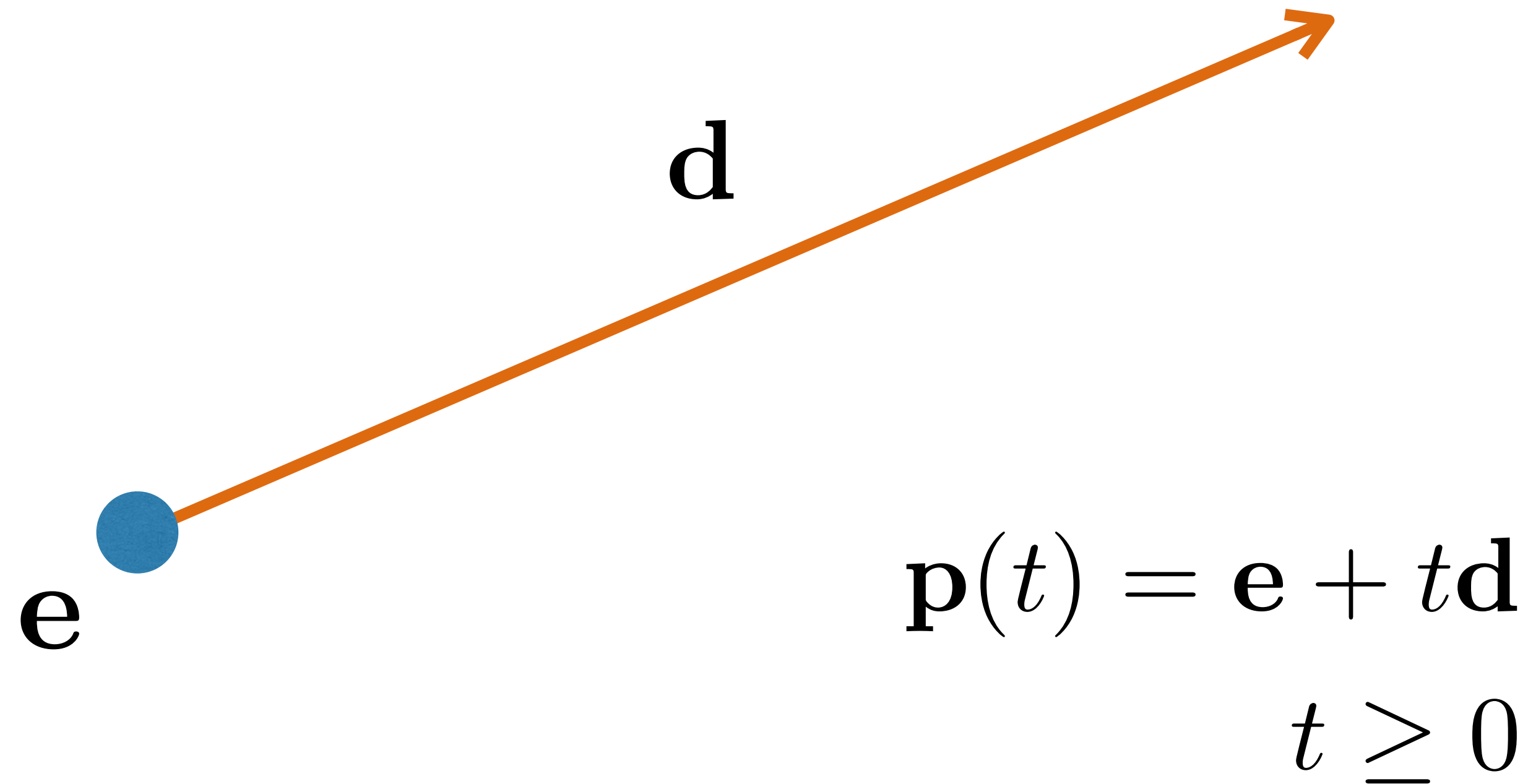
$\mathbf{d}$

$\mathbf{e}$ 

$$\mathbf{p}(t) = \mathbf{e} + t\mathbf{d}$$
$$t \geq 0$$

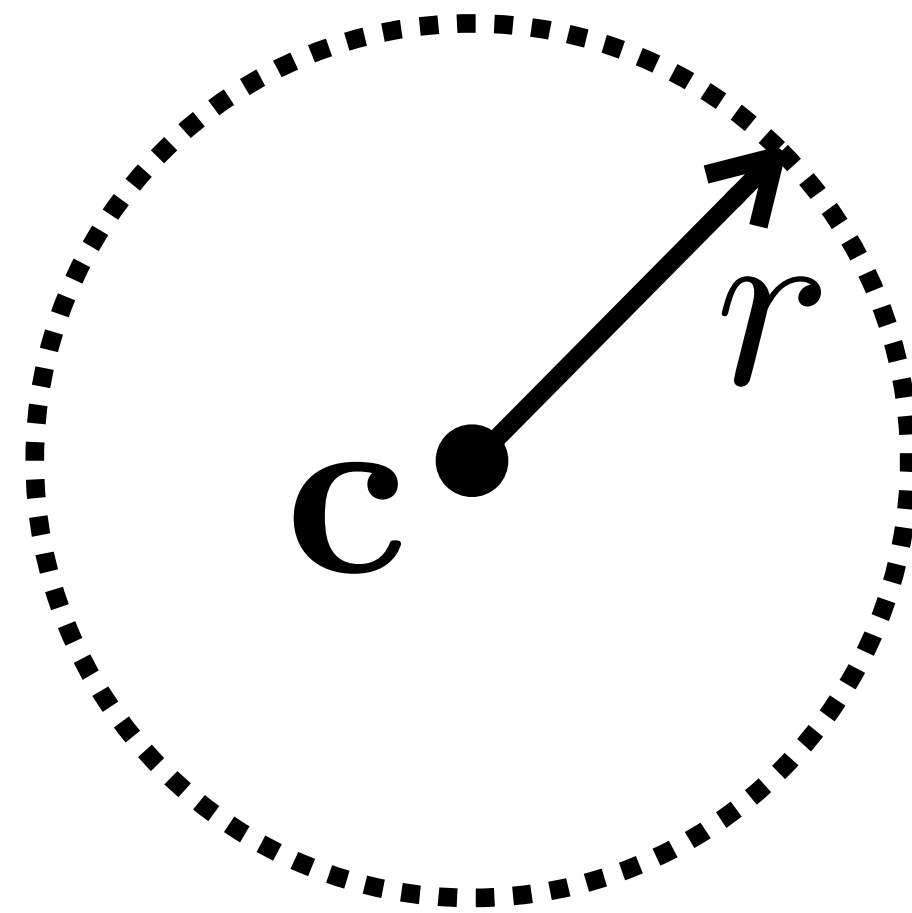
Ray-Sphere Intersection

- We have a ray in parametric form:



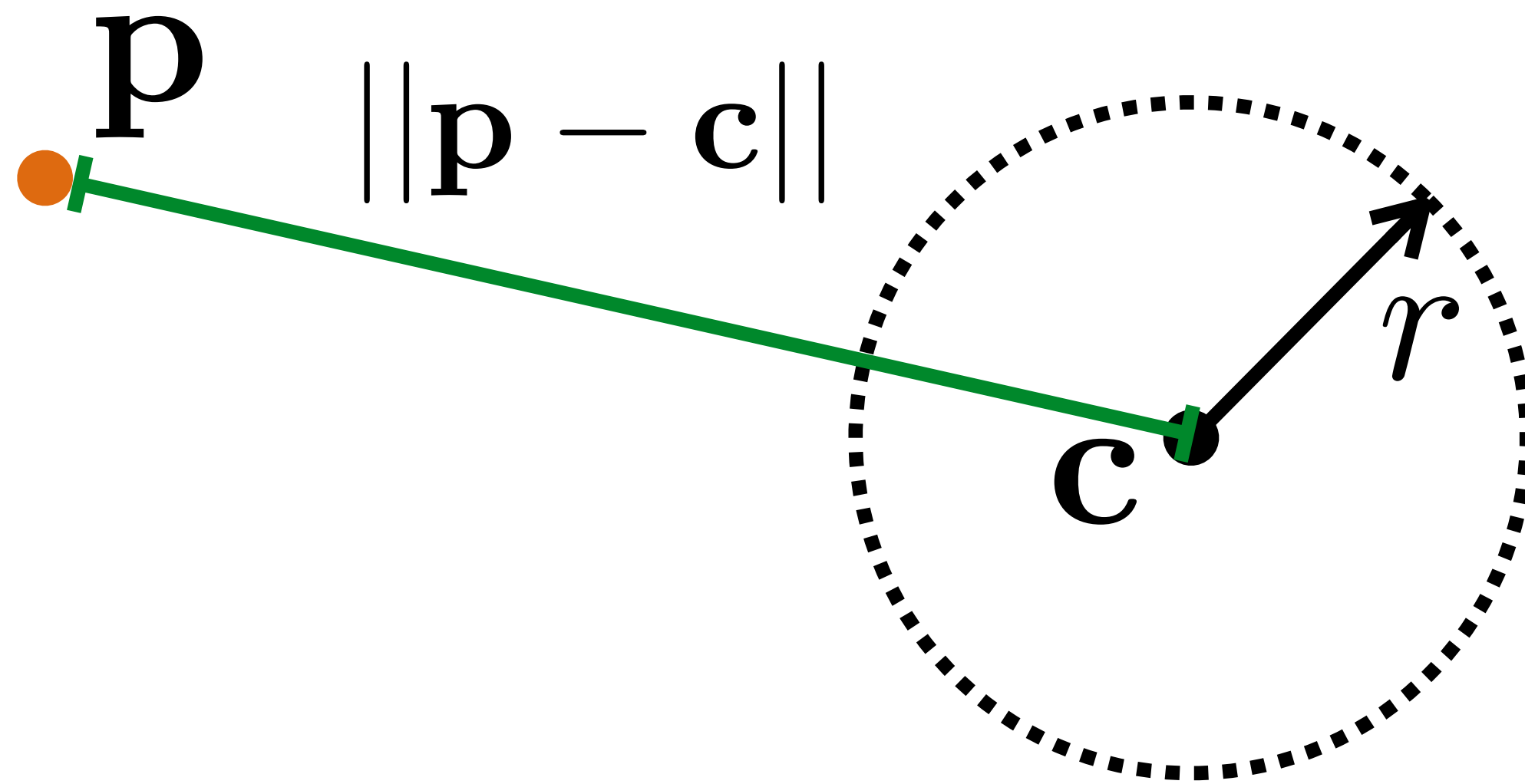
Ray-Sphere Intersection

- A sphere of radius r and center $\mathbf{c}$ in implicit form:



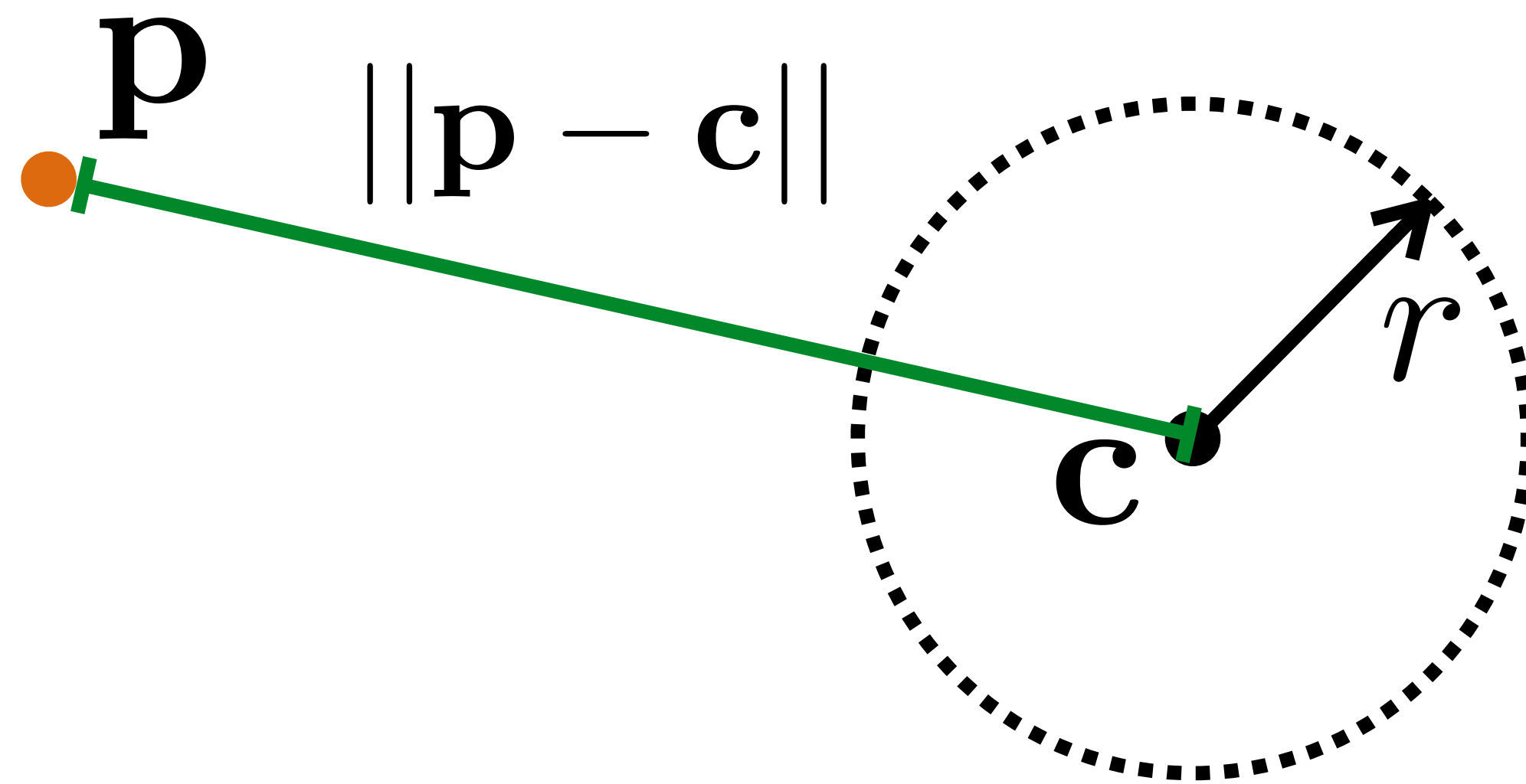
Ray-Sphere Intersection

- A sphere of radius r and center $\mathbf{c}$ in implicit form:



Ray-Sphere Intersection

- A sphere of radius r and center $\mathbf{c}$ in implicit form:

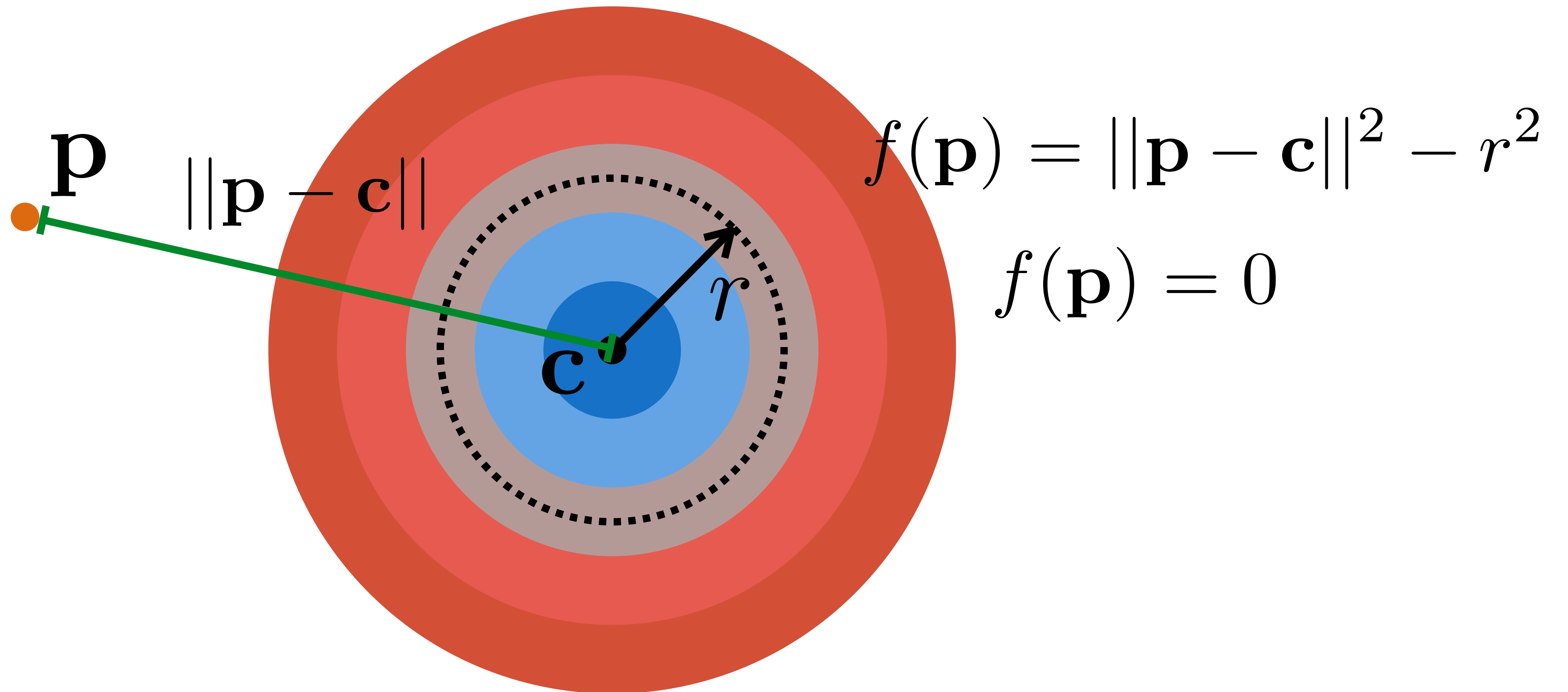


$$f(\mathbf{p}) = ||\mathbf{p} - \mathbf{c}||^2 - r^2$$

$$f(\mathbf{p}) = 0$$

Ray-Sphere Intersection

- A sphere of radius r and center $\mathbf{c}$ in implicit form:



Ray-Sphere Intersection

- A sphere of radius r and center $\mathbf{c}$ in implicit form:

$$\begin{aligned} f(\mathbf{p}) &= ||\mathbf{p} - \mathbf{c}||^2 - r^2 \\ &= (\mathbf{p} - \mathbf{c}) \cdot (\mathbf{p} - \mathbf{c}) - r^2 \end{aligned}$$

Ray-Sphere Intersection

- To find the intersection we need to find the solutions of

$$f(\mathbf{p}(t)) = 0$$

Ray-Sphere Intersection

- To find the intersection we need to find the solutions of

$$f(\mathbf{p}(t)) = 0$$

$$\begin{aligned}\mathbf{p}(t) &= \mathbf{e} + t\mathbf{d} \\ t &\geq 0\end{aligned}$$

$$\begin{aligned}f(\mathbf{p}) &= \|\mathbf{p} - \mathbf{c}\|^2 - r^2 \\ &= (\mathbf{p} - \mathbf{c}) \cdot (\mathbf{p} - \mathbf{c}) - r^2\end{aligned}$$

Do we have a solution in the range of ≥ 0 ?

Ray-Sphere Intersection

- To find the intersection we need to find the solutions of

$$\begin{aligned} f(\mathbf{p}) &= ||\mathbf{p} - \mathbf{c}||^2 - r^2 \\ &= (\mathbf{p} - \mathbf{c}) \cdot (\mathbf{p} - \mathbf{c}) - r^2 \\ &= (\mathbf{e} + t\mathbf{d} - \mathbf{c}) \cdot (\mathbf{e} + t\mathbf{d} - \mathbf{c}) - r^2 \end{aligned}$$

Ray-Sphere Intersection

- To find the intersection we need to find the solutions of

$$\begin{aligned} f(\mathbf{p}) &= ||\mathbf{p} - \mathbf{c}||^2 - r^2 \\ &= (\mathbf{p} - \mathbf{c}) \cdot (\mathbf{p} - \mathbf{c}) - r^2 \\ &= (\mathbf{e} + t\mathbf{d} - \mathbf{c}) \cdot (\mathbf{e} + t\mathbf{d} - \mathbf{c}) - r^2 \\ &= ((\mathbf{e} - \mathbf{c}) + t\mathbf{d}) \cdot ((\mathbf{e} - \mathbf{c}) + t\mathbf{d}) - r^2 \end{aligned}$$

Ray-Sphere Intersection

- To find the intersection we need to find the solutions of

$$\begin{aligned} f(\mathbf{p}) &= ||\mathbf{p} - \mathbf{c}||^2 - r^2 \\ &= (\mathbf{p} - \mathbf{c}) \cdot (\mathbf{p} - \mathbf{c}) - r^2 \\ &= (\mathbf{e} + t\mathbf{d} - \mathbf{c}) \cdot (\mathbf{e} + t\mathbf{d} - \mathbf{c}) - r^2 \\ &= ((\mathbf{e} - \mathbf{c}) + t\mathbf{d}) \cdot ((\mathbf{e} - \mathbf{c}) + t\mathbf{d}) - r^2 \\ &= (\mathbf{e} - \mathbf{c}) \cdot (\mathbf{e} - \mathbf{c}) + (\mathbf{e} - \mathbf{c}) \cdot (t\mathbf{d}) + (t\mathbf{d}) \cdot (\mathbf{e} - \mathbf{c}) + (t\mathbf{d}) \cdot (t\mathbf{d}) - r^2 \end{aligned}$$

Ray-Sphere Intersection

- To find the intersection we need to find the solutions of

$$\begin{aligned} f(\mathbf{p}) &= ||\mathbf{p} - \mathbf{c}||^2 - r^2 \\ &= (\mathbf{p} - \mathbf{c}) \cdot (\mathbf{p} - \mathbf{c}) - r^2 \\ &= (\mathbf{e} + t\mathbf{d} - \mathbf{c}) \cdot (\mathbf{e} + t\mathbf{d} - \mathbf{c}) - r^2 \\ &= ((\mathbf{e} - \mathbf{c}) + t\mathbf{d}) \cdot ((\mathbf{e} - \mathbf{c}) + t\mathbf{d}) - r^2 \\ &= (\mathbf{e} - \mathbf{c}) \cdot (\mathbf{e} - \mathbf{c}) + (\mathbf{e} - \mathbf{c}) \cdot (t\mathbf{d}) + (t\mathbf{d}) \cdot (\mathbf{e} - \mathbf{c}) + (t\mathbf{d}) \cdot (t\mathbf{d}) - r^2 \\ &= (\mathbf{e} - \mathbf{c}) \cdot (\mathbf{e} - \mathbf{c}) + 2t \mathbf{d} \cdot (\mathbf{e} - \mathbf{c}) + t^2 \mathbf{d} \cdot \mathbf{d} - r^2 \end{aligned}$$

Ray-Sphere Intersection

- To find the intersection we need to find the solutions of

$$\begin{aligned} f(\mathbf{p}) &= \|\mathbf{p} - \mathbf{c}\|^2 - r^2 \\ &= (\mathbf{p} - \mathbf{c}) \cdot (\mathbf{p} - \mathbf{c}) - r^2 \\ &= (\mathbf{e} + t\mathbf{d} - \mathbf{c}) \cdot (\mathbf{e} + t\mathbf{d} - \mathbf{c}) - r^2 \\ &= ((\mathbf{e} - \mathbf{c}) + t\mathbf{d}) \cdot ((\mathbf{e} - \mathbf{c}) + t\mathbf{d}) - r^2 \\ &= (\mathbf{e} - \mathbf{c}) \cdot (\mathbf{e} - \mathbf{c}) + (\mathbf{e} - \mathbf{c}) \cdot (t\mathbf{d}) + (t\mathbf{d}) \cdot (\mathbf{e} - \mathbf{c}) + (t\mathbf{d}) \cdot (t\mathbf{d}) - r^2 \\ &= (\mathbf{e} - \mathbf{c}) \cdot (\mathbf{e} - \mathbf{c}) + 2t \mathbf{d} \cdot (\mathbf{e} - \mathbf{c}) + t^2 \mathbf{d} \cdot \mathbf{d} - r^2 \\ &= \underbrace{(\mathbf{d} \cdot \mathbf{d})}_{A} t^2 + \underbrace{2 \mathbf{d} \cdot (\mathbf{e} - \mathbf{c})}_{B} t + \underbrace{(\mathbf{e} - \mathbf{c}) \cdot (\mathbf{e} - \mathbf{c}) - r^2}_{C} \end{aligned}$$



Ray-Sphere Intersection

- To find the intersection we need to find the solutions of

$$f(\mathbf{p}) = \underbrace{(\mathbf{d} \cdot \mathbf{d})}_A t^2 + \underbrace{2 \mathbf{d} \cdot (\mathbf{e} - \mathbf{c})}_B t + \underbrace{(\mathbf{e} - \mathbf{c}) \cdot (\mathbf{e} - \mathbf{c}) - r^2}_C = 0$$
$$At^2 + Bt + C = 0$$

Ray-Sphere Intersection

- To find the intersection we need to find the solutions of

$$f(\mathbf{p}) = \underbrace{(\mathbf{d} \cdot \mathbf{d})}_A t^2 + \underbrace{2 \mathbf{d} \cdot (\mathbf{e} - \mathbf{c})}_B t + \underbrace{(\mathbf{e} - \mathbf{c}) \cdot (\mathbf{e} - \mathbf{c}) - r^2}_C = 0$$
$$At^2 + Bt + C = 0$$

$$t = \frac{-B \pm \sqrt{B^2 - 4AC}}{2A}$$

Ray-Sphere Intersection

- To find the intersection we need to find the solutions of

$$\Delta = B^2 - 4AC$$

$$\Delta < 0 \quad \text{No solution}$$

$$\Delta = 0 \quad \text{One solution}$$

$$\Delta > 0 \quad \text{Two solutions}$$

Ray-Sphere Intersection

- To find the intersection we need to find the solutions of

$$t = \frac{-B \pm \sqrt{B^2 - 4AC}}{2A}$$

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Ray-Sphere Intersection

- To find the intersection we need to find the solutions of

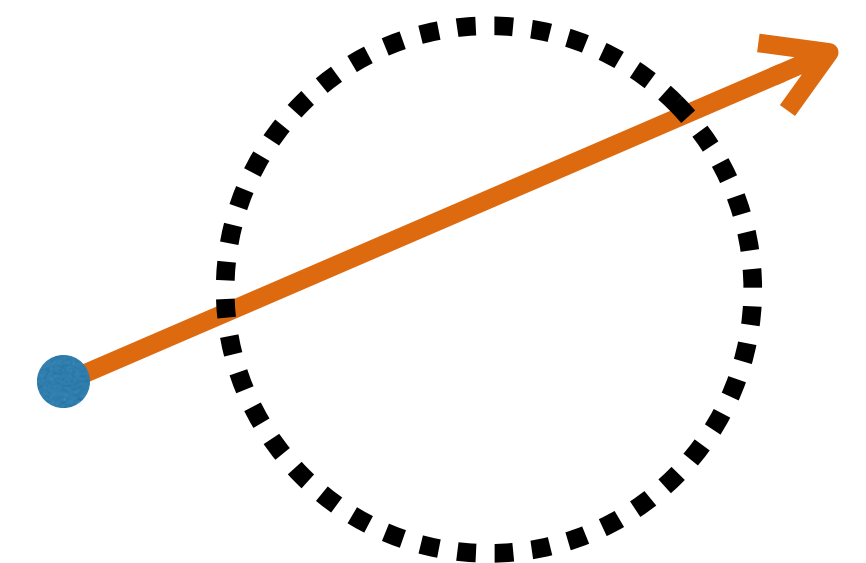
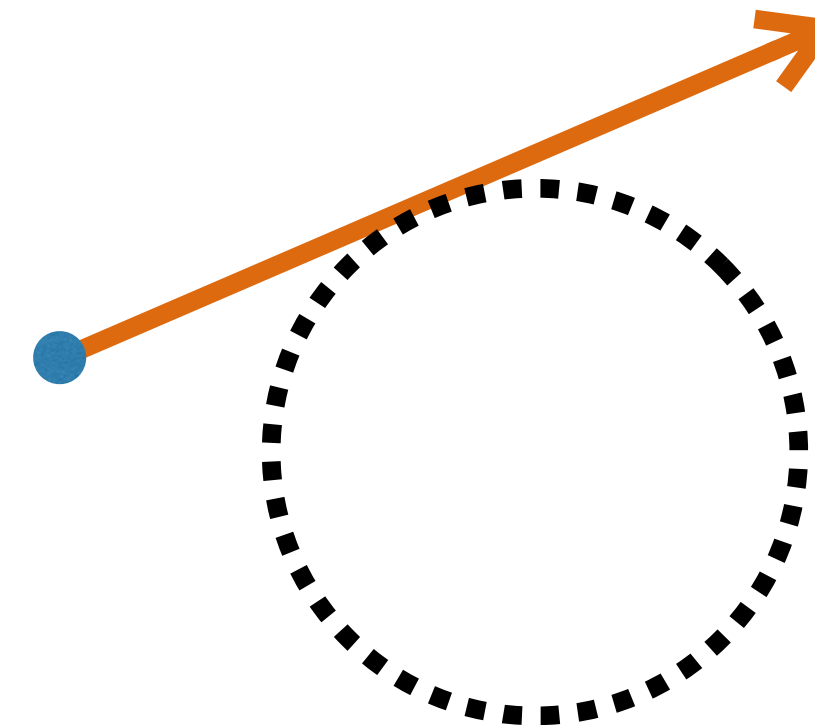
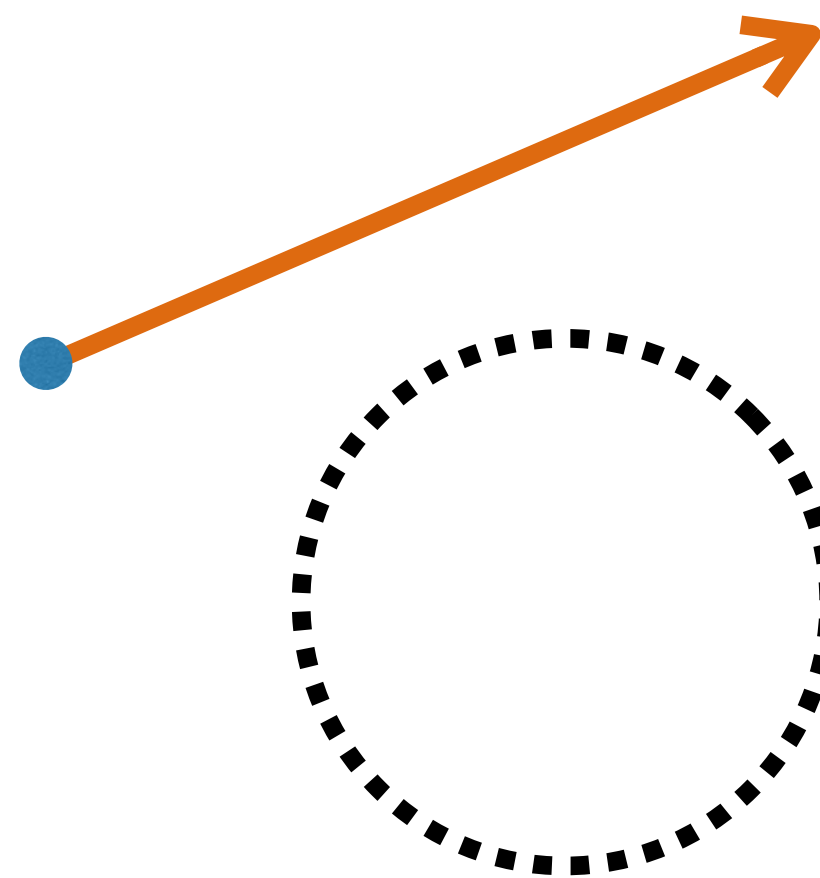
$$t = \frac{-B \pm \sqrt{B^2 - 4AC}}{2A}$$

$$\Delta = B^2 - 4AC$$

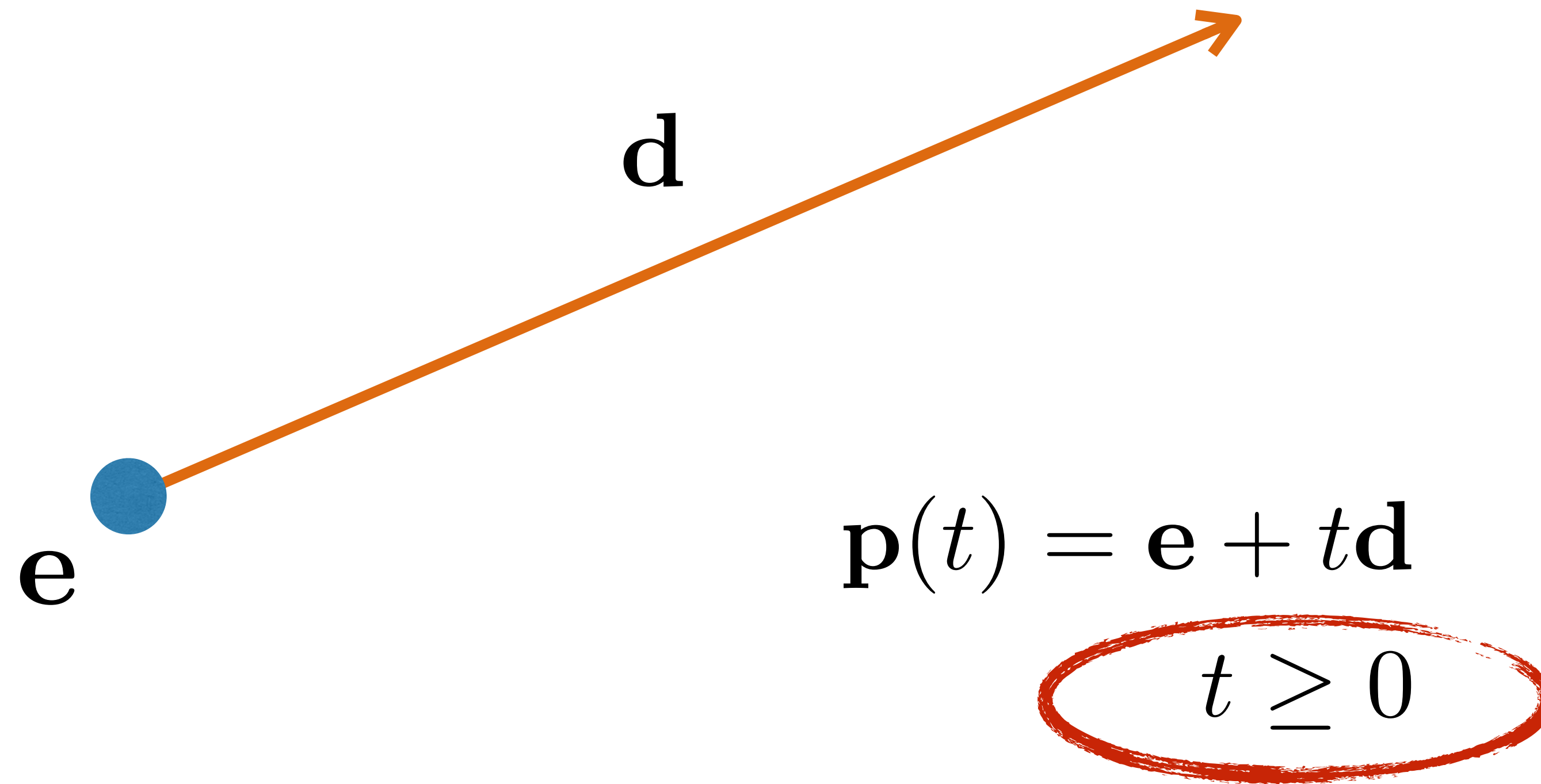
$$\Delta < 0 \quad \text{No solution}$$

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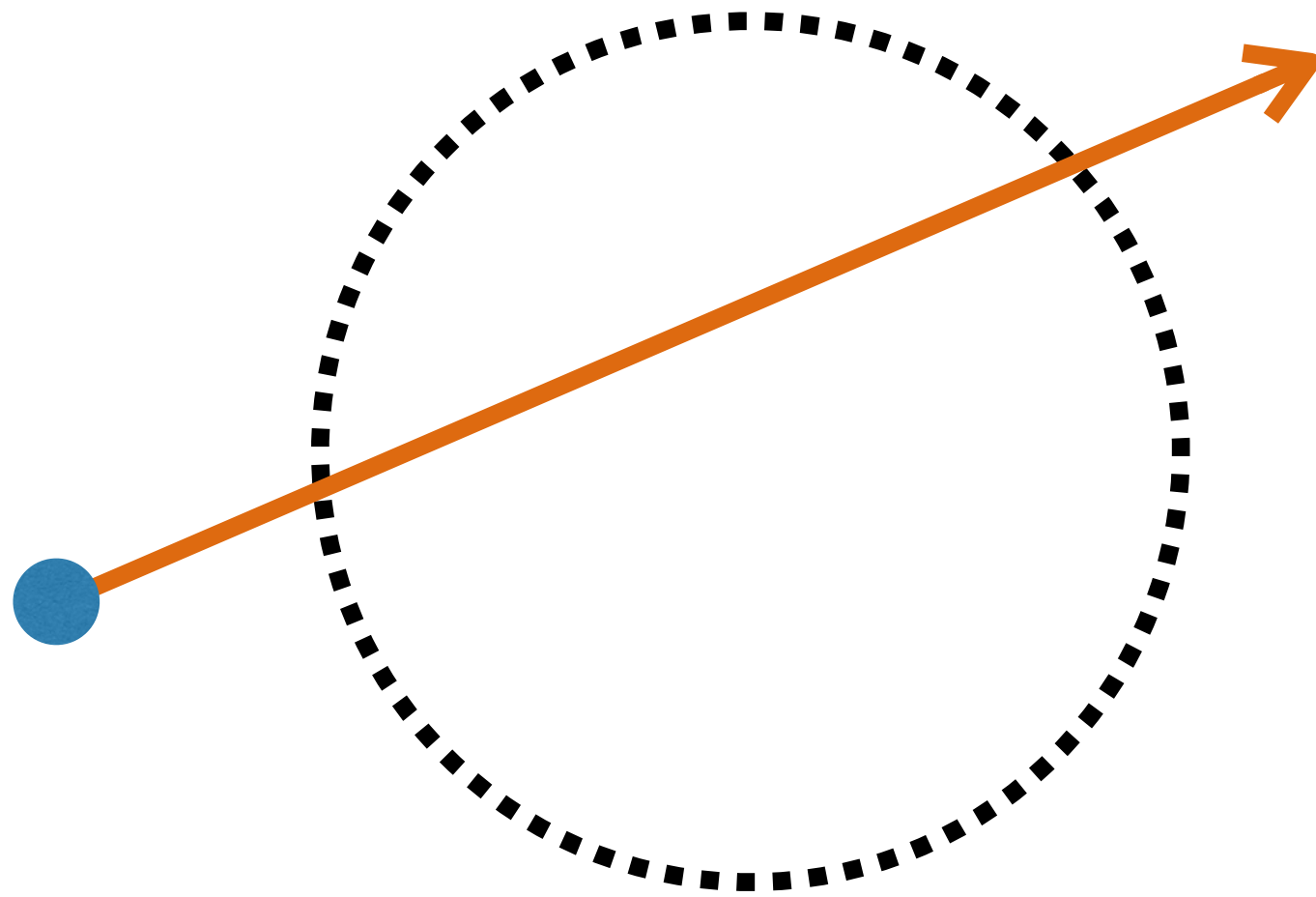
$$\Delta > 0 \quad \text{Two solutions}$$



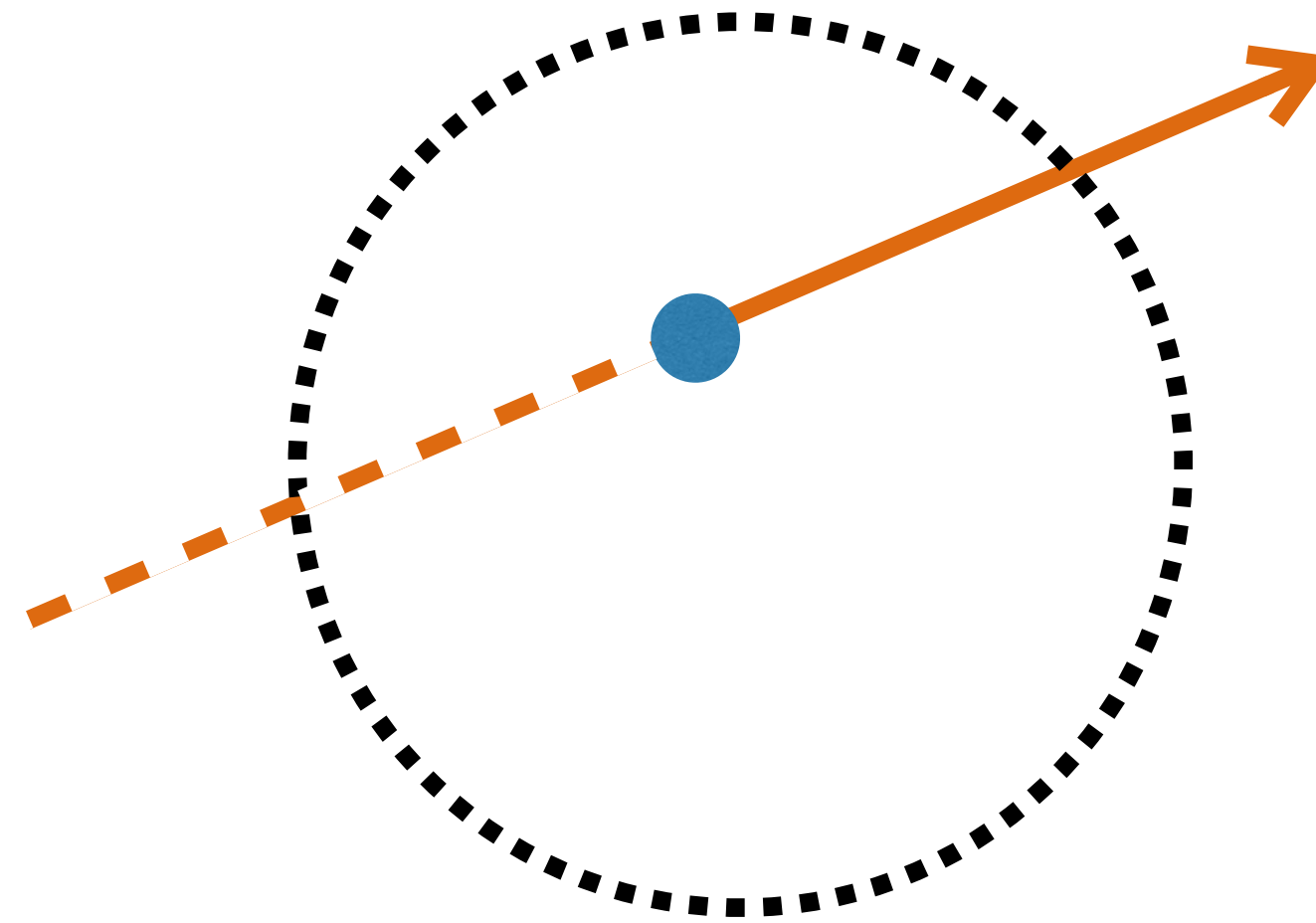
Ray-Sphere Intersection



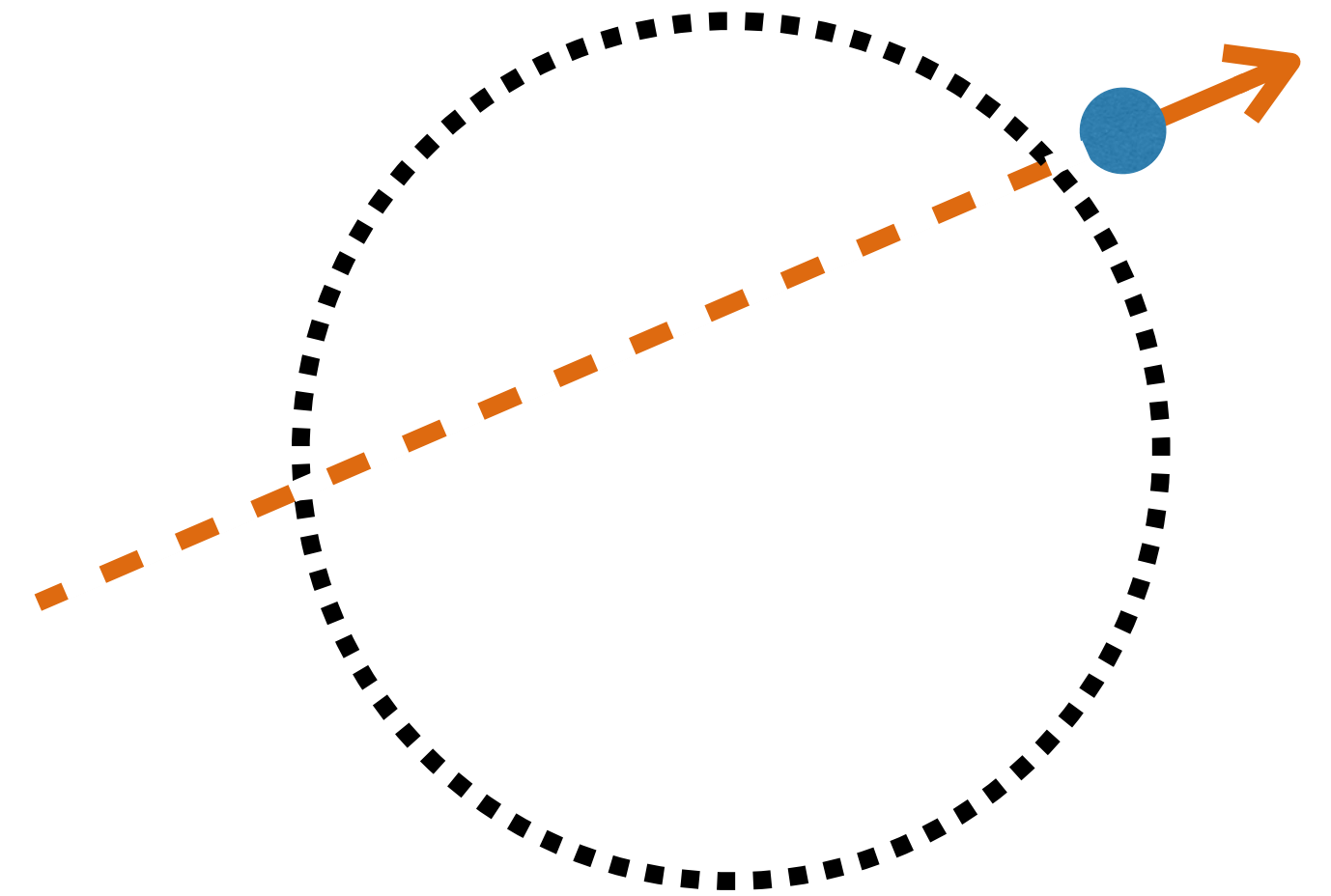
Ray-Sphere Intersection



$$t_1, t_2 \in [0, +\infty)$$



$$t_1 < 0, t_2 \in [0, +\infty)$$



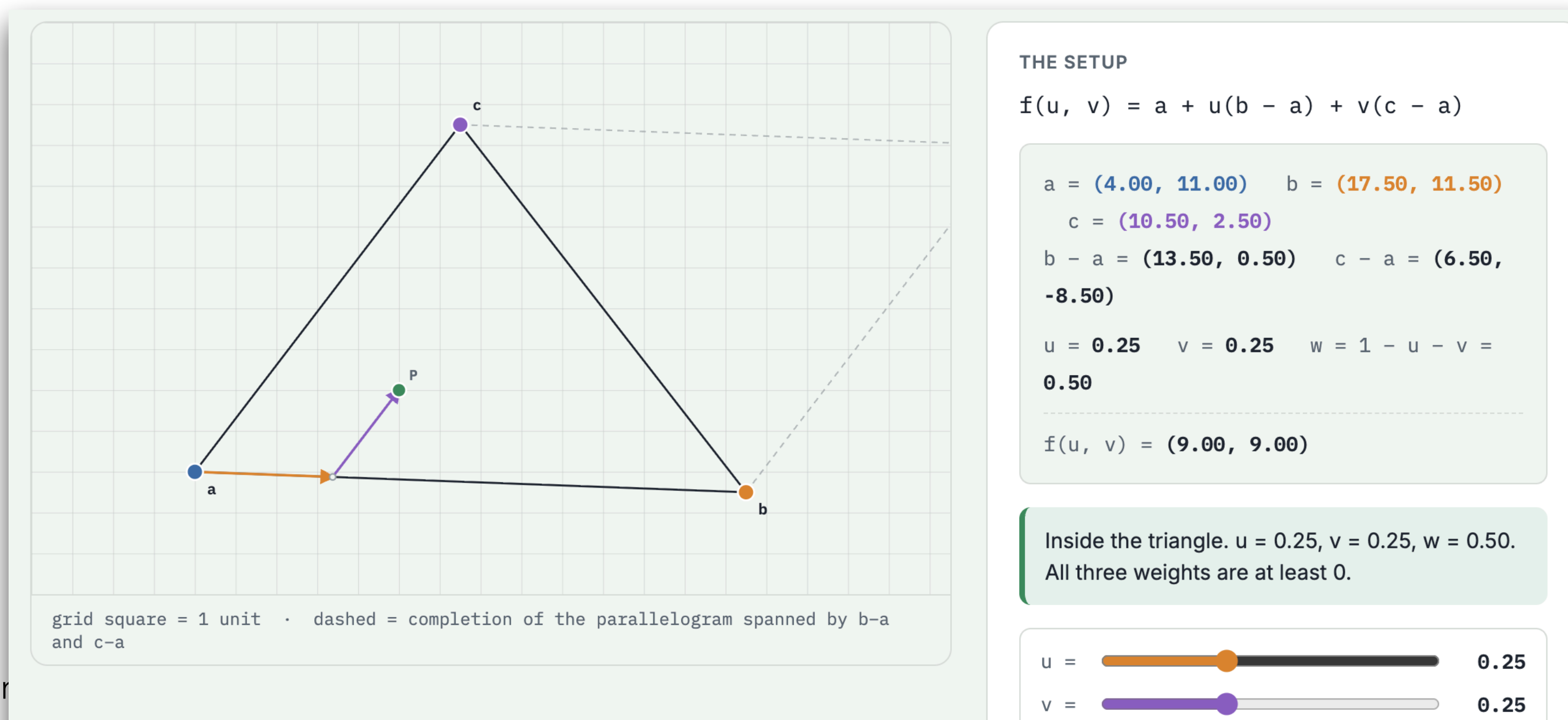
$$t_1, t_2 < 0$$

2. Ray-Triangle Intersection

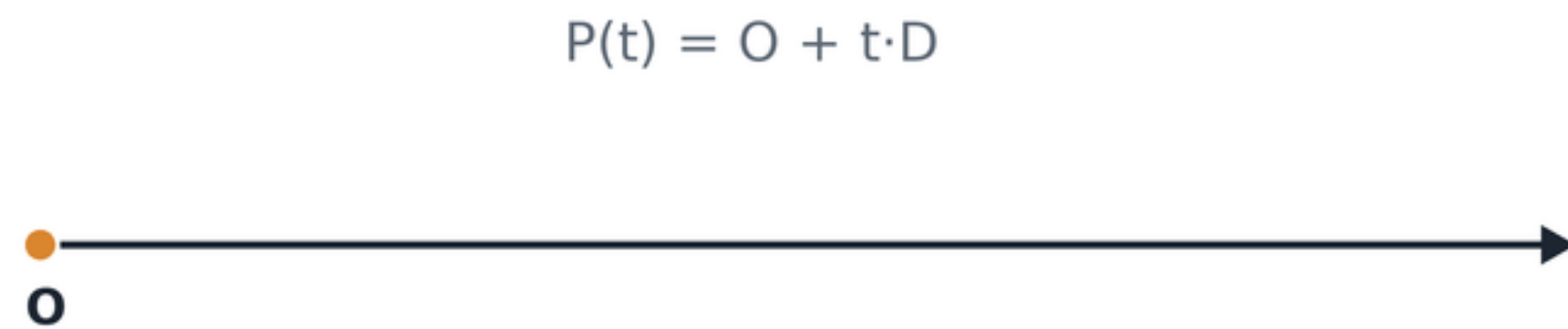
- Explicit parametrization of a triangle with vertices **a**, **b**, **c**:

$$\mathbf{f}(u, v) = \mathbf{a} + u(\mathbf{b} - \mathbf{a}) + v(\mathbf{c} - \mathbf{a})$$

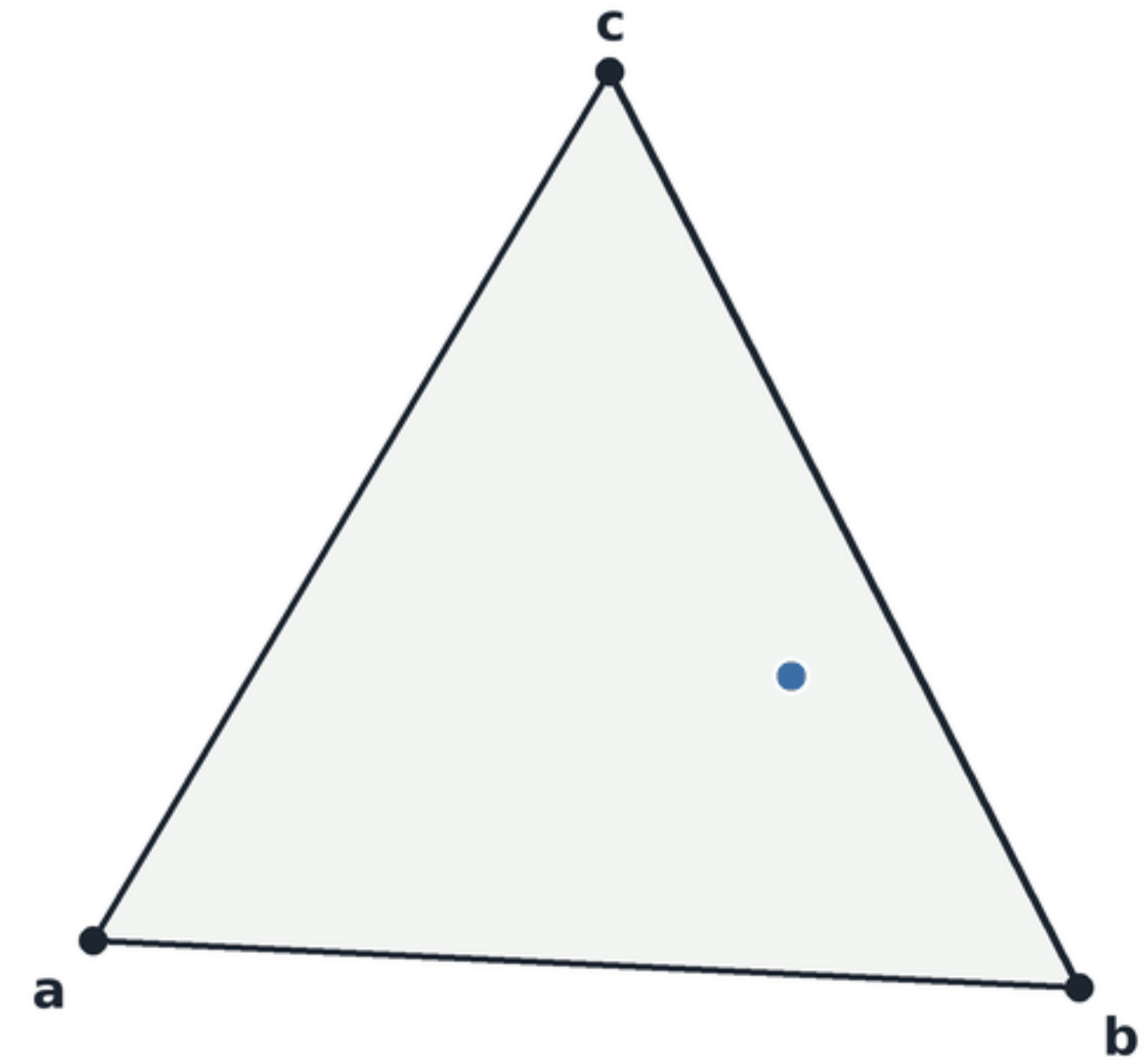
$$0 \leq u, v \quad u + v \leq 1$$



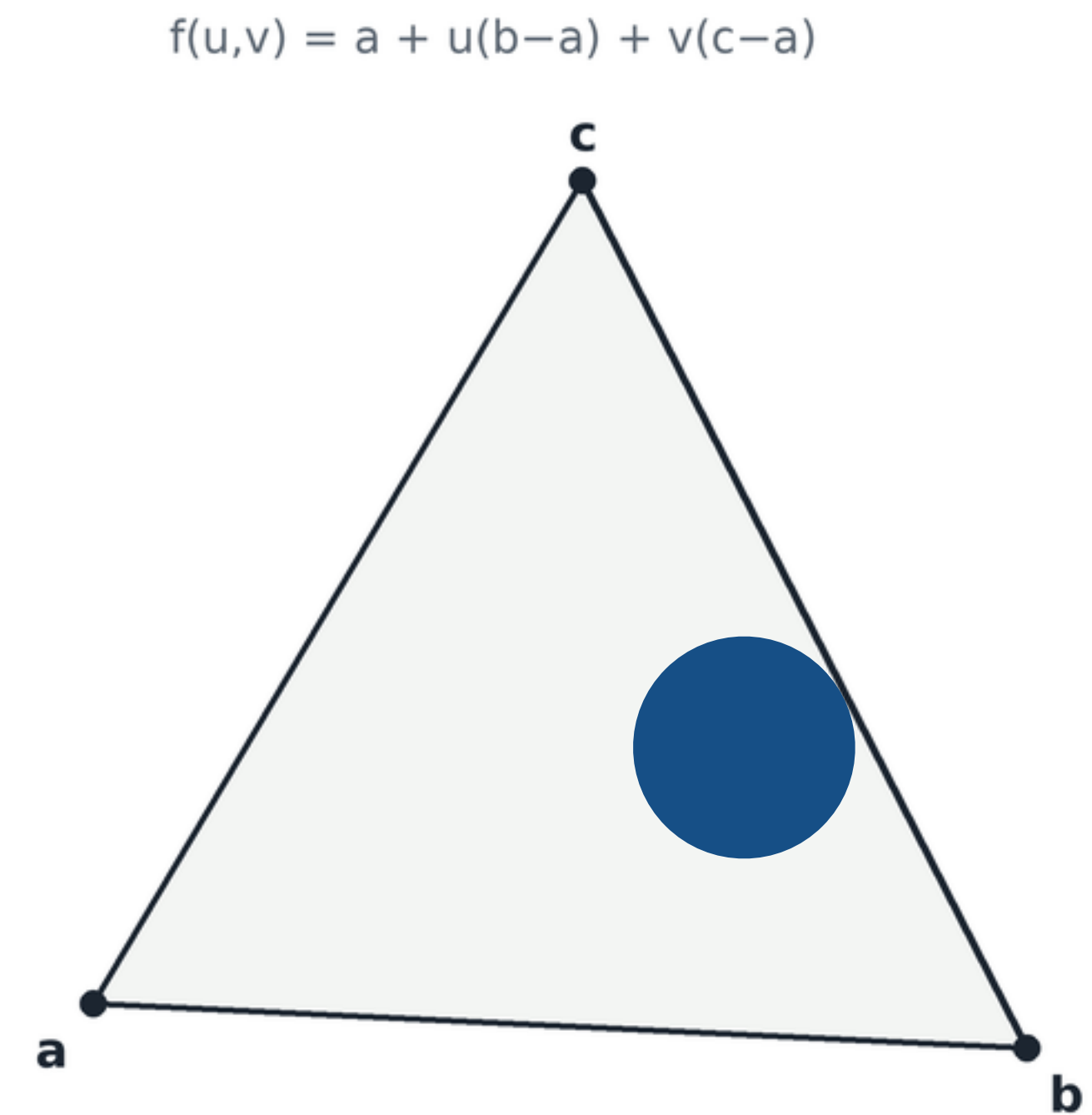
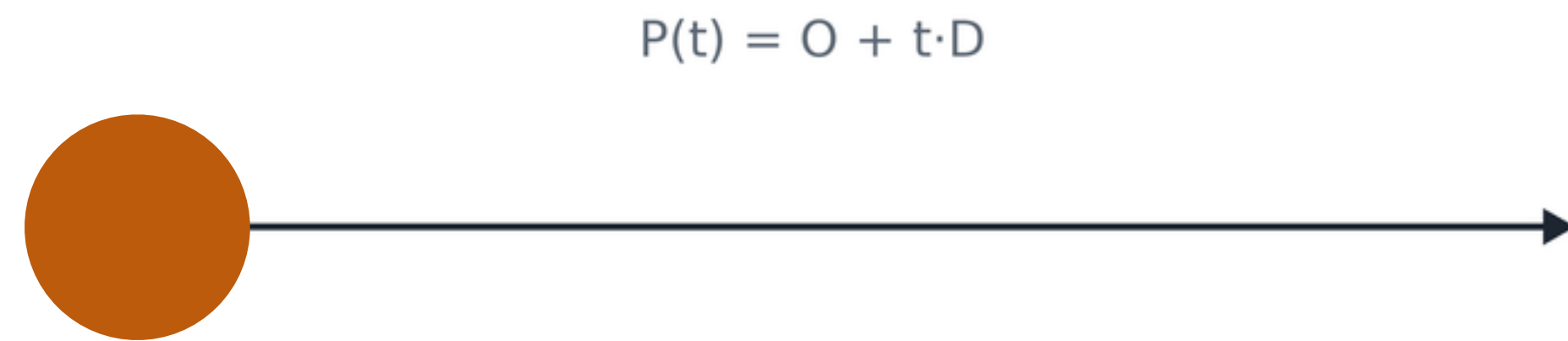
Ray-Triangle Intersection



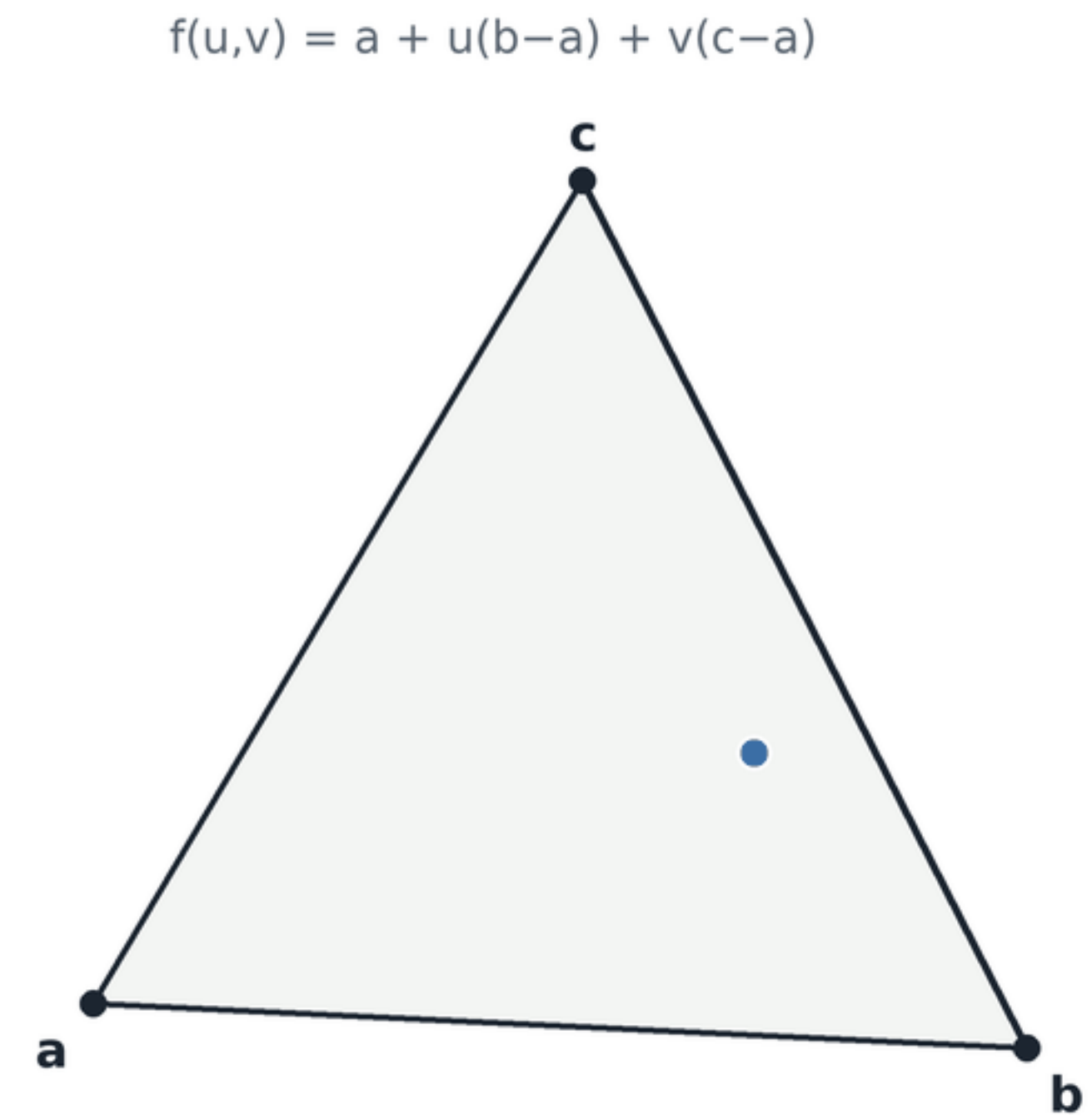
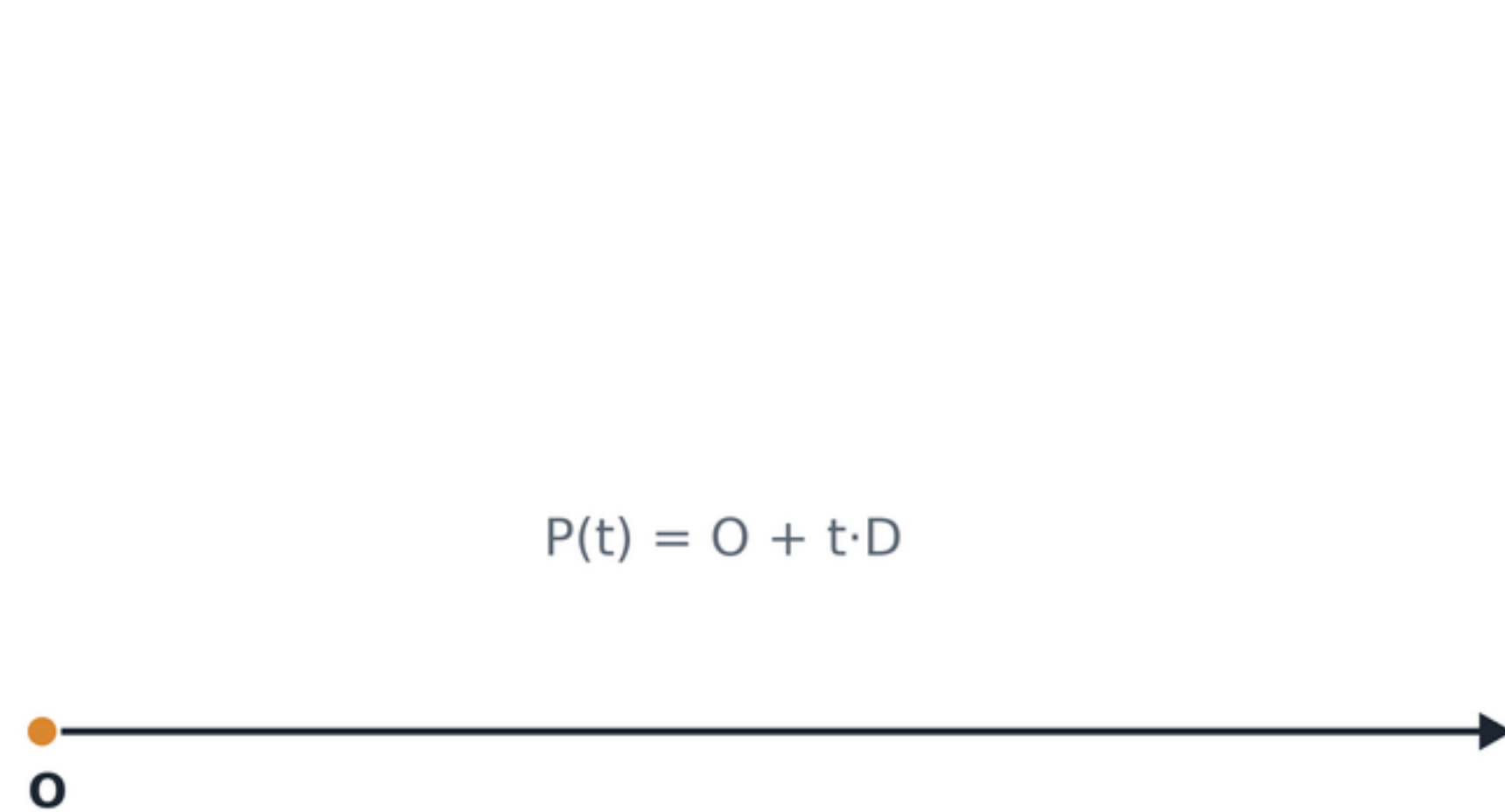
$$f(u,v) = a + u(b-a) + v(c-a)$$



Ray-Triangle Intersection



Ray-Triangle Intersection



Ray-Triangle Intersection

- Does there exist a location where both a point on the ray and a point on the triangle can be?



$$\mathbf{p}(t) = \mathbf{e} + t\mathbf{d}$$
$$t \geq 0$$

$$\mathbf{f}(u, v) = \mathbf{a} + u(\mathbf{b} - \mathbf{a}) + v(\mathbf{c} - \mathbf{a})$$
$$0 \leq u, v \quad u + v \leq 1$$

$$\mathbf{p}(t) = \mathbf{f}(u, v)$$

Ray-Triangle Intersection

$$\mathbf{p}(t) = \mathbf{e} + t\mathbf{d}$$
$$t \geq 0$$

$$\mathbf{f}(u, v) = \mathbf{a} + u(\mathbf{b} - \mathbf{a}) + v(\mathbf{c} - \mathbf{a})$$
$$0 \leq u, v \quad u + v \leq 1$$

$$\mathbf{e} + \mathbf{d}t = \mathbf{a} + u(\mathbf{b} - \mathbf{a}) + v(\mathbf{c} - \mathbf{a})$$

$$(\mathbf{a} - \mathbf{b})u + (\mathbf{a} - \mathbf{c})v + \mathbf{d}t = \mathbf{a} - \mathbf{e}$$

In 3D, 3 equations and 3 unknowns (u, v, t)

Ray-Triangle Intersection

$$(\mathbf{a} - \mathbf{b}) u + (\mathbf{a} - \mathbf{c}) v + \mathbf{d} t = \mathbf{a} - \mathbf{e}$$

$$\begin{bmatrix} | \\ \mathbf{y} \\ | \end{bmatrix} = \begin{bmatrix} | & | & | \\ \mathbf{c}_1 & \mathbf{c}_2 & \mathbf{c}_3 \\ | & | & | \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$\mathbf{y} = x_1 \mathbf{c}_1 + x_2 \mathbf{c}_2 + x_3 \mathbf{c}_3$$

Weighted sum of the columns

Ray-Triangle Intersection

$$(\mathbf{a} - \mathbf{b}) u + (\mathbf{a} - \mathbf{c}) v + \mathbf{d} t = \mathbf{a} - \mathbf{e}$$

$$\begin{bmatrix} (a - b)_x & (a - c)_x & d_x \\ (a - b)_y & (a - c)_y & d_y \\ (a - b)_z & (a - c)_z & d_z \end{bmatrix} \begin{bmatrix} u \\ v \\ t \end{bmatrix} = \begin{bmatrix} (a - e)_x \\ (a - e)_y \\ (a - e)_z \end{bmatrix}$$

Ray-Triangle Intersection

$$\begin{bmatrix} (a-b)_x & (a-c)_x & d_x \\ (a-b)_y & (a-c)_y & d_y \\ (a-b)_z & (a-c)_z & d_z \end{bmatrix} \begin{bmatrix} u \\ v \\ t \end{bmatrix} = \begin{bmatrix} (a-e)_x \\ (a-e)_y \\ (a-e)_z \end{bmatrix}$$

How to solve?

1. Direct solver

$$\mathbf{Ax} = \mathbf{b}$$

$$\mathbf{x} = \mathbf{A}^{-1}\mathbf{b}$$

```
Matrix3f A;
Vector3f b;
A << 5,3,-7, -3,5,12, 9,-2,-2;
b << 4, 9, -3;
cout << "Here is the matrix A:\n" << A << endl;
cout << "Here is the vector b:\n" << b << endl;
Vector3f x = A.colPivHouseholderQr().solve(b);
cout << "The solution is:\n" << x << endl;
```

2. Cramer's rule

The rules for 3×3 matrices are similar. Given

$$\begin{cases} a_1x + b_1y + c_1z = d_1 \\ a_2x + b_2y + c_2z = d_2 \\ a_3x + b_3y + c_3z = d_3 \end{cases}$$

which in matrix format is

$$\begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix}$$

Then the values of x , y and z can be found as follows:

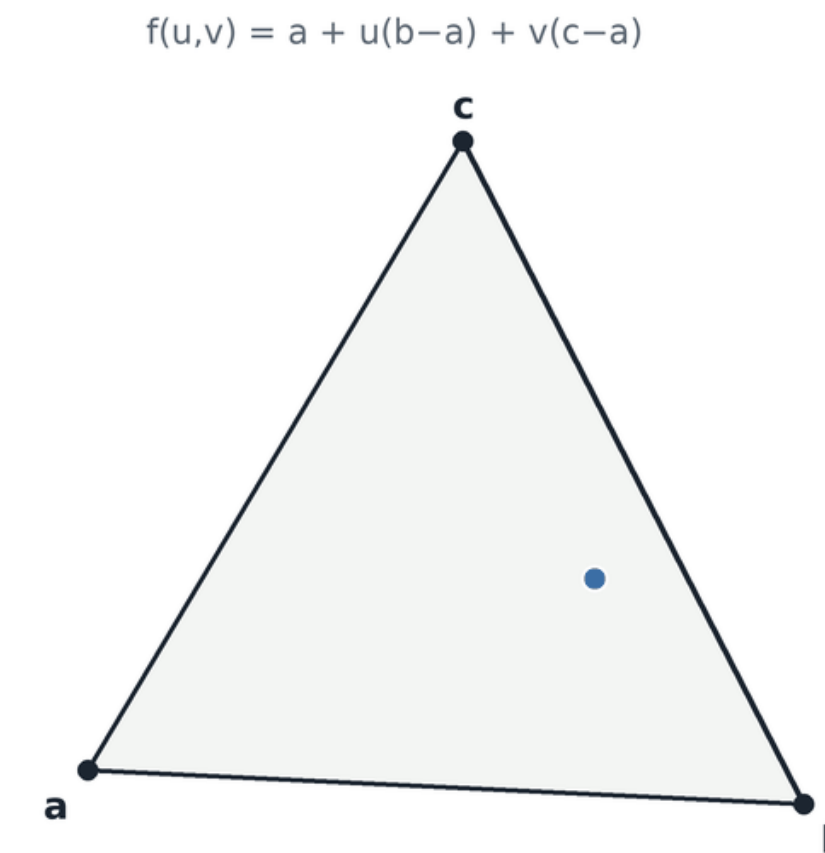
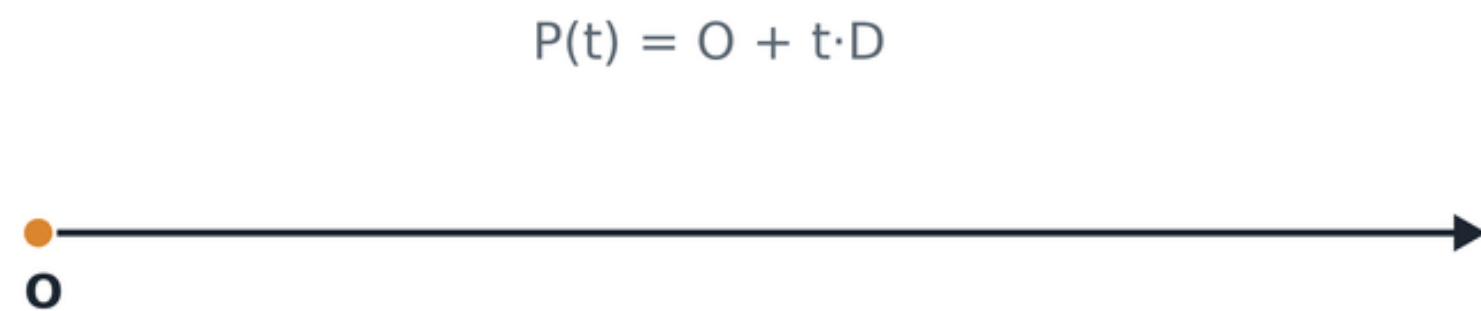
$$x = \frac{\begin{vmatrix} d_1 & b_1 & c_1 \\ d_2 & b_2 & c_2 \\ d_3 & b_3 & c_3 \end{vmatrix}}{\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}}, \quad y = \frac{\begin{vmatrix} a_1 & d_1 & c_1 \\ a_2 & d_2 & c_2 \\ a_3 & d_3 & c_3 \end{vmatrix}}{\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}}, \quad z = \frac{\begin{vmatrix} a_1 & b_1 & d_1 \\ a_2 & b_2 & d_2 \\ a_3 & b_3 & d_3 \end{vmatrix}}{\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}}$$



Ray-Triangle Intersection

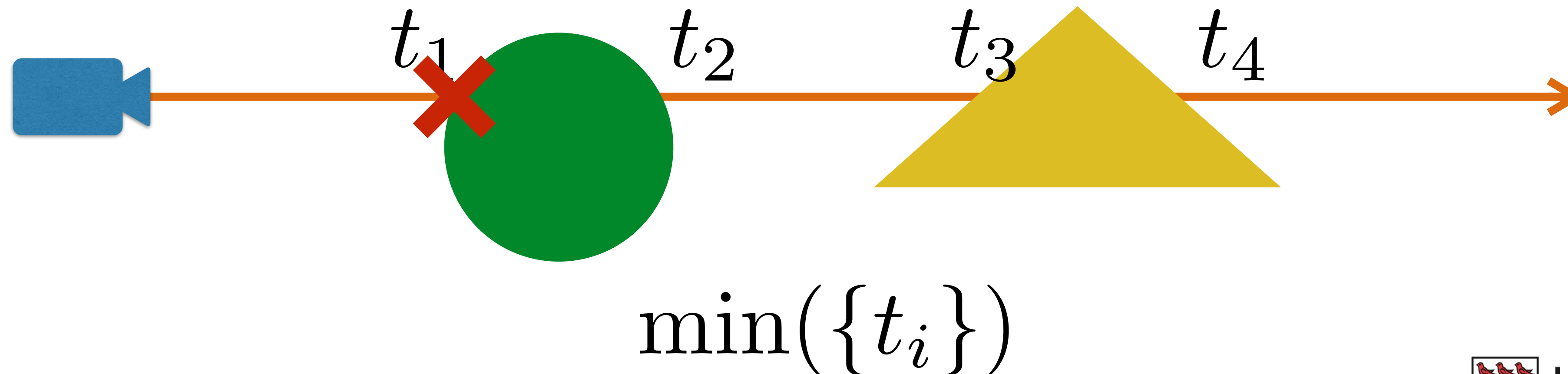
$$\begin{bmatrix} (a - b)_x & (a - c)_x & d_x \\ (a - b)_y & (a - c)_y & d_y \\ (a - b)_z & (a - c)_z & d_z \end{bmatrix} \begin{bmatrix} u \\ v \\ t \end{bmatrix} = \begin{bmatrix} (a - e)_x \\ (a - e)_y \\ (a - e)_z \end{bmatrix}$$

$$t \geq 0, \quad u \geq 0, \quad v \geq 0, \quad u + v \leq 1$$



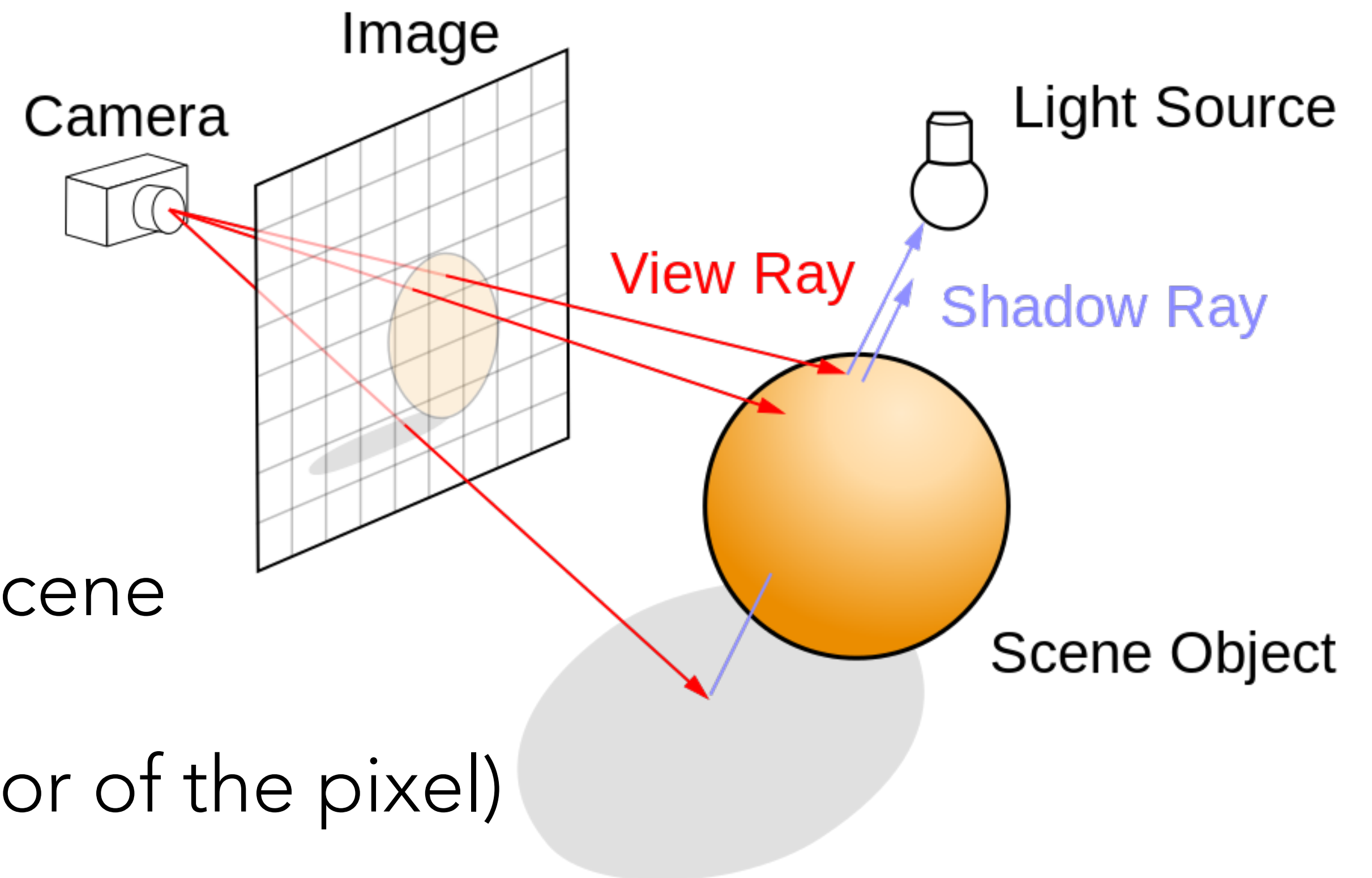
Multiple Objects

- It is simple, intersect it with all of them and only keep the closest intersection
- To speed up computation, you can use a spatial data structure to prune the number of collisions that you need to check



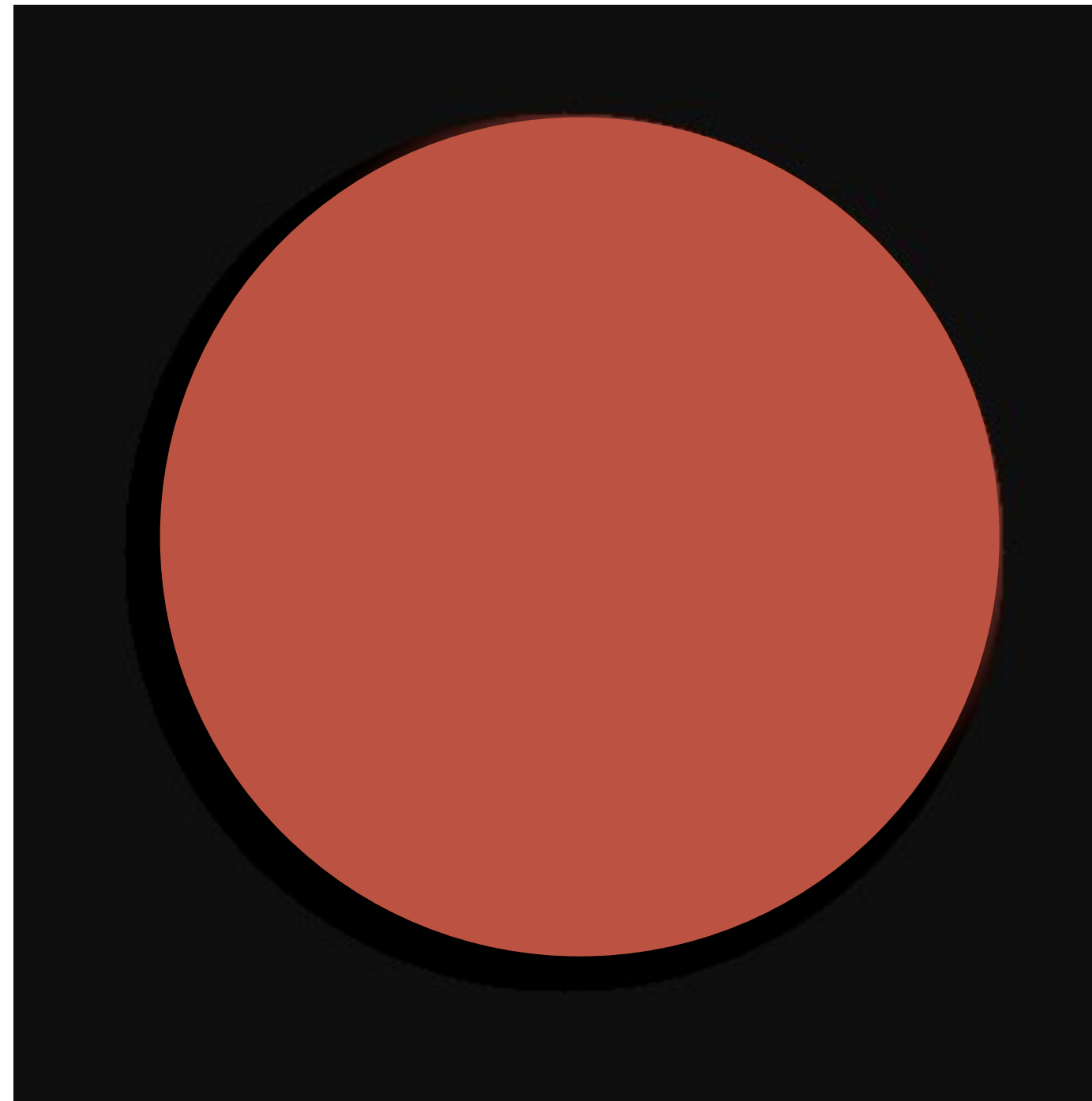
Basic Raytracing

1. Generation of rays
2. Intersection with objects in the scene
3. Shading (computation of the color of the pixel)

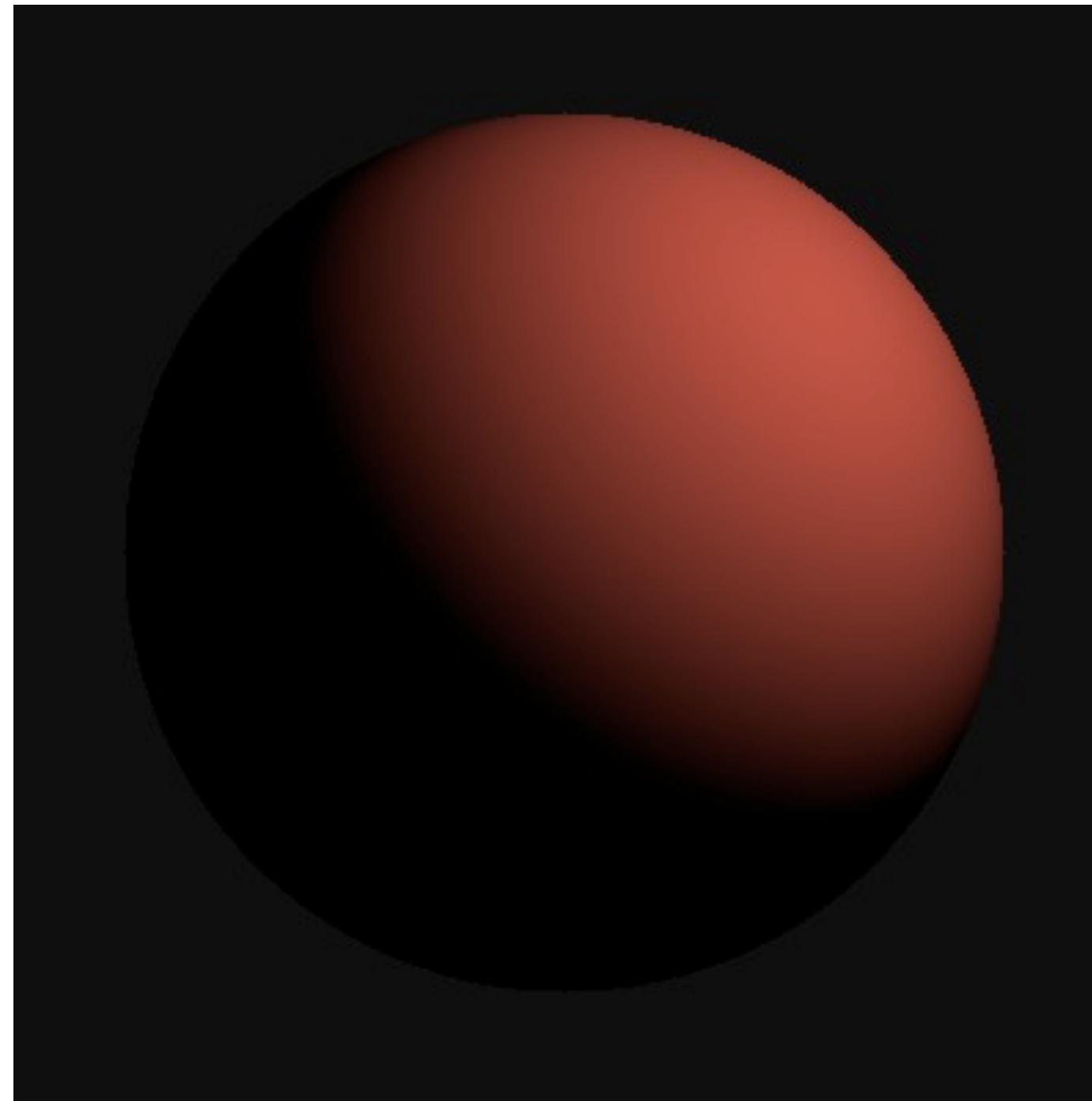


By Henrik - Own work, GFDL, <https://commons.wikimedia.org/w/index.php?curid=3869326>

Shading

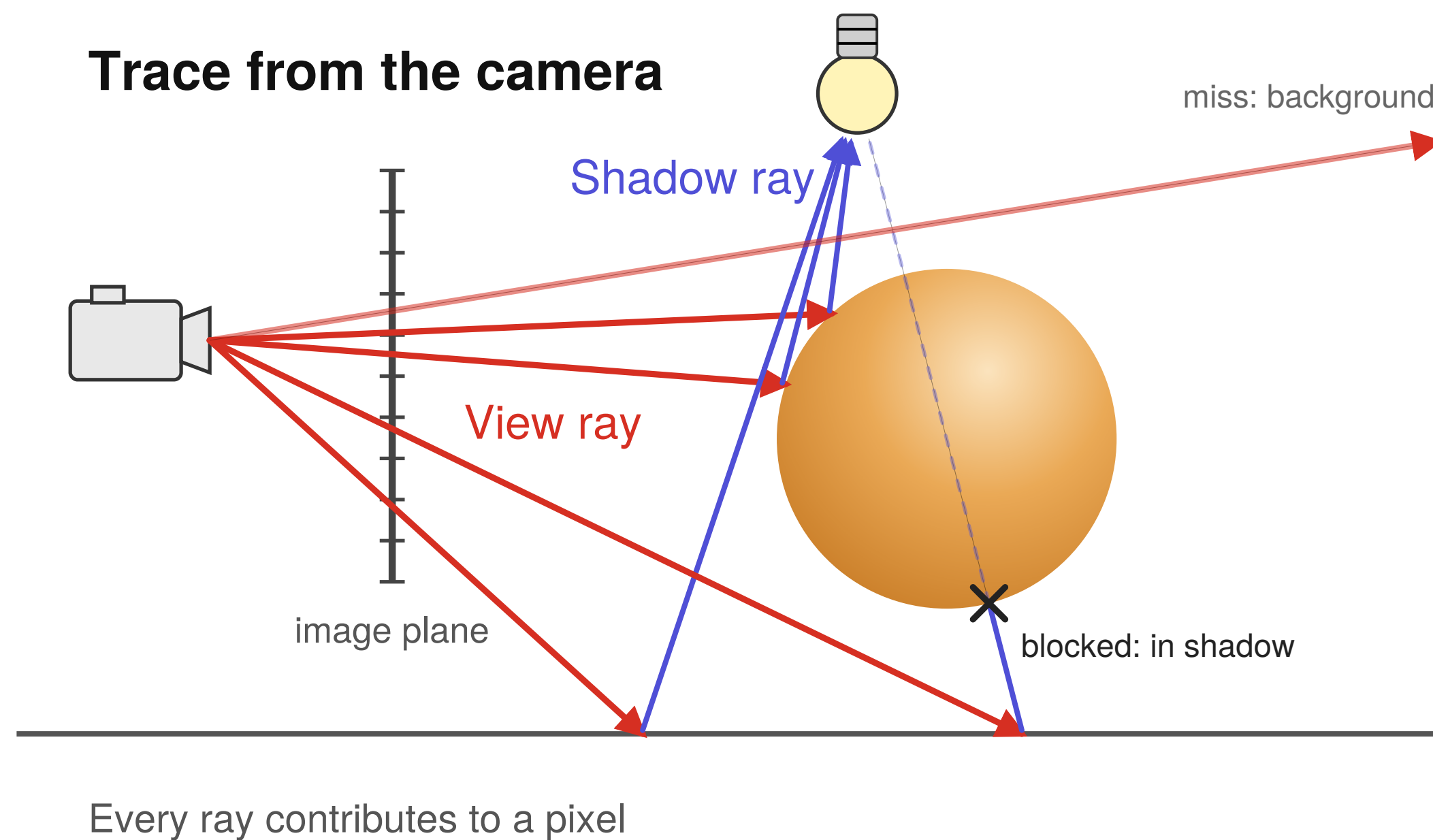


Shading

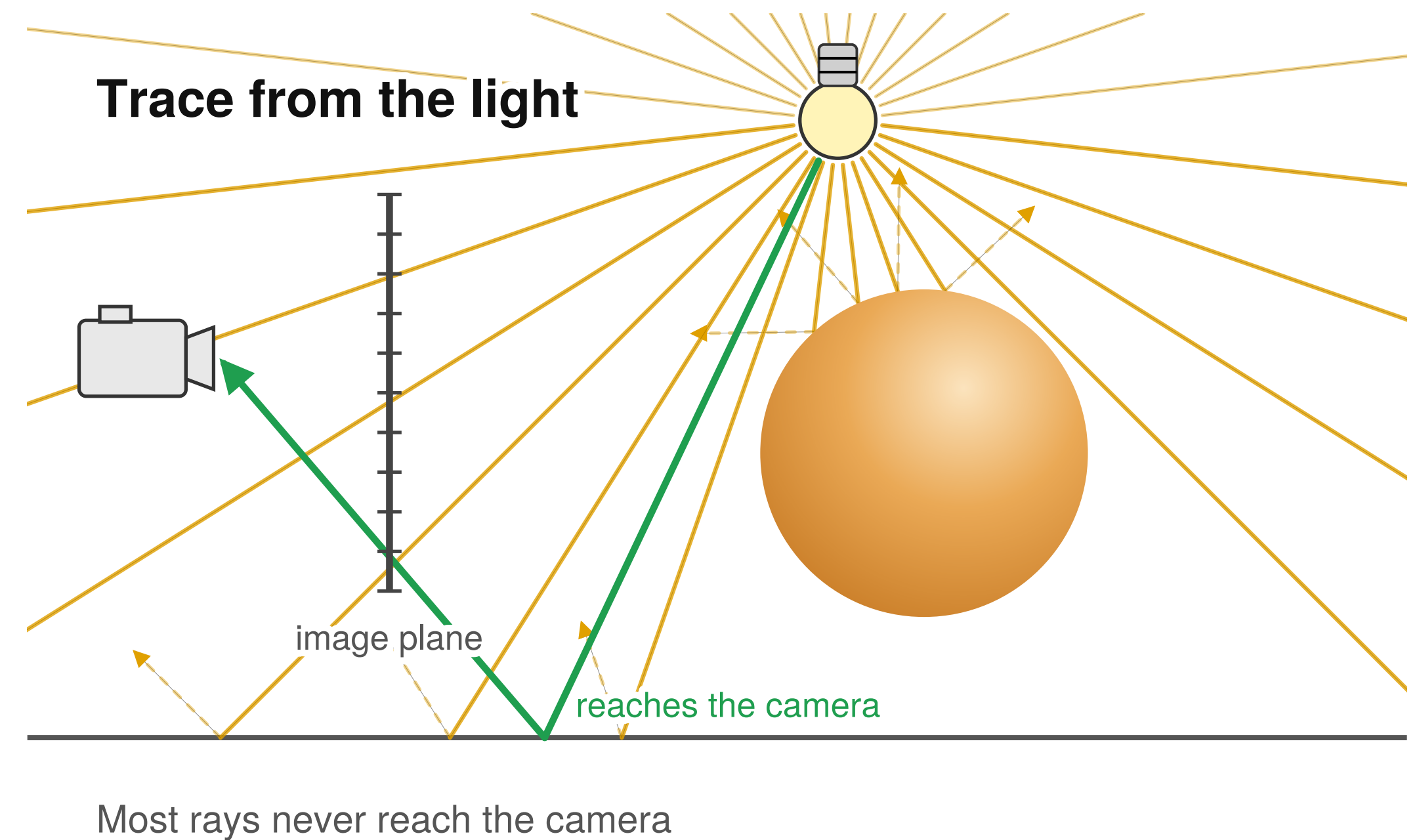


Shading

- Light paths are reversible



All rays are useful

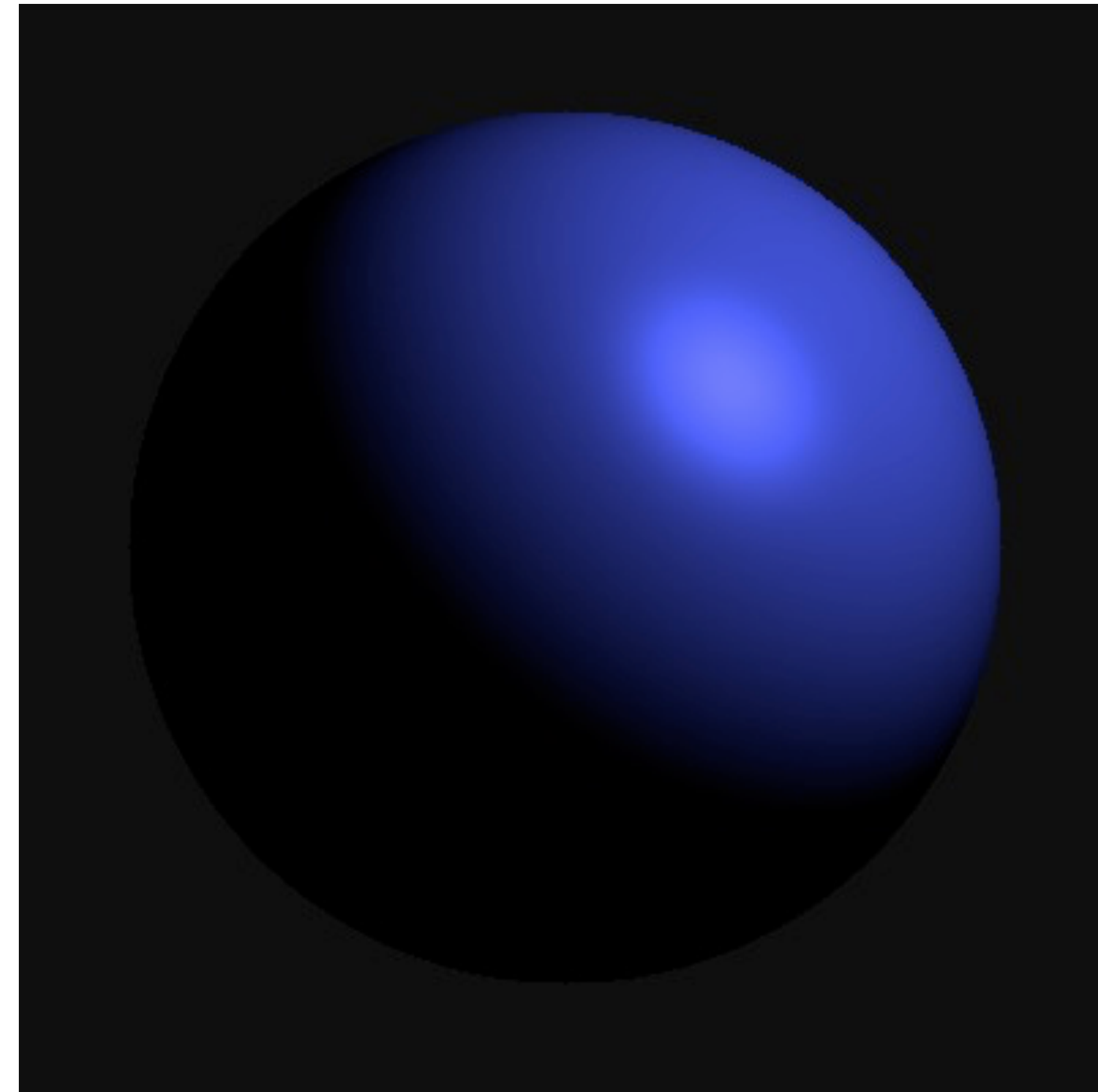
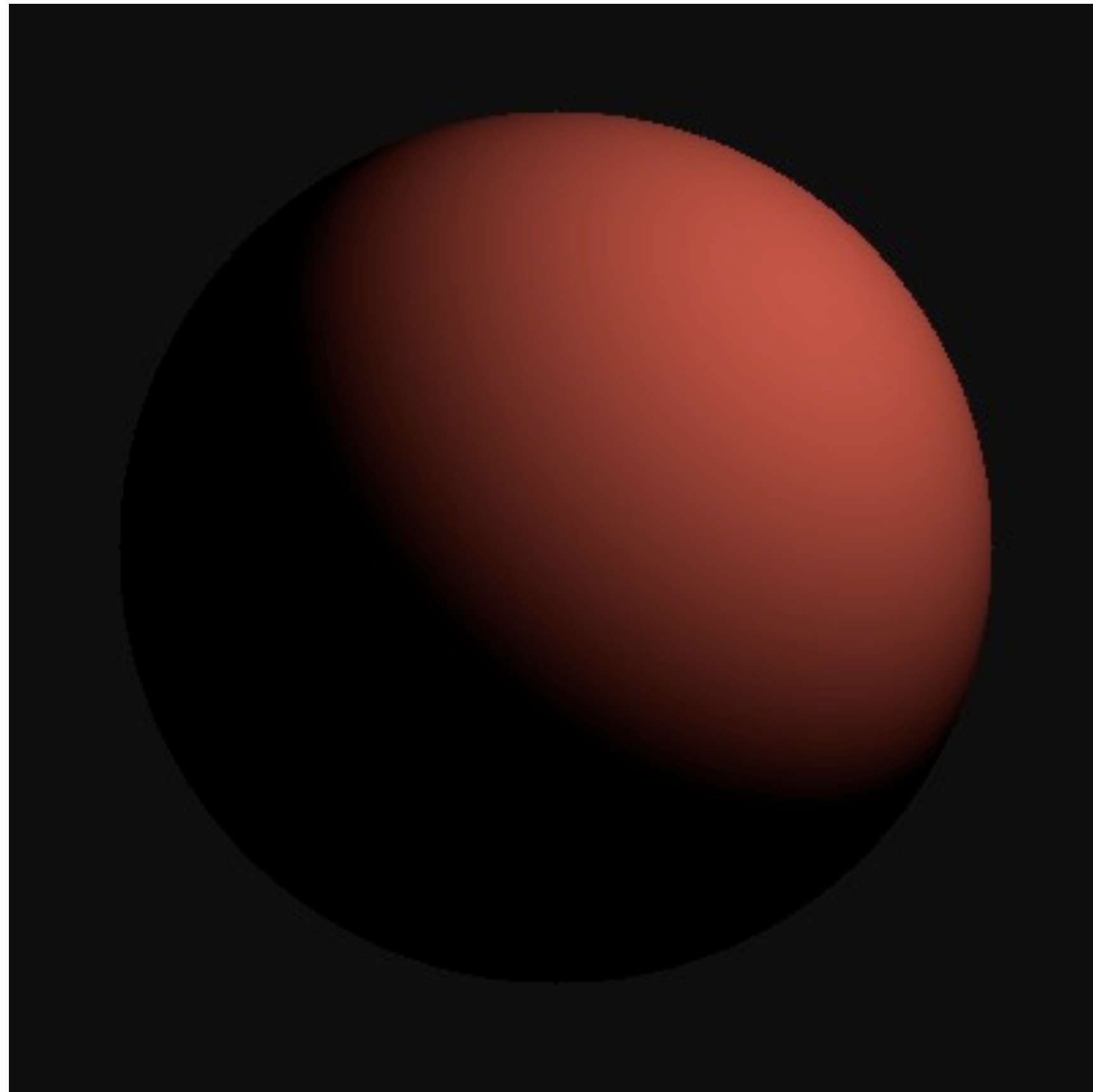


Only rays hit camera are useful;
Others are wasted

Shading

- Modeling accurately the behavior of light is difficult and computationally expensive
- We will use an approximation that is simple and efficient. It is divided in 3 parts:
 - Diffuse (Lambertian) Shading
 - Specular (Blinn-Phong) Shading
 - Ambient Shading
- The three terms will be summed together to obtain the final color

Diffuse and Specular

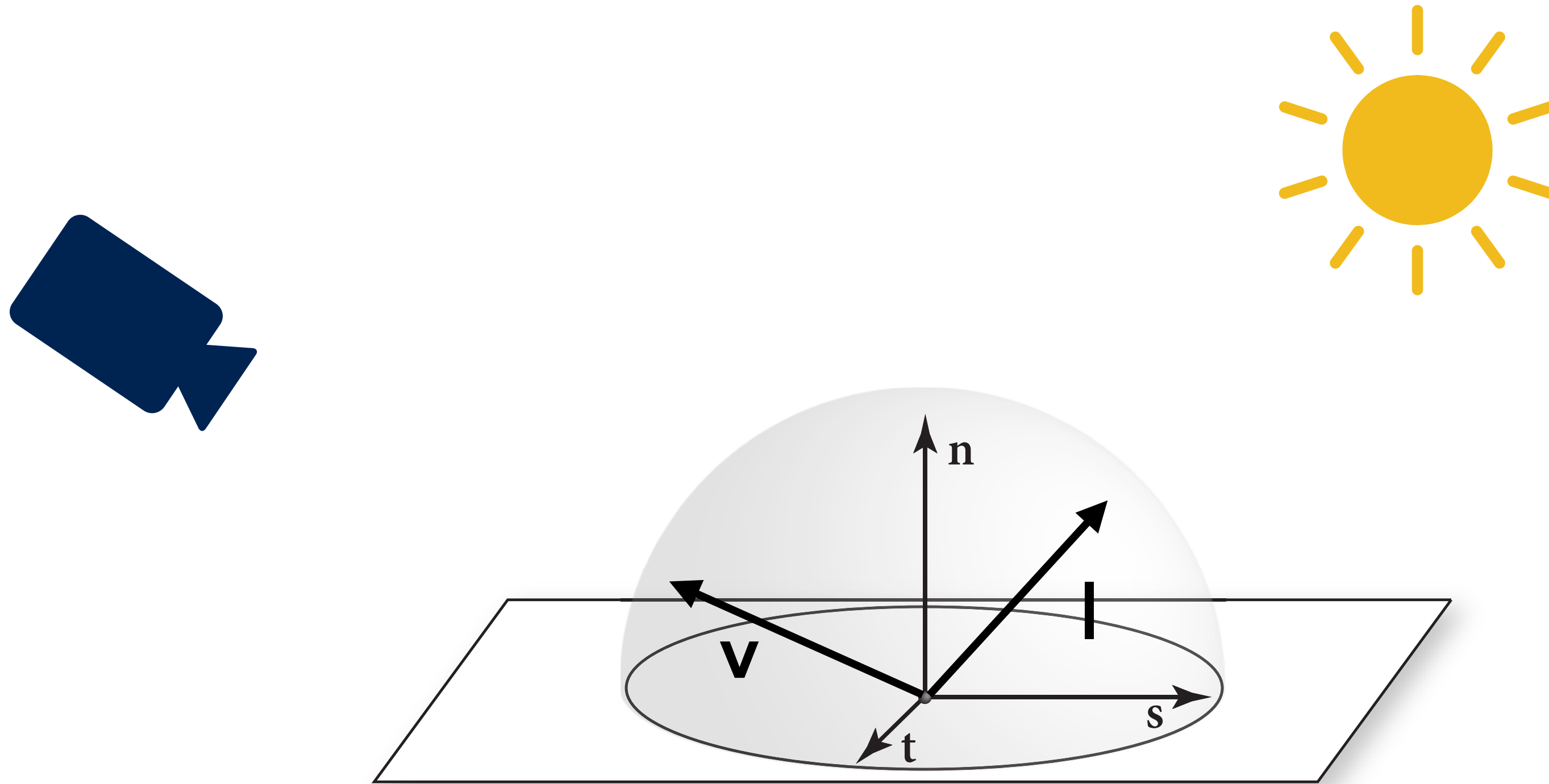


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Shading Variables

- The shading depends on the entire scene, the light can bounce, reflect and be absorbed by anything that it encounters => Physically based
- We will simplify it => Non-physically based

Shading Variables

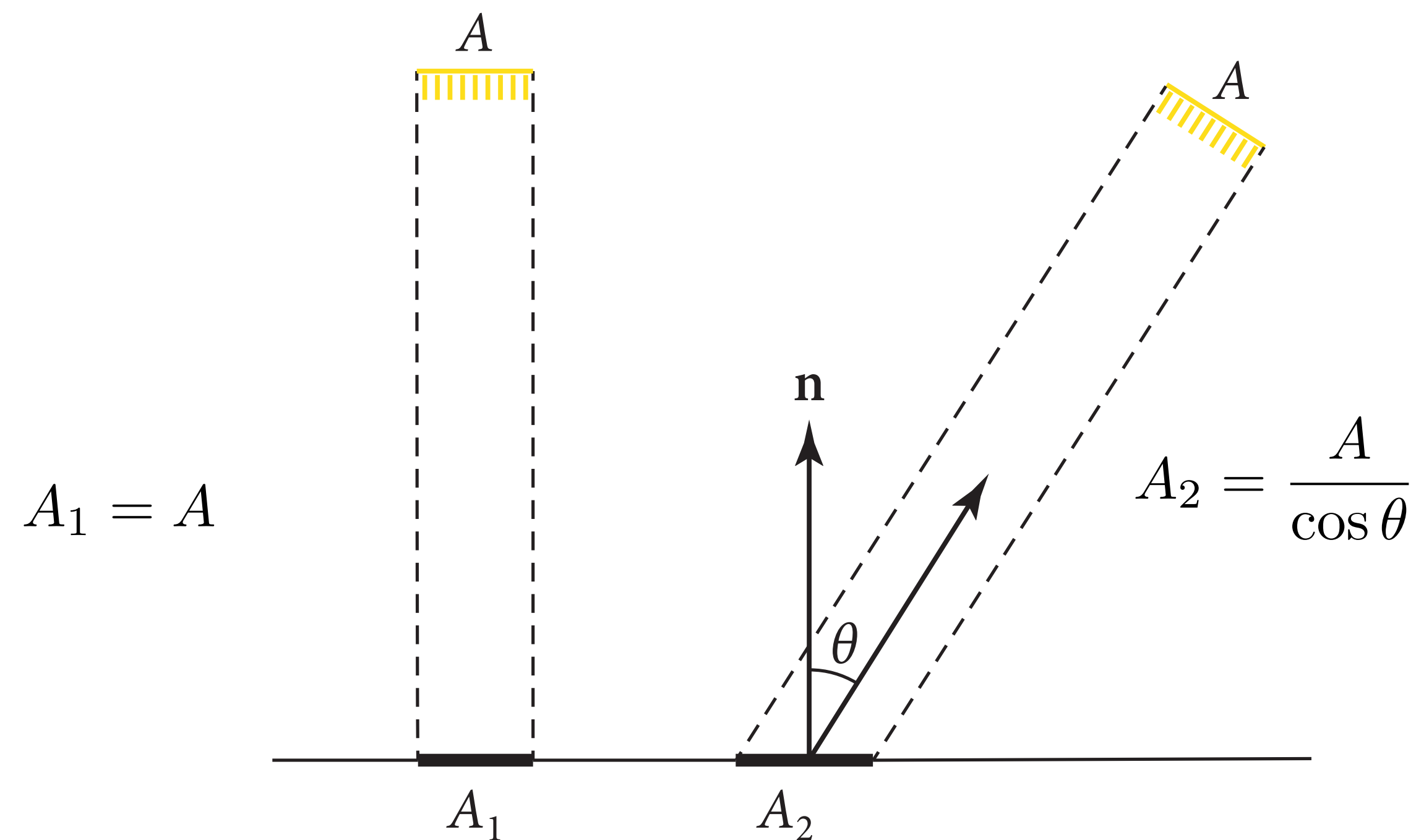


Pharr, Matt, et al. Physically based rendering. Morgan Kaufmann Publisher, 2016.

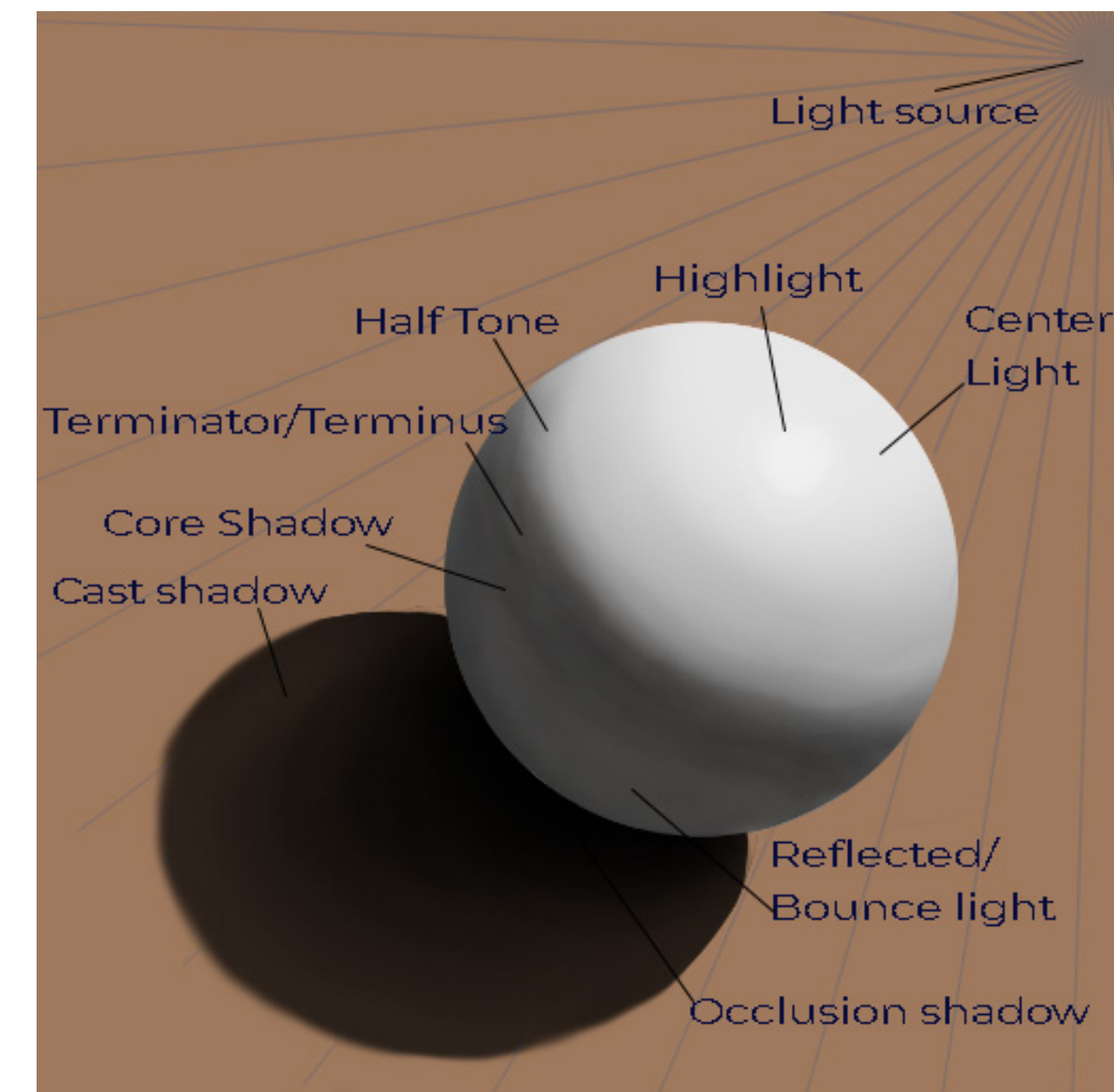
- The light direction $\mathbf{l}$ (a *unit* vector pointing to the light source)
- The view direction $\mathbf{v}$ (a *unit* vector pointing toward the camera)
- The surface normal $\mathbf{n}$ (a unit vector perpendicular to the surface at the point of intersection)

Diffuse Shading

- Lambert (18th century) observed that the amount of energy from a light source that falls on an area of surface depends on the angle of the surface to the light



Pharr, Matt, et al. Physically based rendering. Morgan Kaufmann Publisher, 2016.



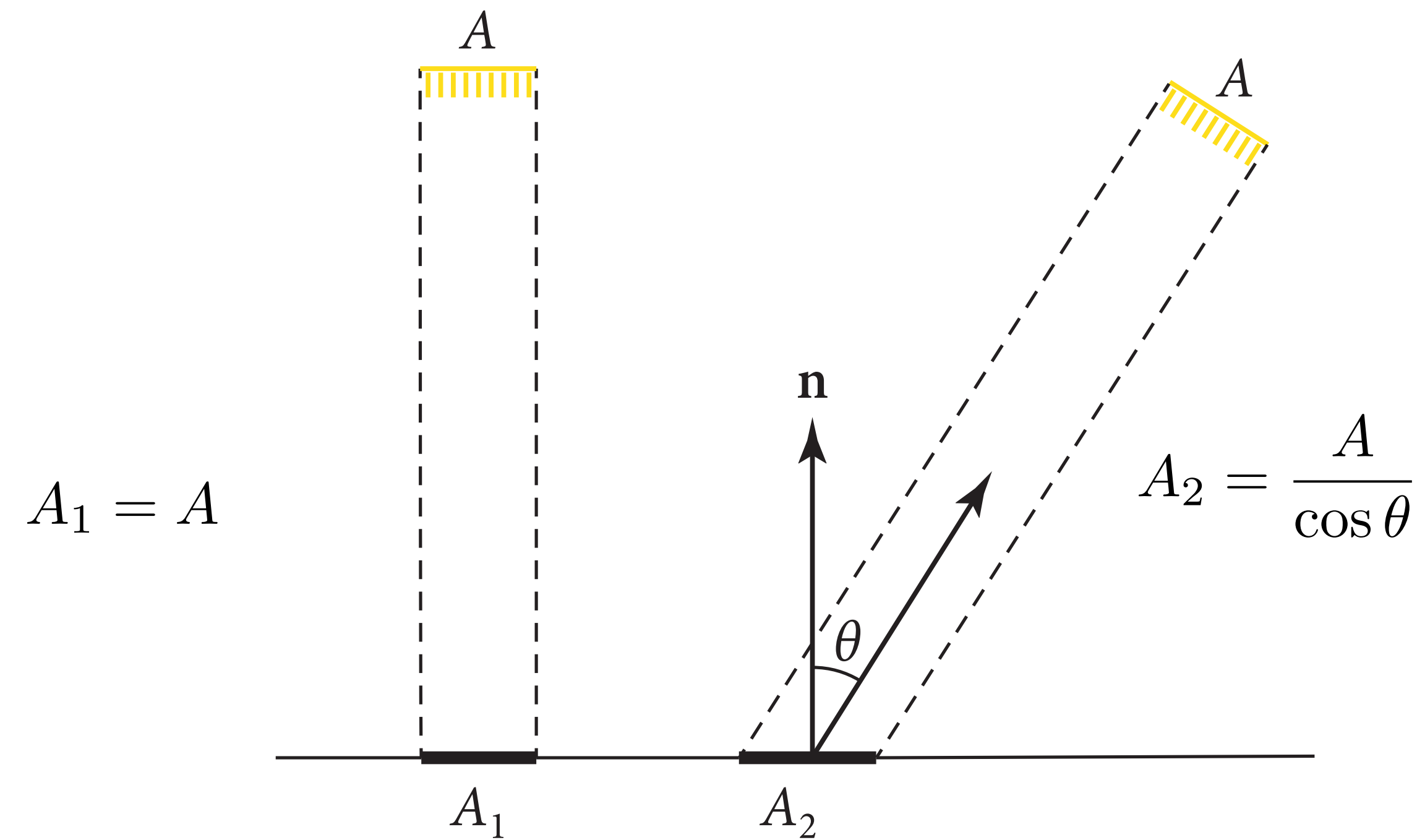
<https://cecelyv.com/how-to-draw-a-sphere/>



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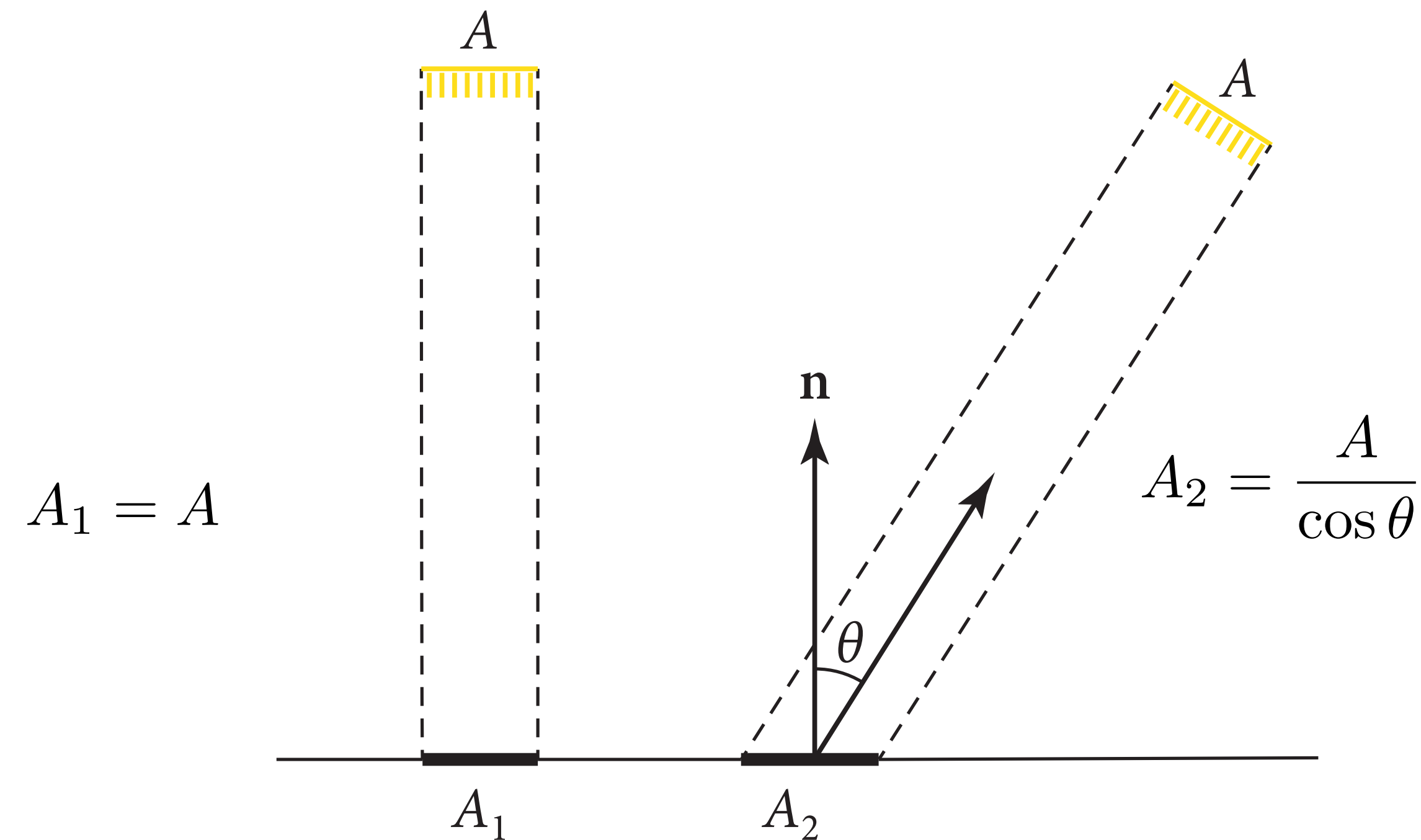
Diffuse Shading



Pharr, Matt, et al. Physically based rendering. Morgan Kaufmann Publisher, 2016.

$$\text{Radiance} \quad L \propto \frac{I}{A_{\text{surface}}} \quad \begin{matrix} \text{Intensity} \\ \text{Surface area lit} \end{matrix}$$

Diffuse Shading



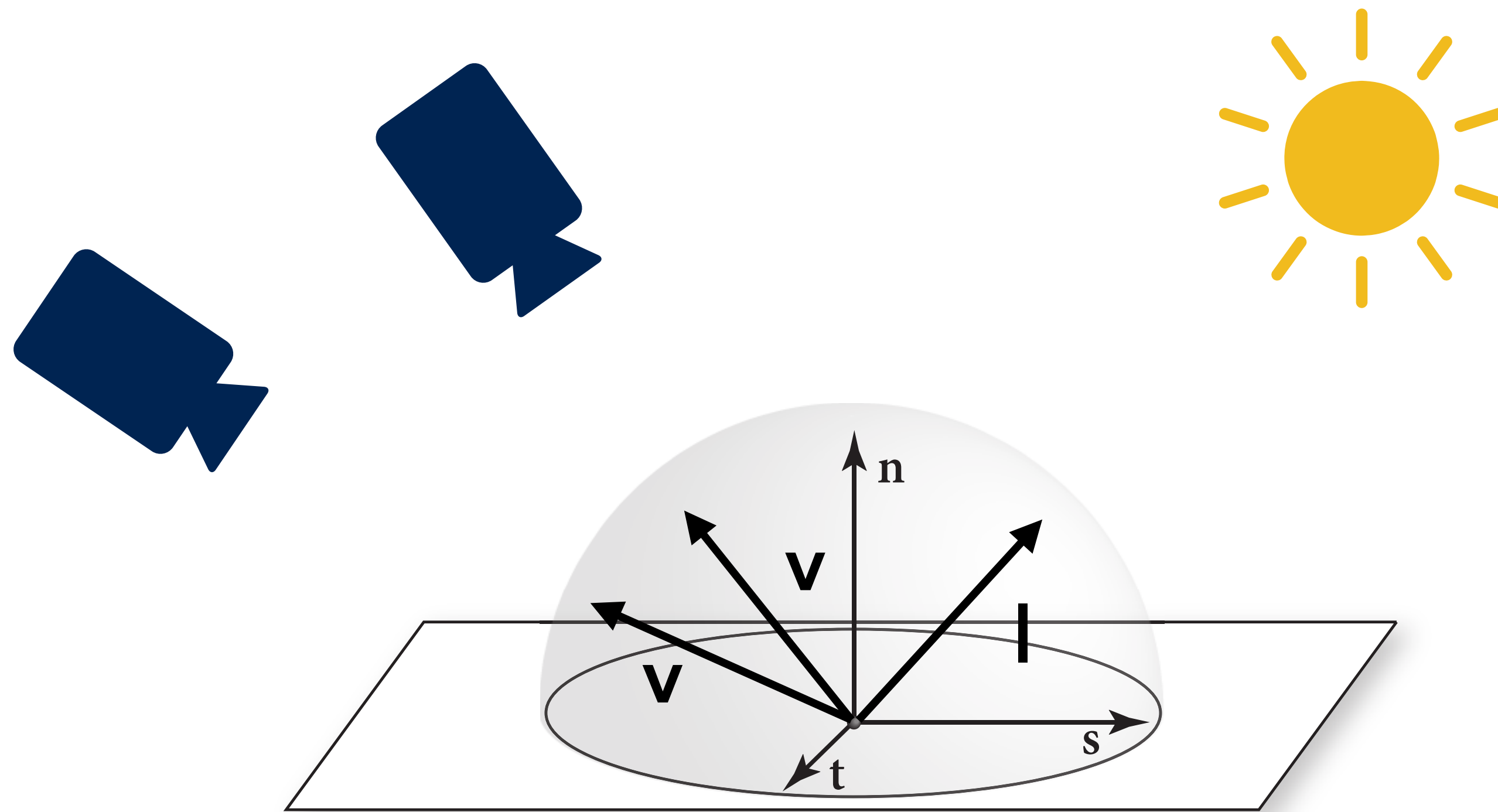
Pharr, Matt, et al. Physically based rendering. Morgan Kaufmann Publisher, 2016.

diffuse coefficient

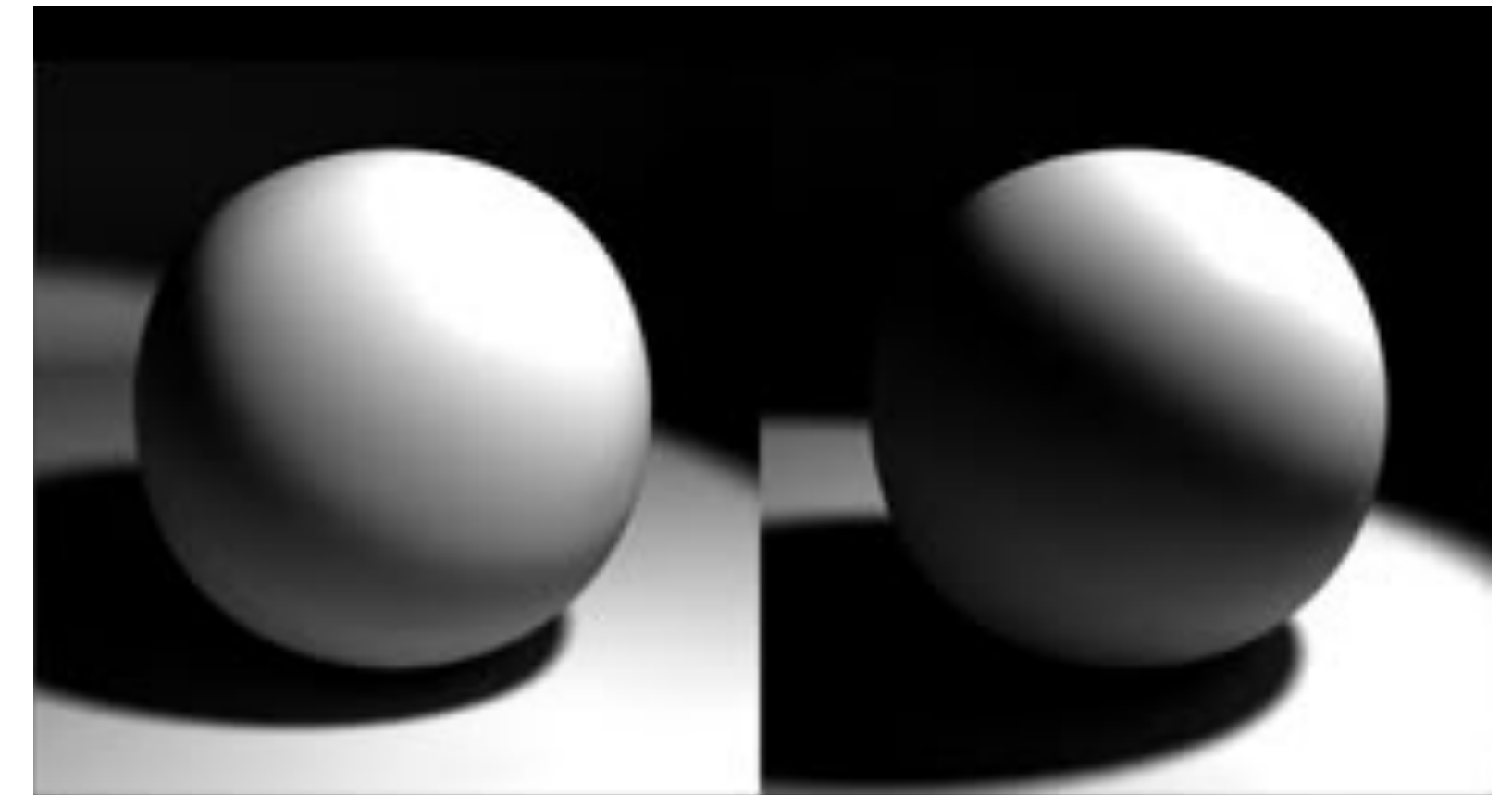
$\cos \theta$ if facing the light

$$L = k_d I \max(0, \mathbf{n} \cdot \mathbf{l})$$

Diffuse Shading



Pharr, Matt, et al. Physically based rendering. Morgan Kaufmann Publisher, 2016.

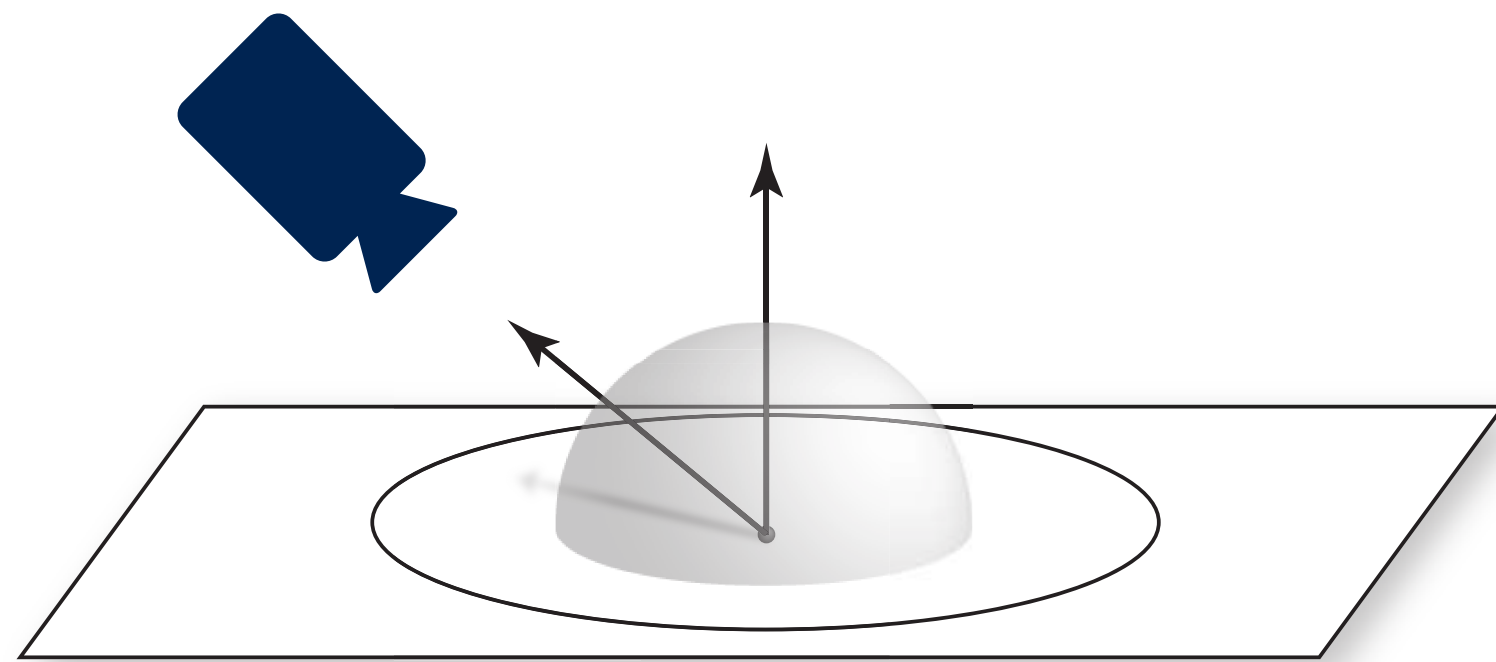


<https://xing2lee.tumblr.com/post/110596023836/im-going-to-be-doing-some-lit-sphere-object/amp>

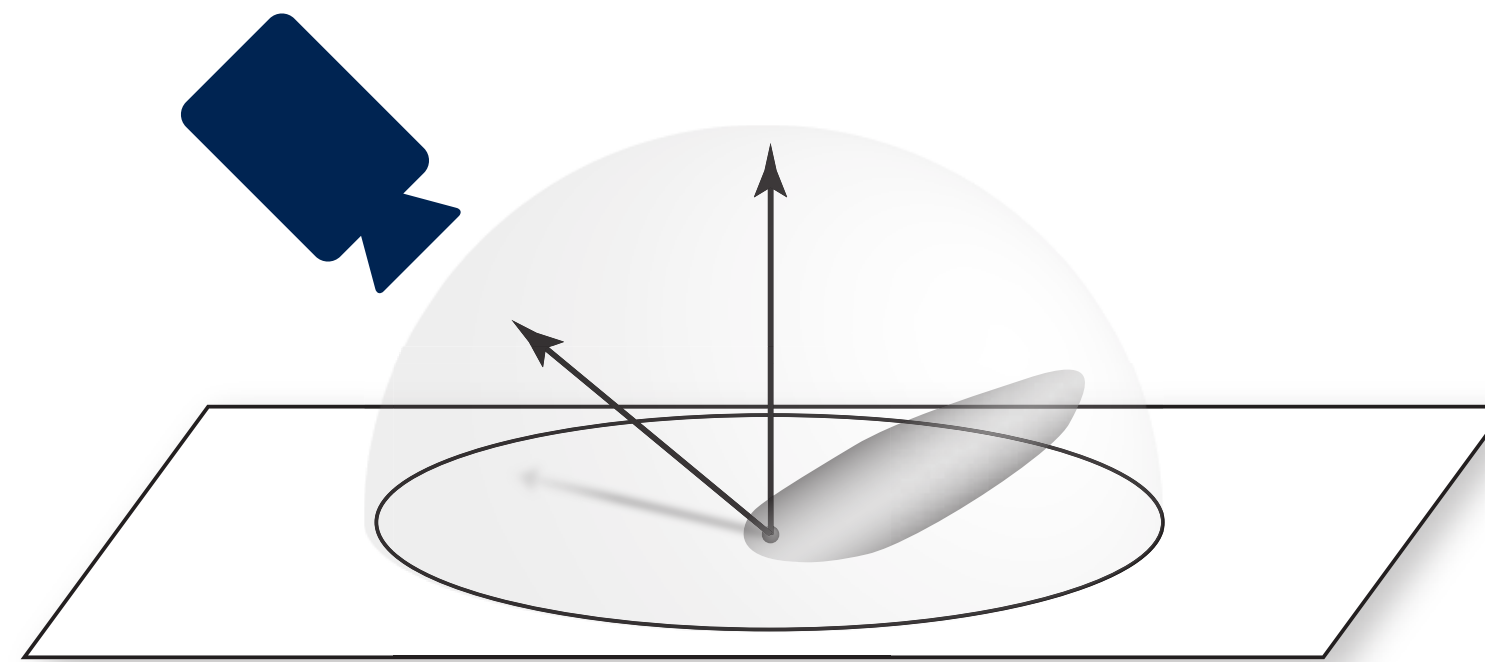
Same amount of light reflected to all (camera) directions

Specular Shading

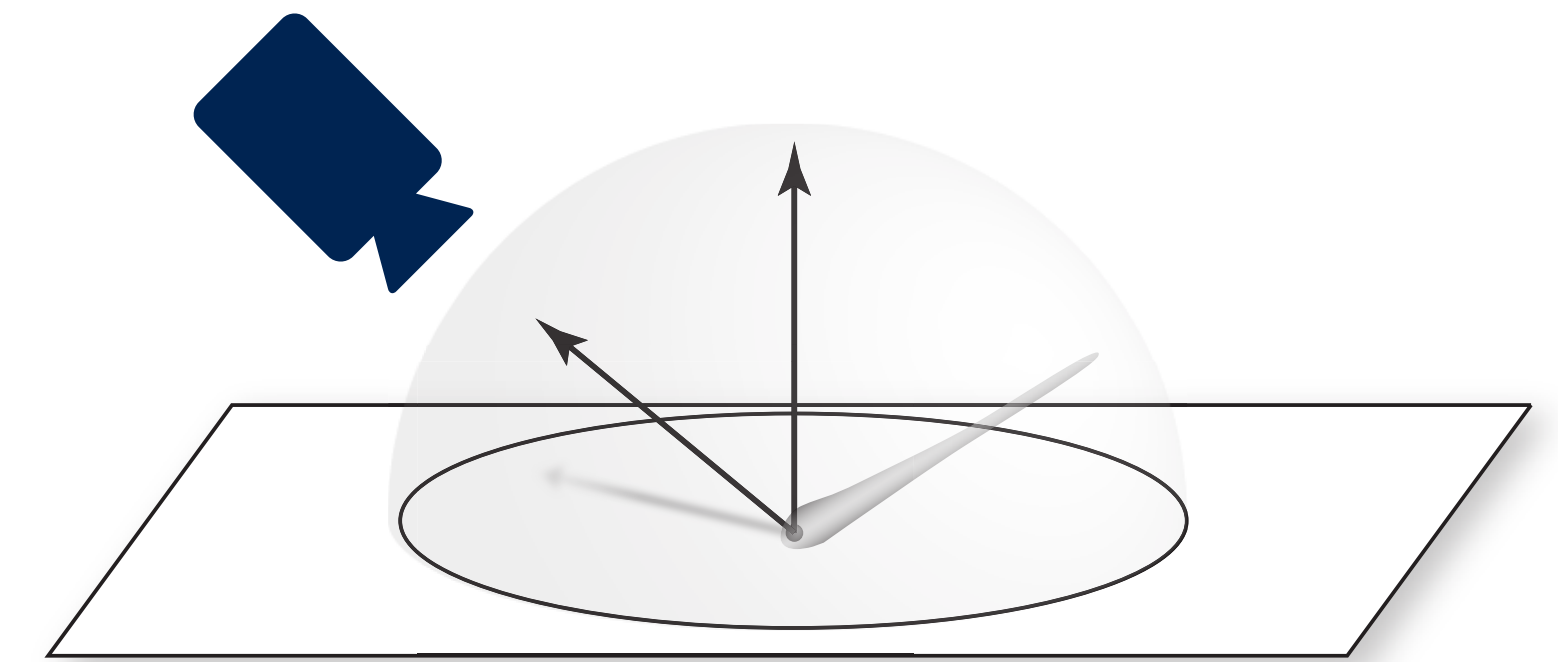
- Specular highlights depend on the position of the viewer



Diffuse



Glossy specular

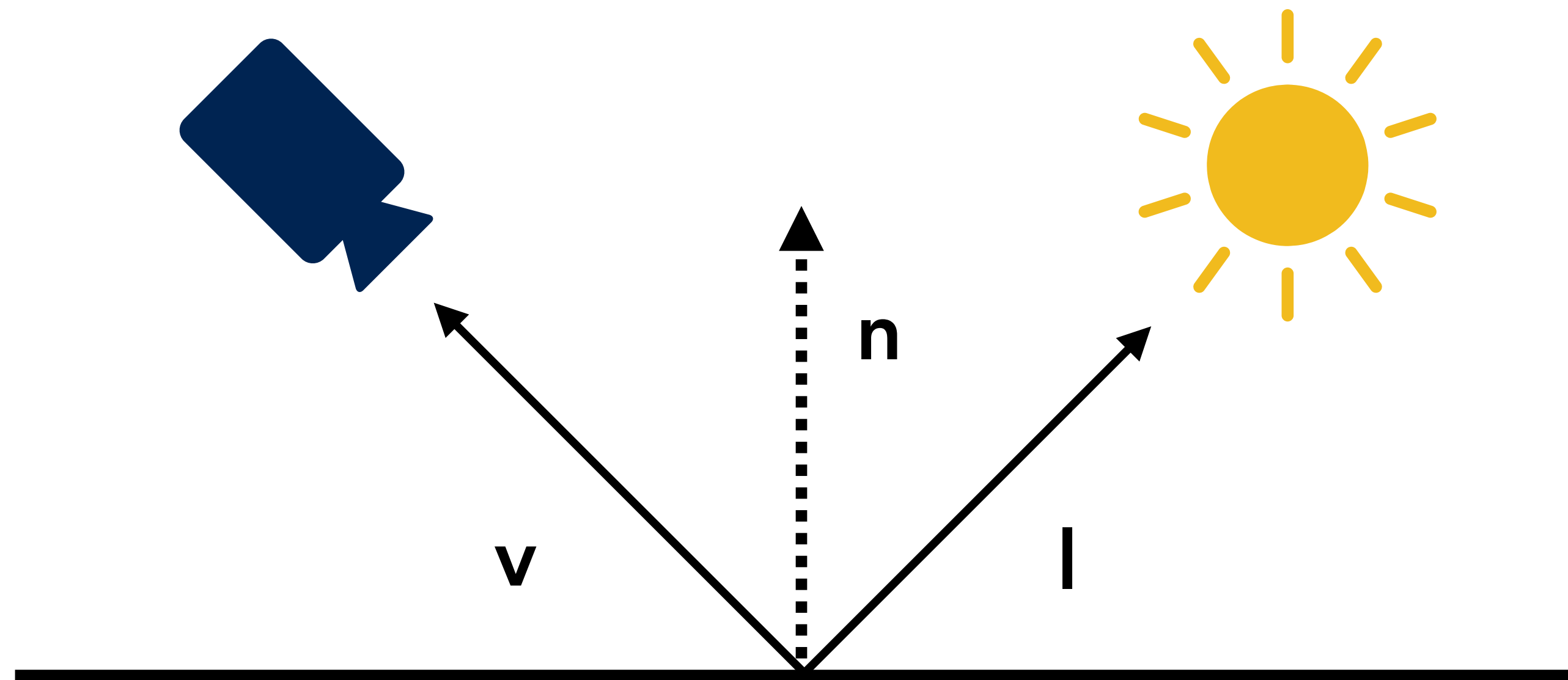


(Nearly) perfect specular
Mirror

<https://gurneyjourney.blogspot.com/2013/03/highlights-and-specularity-part-1.html>

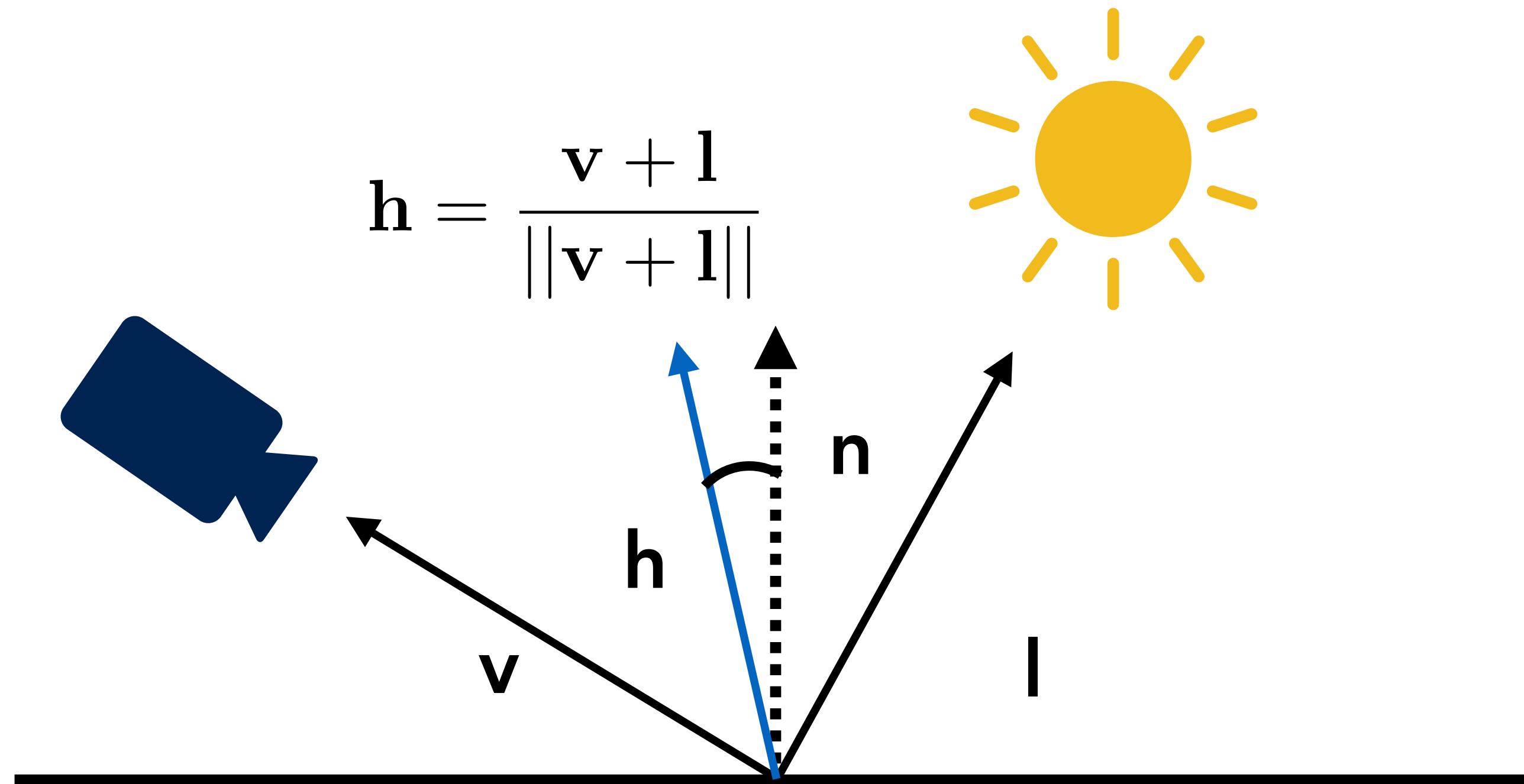
Specular Shading

- A simple and effective model to model them has been proposed by Phong (1975) and refined by Blinn (1976)
- The idea is to produce a reflection that is bright if $\mathbf{v}$ and $\mathbf{l}$ are symmetric wrt $\mathbf{n}$



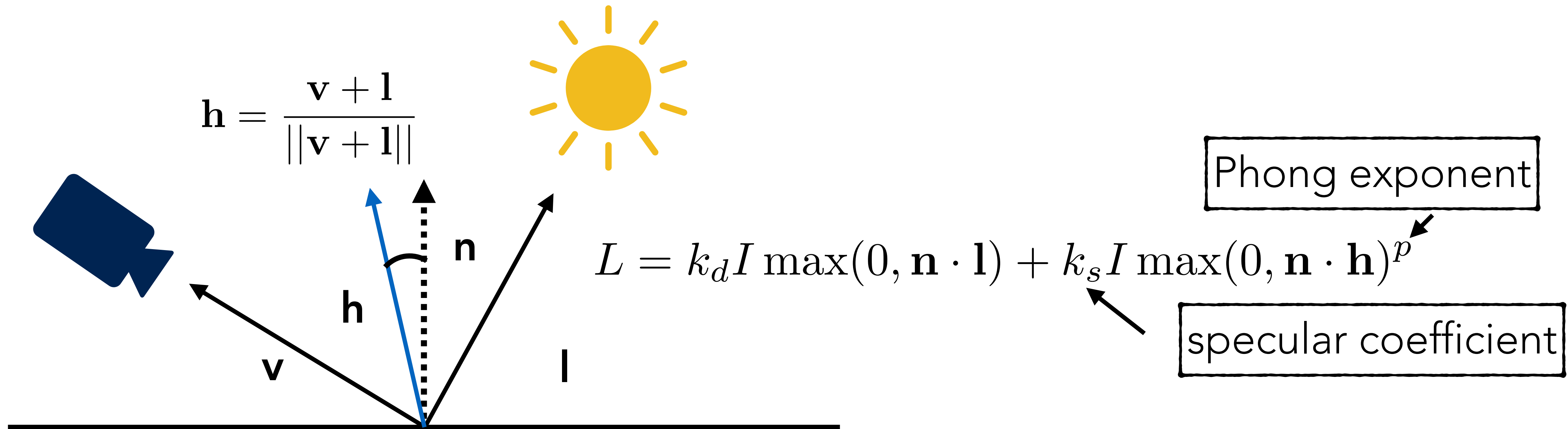
Specular Shading

- To measure the asymmetry we measure the angle between **h** (the bisector of **v** and **l**) and **n**



Specular Shading

- To measure the asymmetry we measure the angle between **h** (the bisector of **v** and **l**) and **n**

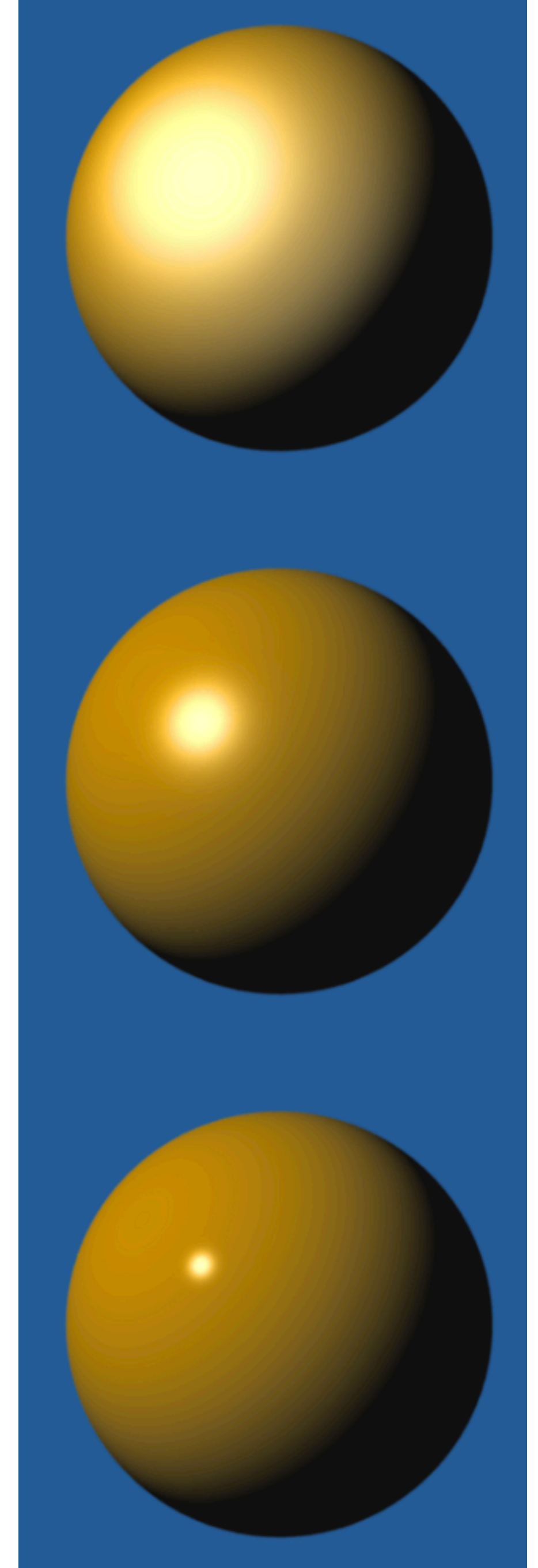
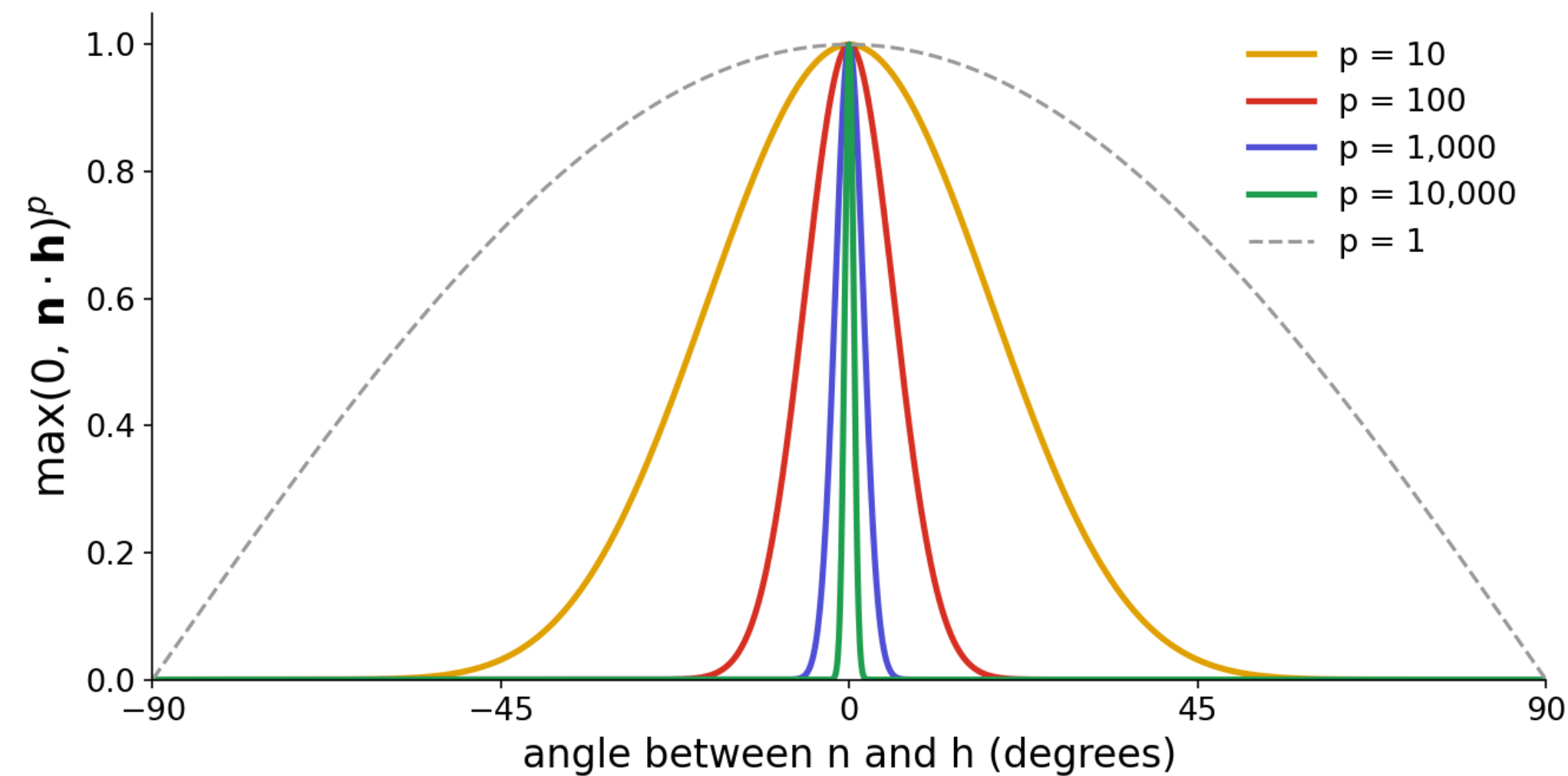


Specular Shading

- p : Phong exponent controls the size of the highlight

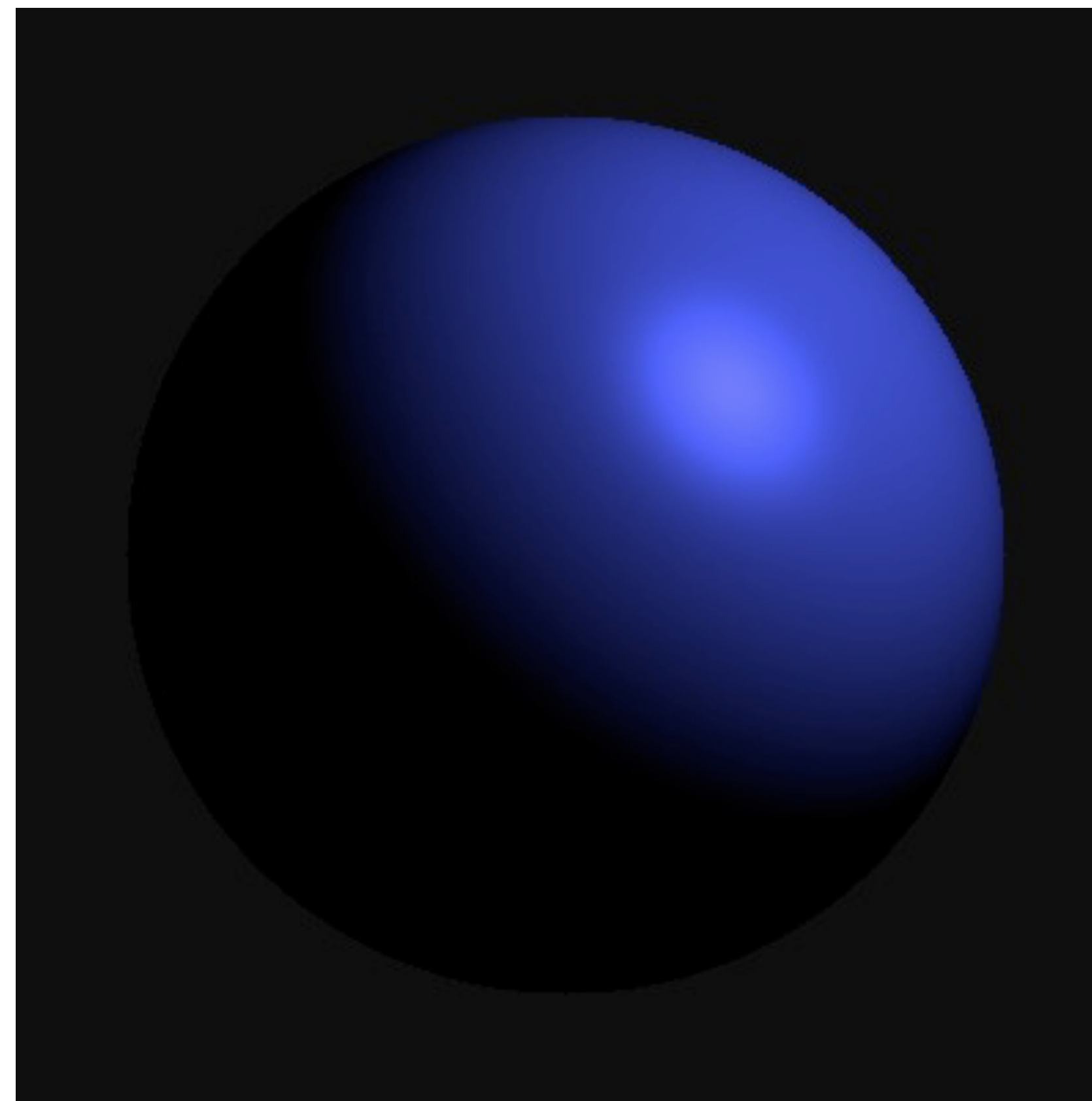
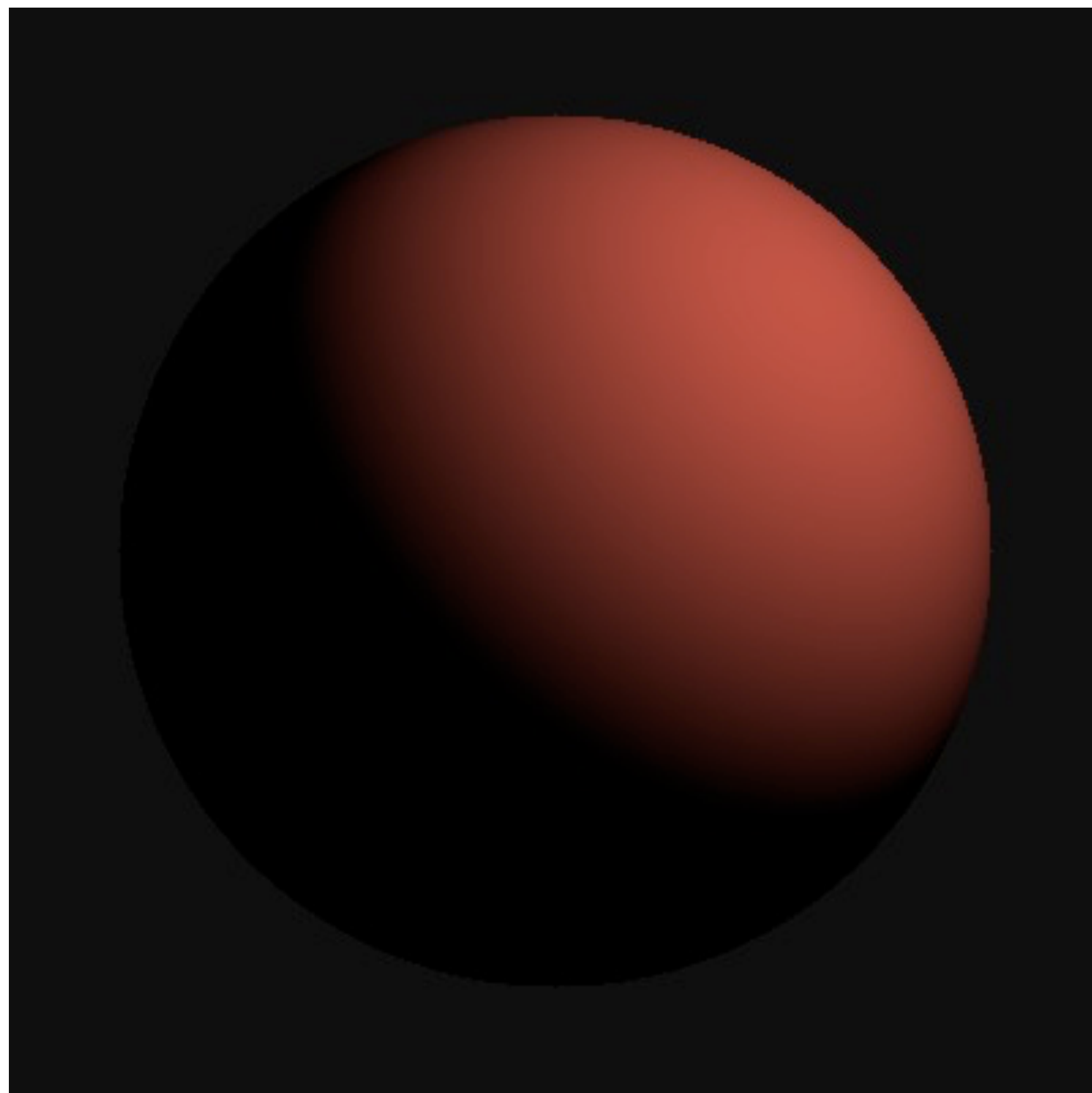
$$\max(0, \mathbf{n} \cdot \mathbf{h})^p$$

$p = 100 \rightarrow$ Shiny
 $p = 1000 \rightarrow$ Glossy
 $p > 10,000 \rightarrow$ Mirror



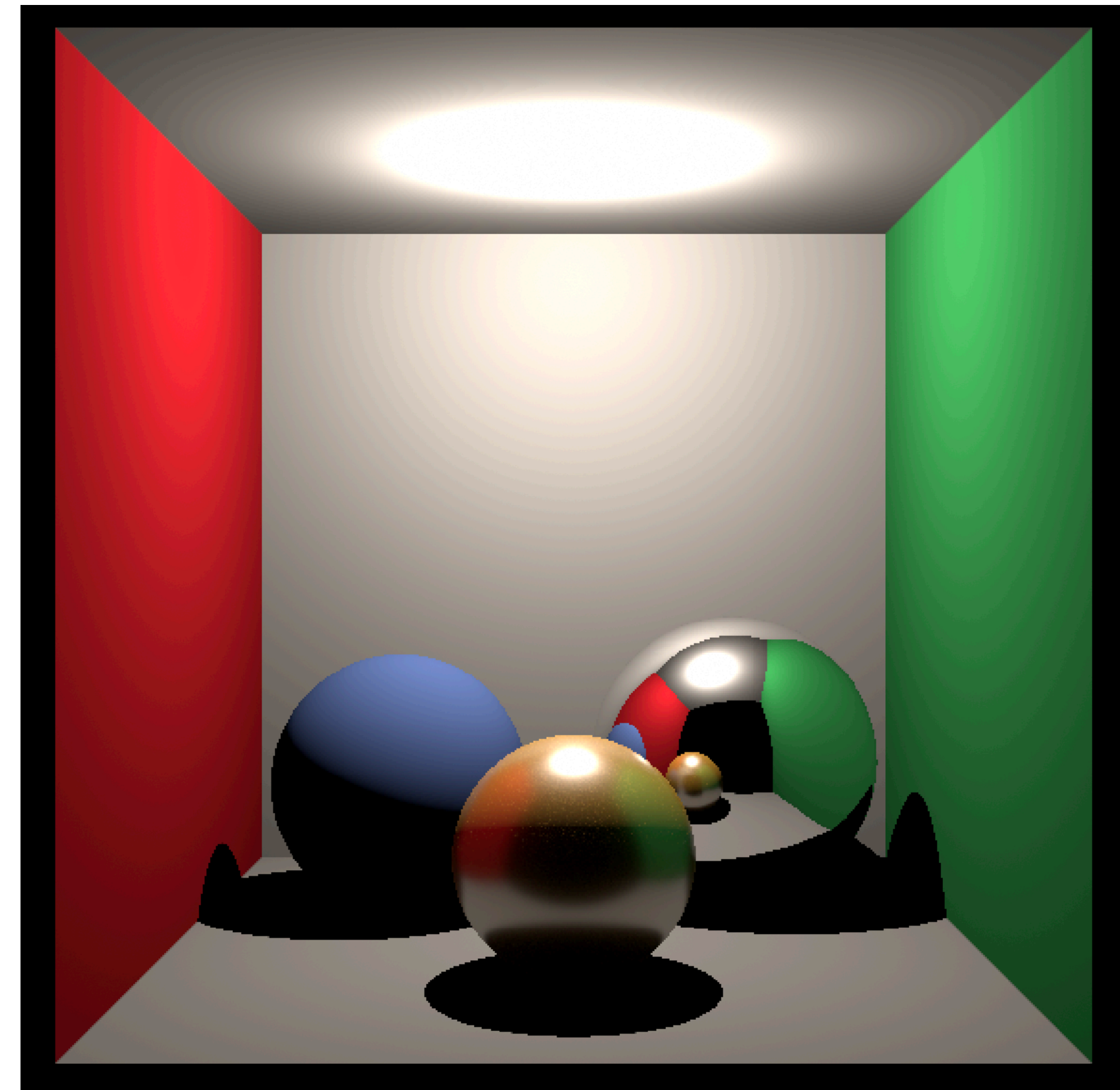
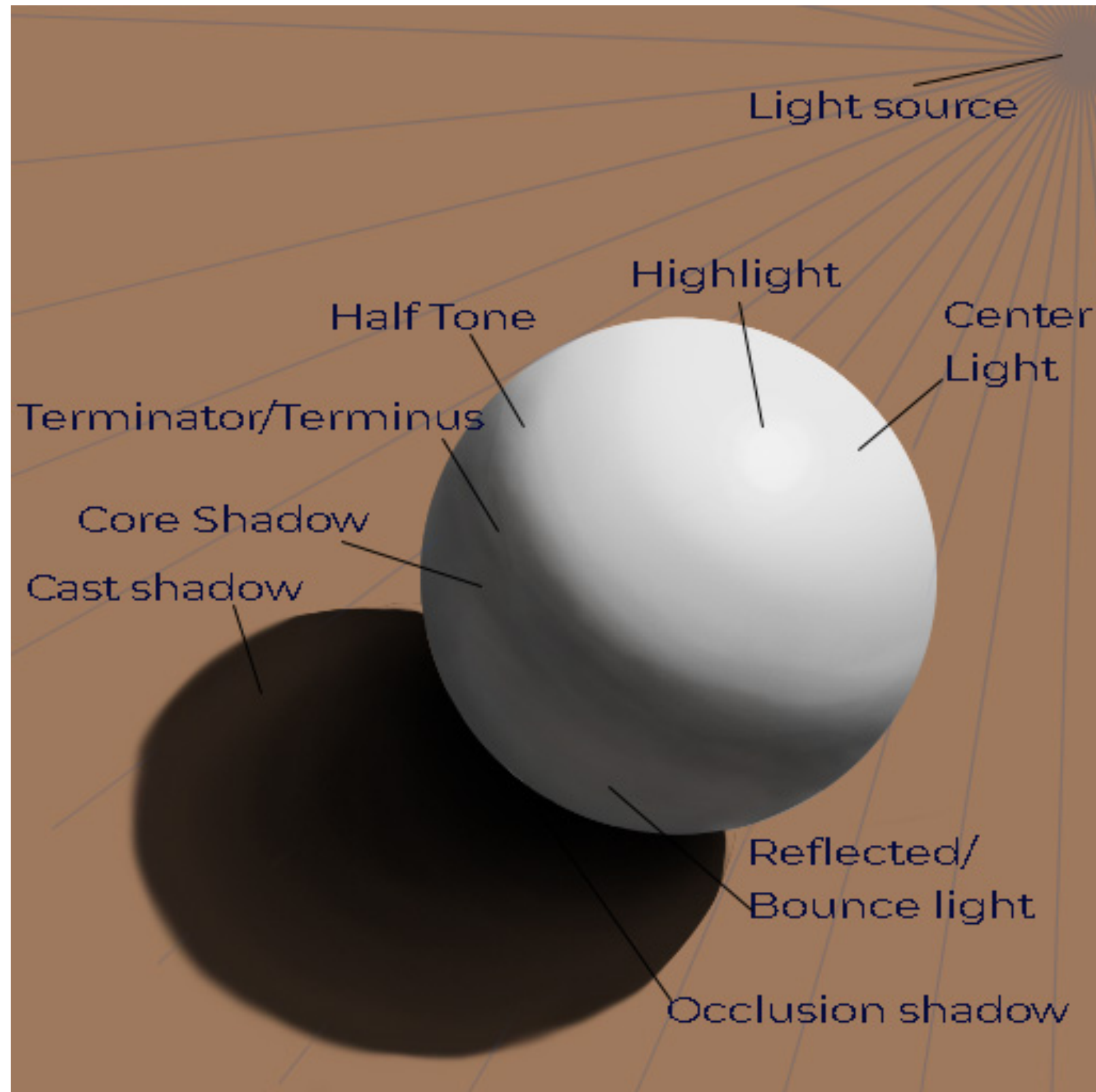
<https://commons.wikimedia.org/wiki/File:Specularity-phong-showcase.png>

Diffuse and Specular



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Final Shading Equation



<https://cecelyv.com/how-to-draw-a-sphere>

What is missing here?

Final Shading Equation

$$L = k_a I_a + k_d I \max(0, \mathbf{n} \cdot \mathbf{l}) + k_s I \max(0, \mathbf{n} \cdot \mathbf{h})^p$$



Ambient



Diffuse

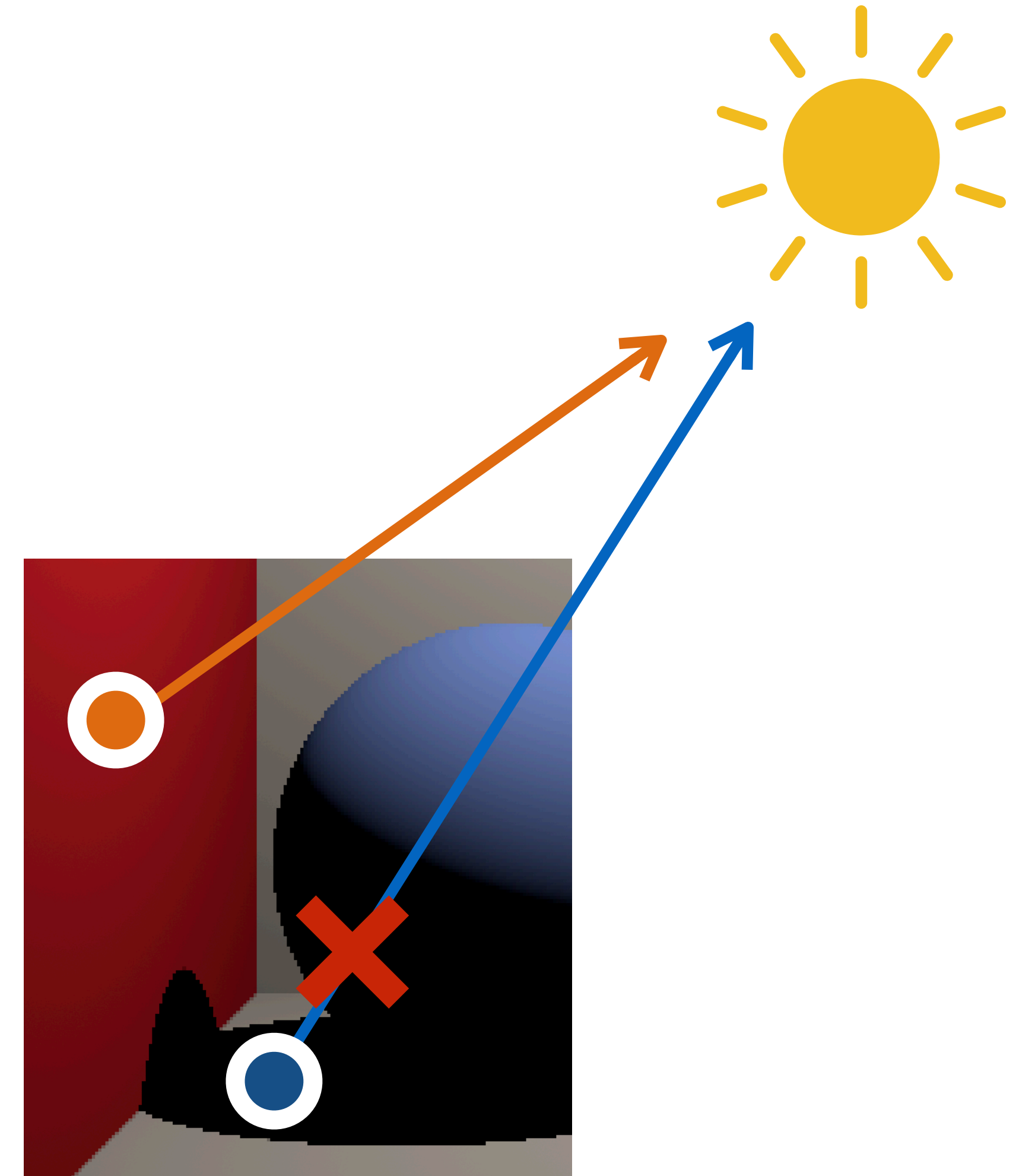


Specular

If you have multiple lights, simply sum them up all together.
Note that the ambience light should be considered only once.

Shadows

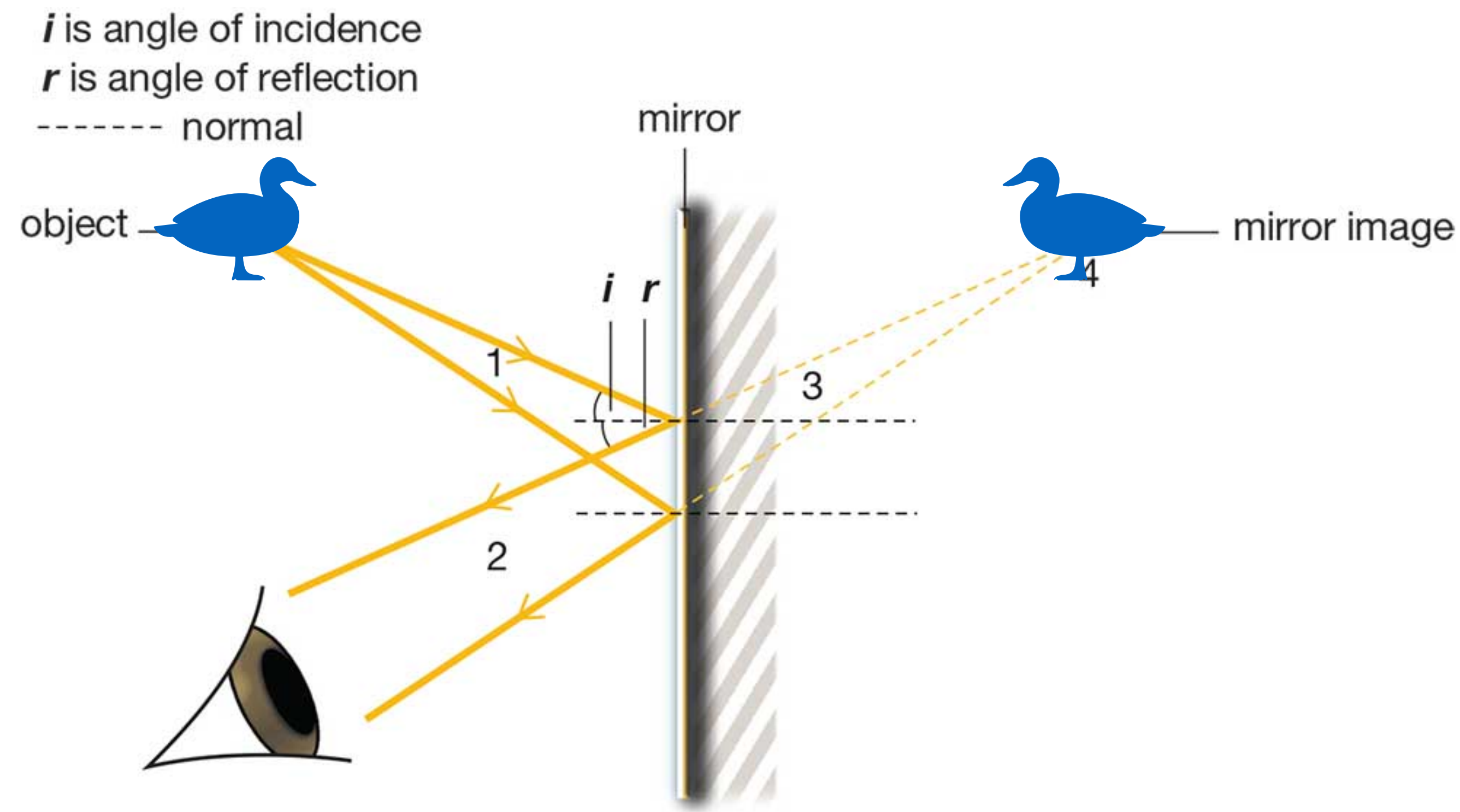
- The blue point does not receive light, while the orange one does
- To check it, cast a ray from each point to the light — if you intersect something (before reaching the light) then it is in a shadow area, and the light should not contribute to its color
- These rays are usually called **shadow rays**



The shadow rays should be casted an epsilon away from the source

Ideal Reflections

- It is easy to add ideal reflections (also called mirror reflections) to your ray tracing program

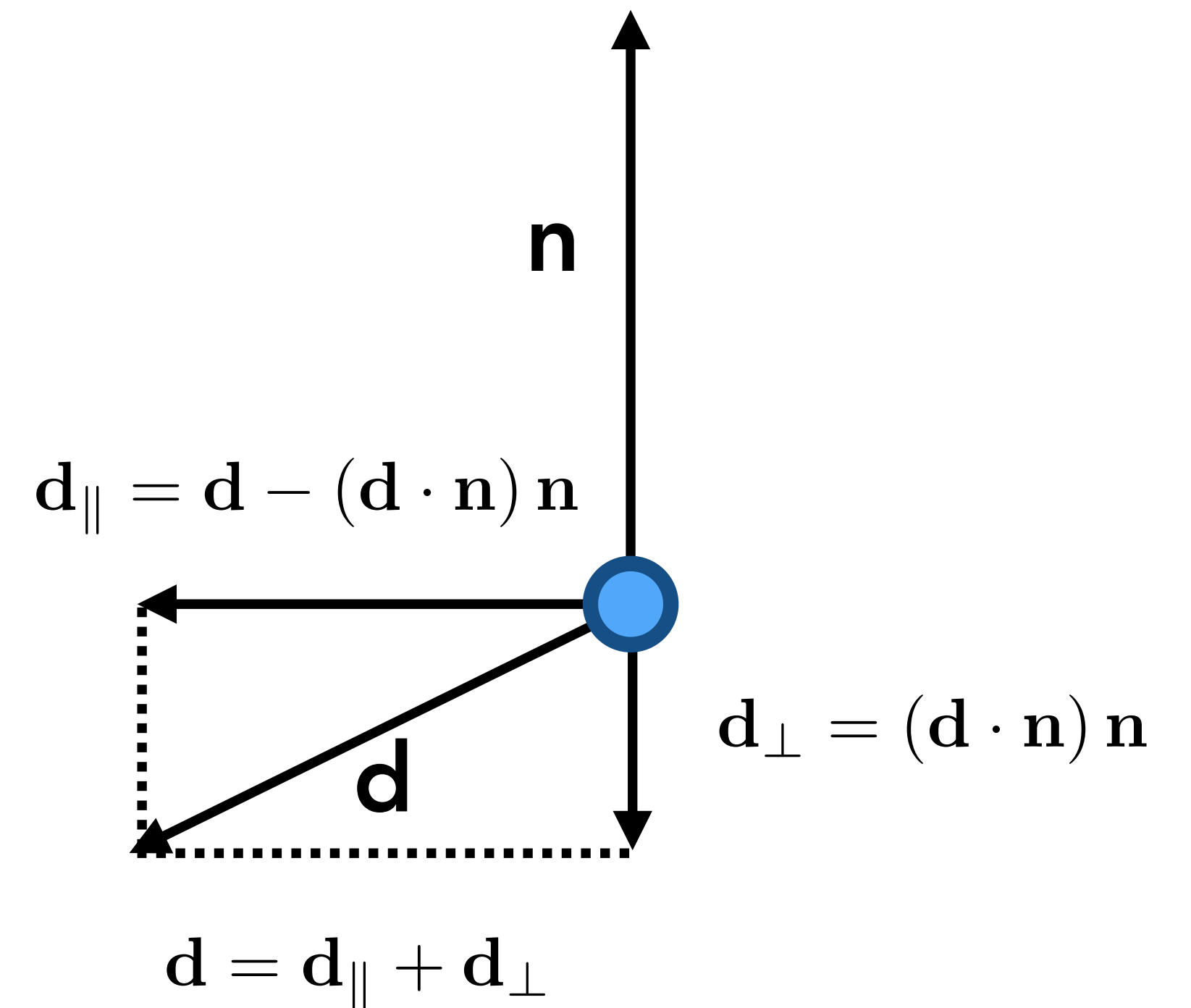
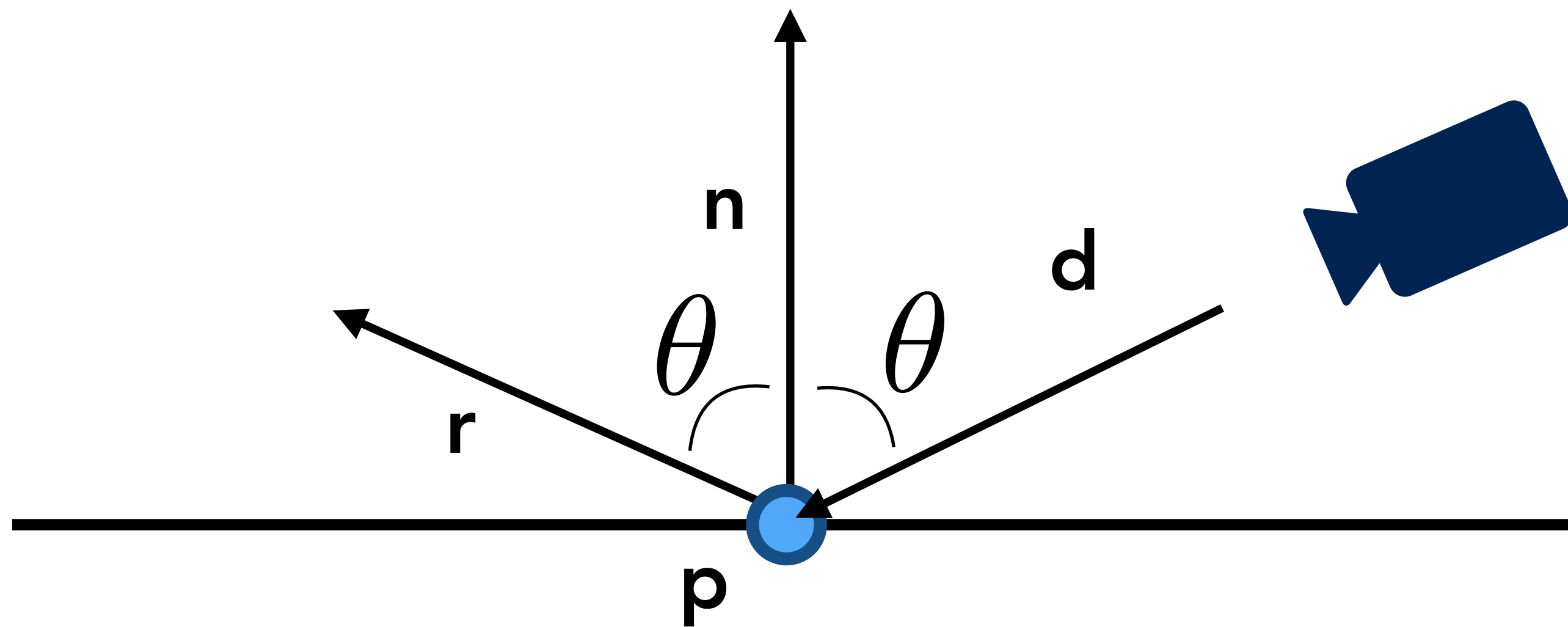


By Prabhu B - <http://www.flickr.com/photos/kshathriya/851429608/sizes/l/>, CC BY 2.0, <https://commons.wikimedia.org/w/index.php?curid=5779465>

https://www.igcsephysics.com/cambridge/sample/waves_light_and_sound/page132.html

Ideal Reflections

- It is easy to add ideal reflections (also called mirror reflections) to your ray tracing program



$$\mathbf{r} = \mathbf{d}_{\parallel} - \mathbf{d}_{\perp}$$

$$\mathbf{r} = \mathbf{d} - 2(\mathbf{d} \cdot \mathbf{n}) \mathbf{n}$$

Ideal Reflections

- It is easy to add ideal reflections (also called mirror reflections) to your ray tracing program

$$\mathbf{r} = \mathbf{d} - 2(\mathbf{d} \cdot \mathbf{n})\mathbf{n}$$

$$L = L_{\text{local}} + k_m \text{raycolor}(\mathbf{p} + s\mathbf{r}), \quad s \in [\epsilon, \infty)$$

Mirror reflectance

Limit the recursion depth!

A simple ray-tracing program

- The source code of Assignment 2 is a simple ray tracer

Conclusions

- Ray tracing is an effective way to render images
- Specular reflection and Shadows are straightforward to implement
- It is ubiquitously used even in rasterization pipelines to “pick” objects

References

Fundamentals of Computer Graphics, Fourth Edition
4th Edition **by Steve Marschner, Peter Shirley**

Chapters 4, 10, 13